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Out of the League
Family Medicine

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TOPIC: Alcohol problem drinking:

Definition

- The Chief Medical Officer's low risk drinking guideline advises that:
    - It is safest for both men and women not to drink more than 14 units a week on a regular basis.
    - People who drink as much as 14 units per week should spread this evenly over 3 days or more.
    - During pregnancy, the safest approach is to avoid drinking alcohol.

Whom to screen

- If screening everyone is not feasible or practicable, focus on people who have an alcohol-related condition or who are at increased risk of harm from alcohol. This includes people:
    - With relevant physical conditions (such as hypertension, gastrointestinal disorders, or liver disorders).
    - With relevant mental health problems (such as anxiety, depression, or other mood disorders).
    - Who have been assaulted.
    - At risk of self-harm.
    - Who regularly experience accidents or minor traumas.
    - Who regularly attend genito-urinary medicine clinics or seek emergency contraception.
- Always screen people who make an active call for help, or in whom an alcohol problem is strongly indicated as an incidental finding.
    - An active request for help (either from the person, or through friends or relatives) is more likely in people with harmful drinking or alcohol dependence. People who are drinking levels below this but above the low risk drinking guidelines will not usually seek medical help, even though they may be aware that their drinking is putting them at risk.
    - Incidental findings that raise suspicion of problem drinking may include:
        - Abnormal blood tests such as a raised gamma-glutamyl transferase (GGT) and mean corpuscular volume (MCV).
        - Signs of an alcohol problem, such as dilated facial capillaries, bloodshot eyes, or hand tremor.
        - Symptoms suggestive of a possible alcohol problem — professionals should be aware of the complications and comorbid diseases associated with an alcohol problem, or have a raised suspicion of an alcohol problem based on an individual's behaviour (for example use of illicit drugs, smelling of alcohol in consultations, numerous accidents, or requesting numerous sick notes). Recognizing these cues should initiate further questioning and the opportunistic (targeted) detection of a potential alcohol problem.

Screening tool:

Use the AUDIT (Alcohol Use Disorders Identification Test) questionnaire to routinely assess the nature and severity of alcohol misuse. AUDIT scores are interpreted as:

- 0-7 indicates low-risk.
- 8-15 indicates increasing risk.
- 16-19 indicates higher risk.
- 20 or more indicates possible dependence.

The AUDIT-C questionnaire consists of the first 3 questions from AUDIT and can be used where time is limited.

- If the score is 5 or more, and time permits, complete the remaining AUDIT questions to obtain a full score.

If the result of the AUDIT questionnaire suggests alcohol dependence, consider assessing the severity using the SADQ (Severity of Alcohol Dependence Questionnaire) or LDQ (Leeds Dependence Questionnaire).

Diagnosis

- The Diagnostic and Statistical Manual of Mental Disorders Text Revision (DSM-5-TR) defines AUD as a problematic pattern of alcohol use leading to clinically significant impairment or distress, when manifested by at least two of the following, occurring within a 12-month period:
  - Alcohol is often taken in larger amounts or over a longer period than was intended.
  - There is a persistent desire, or unsuccessful efforts, to cut down or control alcohol use.
  - A great deal of time is spent in activities necessary to obtain alcohol, use alcohol, or recover from its effects.
  - Craving, or a strong desire or urge to use alcohol.
  - Recurrent alcohol use resulting in a failure to fulfill major role obligations at work, school, or home.
  - Continued alcohol use despite having persistent or recurrent social or interpersonal problems caused or exacerbated by the effects of alcohol.
  - Important social, occupational, or recreational activities are given up or reduced because of alcohol use.
  - Recurrent alcohol use in situations in which it is physically hazardous.
  - Alcohol use is continued, despite knowledge of having a persistent or recurrent physical or psychological problem that is likely to have been caused or exacerbated by alcohol.
  - Tolerance, as defined by either of the following:
    - A need for markedly increased amounts of alcohol to achieve intoxication or the desired effect.
    - A markedly diminished effect with continued use of the same amount of alcohol.
  - Withdrawal, as manifested by either of the following:
    - The characteristic withdrawal syndrome for alcohol.
    - Alcohol (or a closely related substance, such as a benzodiazepine) is taken to relieve or avoid withdrawal symptoms.
- AUD can be:
  - Mild — 2 to 3 symptoms
  - Moderate — 4 to 5 symptoms.
  - Severe — 6 or more symptoms.

Withdrawal symptoms

- Common mild or moderate symptoms include:
    - Anxiety.
    - Nausea and vomiting.
    - Autonomic dysfunction — tremor, tachycardia, sweating, palpitations.
        - Mild tremor may not be seen, but can be felt fingertip to fingertip.
        - Moderate tremor can be seen with arms extended.
        - Severe tremor can be seen even without arms extended.
    - Insomnia.
- Features of mild or moderate withdrawal typically last for 3 to 7 days after cessation of alcohol use.
- Severe symptoms include:
    - Delirium (delirium tremens) — this occurs in around 2% of cases. It has a rapid onset, is difficult to control, and is a medical emergency. Symptoms tend to appear 48-72 hours after the last alcoholic drink and may include:
        - Profound confusion/delirium.
        - Visual, auditory, and tactile hallucinations — this affects up to 25% of people.
        - Coarse tremor.
        - Features of clinical instability, such as tachycardia, fever, ketoacidosis, and circulatory collapse.
    - Seizures — these may be the first manifestation of alcohol withdrawal in some people, and normally occur in the first 12-24 hours after the person's last drink. Generalised tonic-clonic seizures occur in 1-3% of cases, and require urgent treatment.
- Other uncommon symptoms include:
    - Agitation.
    - Anorexia.
    - Depression.
    - Fever or hypothermia.
    - Headache.
    - Hypertension or hypotension.

- For people in acute alcohol withdrawal, or who are assessed to be at high risk of developing alcohol withdrawal seizures or delirium tremens — offer admission to hospital for immediate medically assisted alcohol withdrawal.
    - Consider a lower threshold for admission for people who are frail, have cognitive impairment or multiple comorbidities, lack social support, or have learning difficulties.
- Urgently admit people with clinical features of Wernicke's encephalopathy (ataxia, ophthalmoplegia, nystagmus, acute confusional state, or more rarely hypotension or hypothermia) for treatment with parenteral thiamine.
- Urgently refer people who are assessed to be at high risk of suicide to mental health services.
- Refer all people with alcohol misuse or suspected alcohol misuse and a coexisting psychotic disorder to secondary care mental health services for assessment and further management.
- For people who are alcohol dependent but do not need to be admitted to hospital — offer advice to avoid a sudden reduction in alcohol intake and information about how to contact local alcohol support services.
- Note: People who are dependent on alcohol should be advised that they are required by law to notify the DVLA and will be required to surrender their driving license for a period. Healthcare professionals involved in their care may be obliged to disclose relevant medical information about them, in confidence, to the DVLA if they continue to drive when they are not fit to do so.

- For people being managed in primary care:
    - Offer a session of structured brief adviceabout alcohol consumption. If this cannot be offered immediately, offer an additional appointment as soon as possible. Routinely monitor their progress in reducing their alcohol consumption to a low-risk level. Where required, offer an additional session of structured brief advice.
    - If the person has not responded to structured brief advice, offer an extended brief intervention. All people receiving an extended brief intervention should be followed-up, and up to five sessions may be given.
        - An extended brief intervention may also be helpful for people who are alcohol dependent but are reluctant to accept a referral to specialist alcohol services.
    - For harmful drinkers and people with mild alcohol dependence, offer a psychological intervention focused specifically on alcohol-related cognitions, behaviour, problems, and social networks. Behavioural couples therapy should be offered if the person has a regular partner who is willing to participate in treatment.
    - Offer prophylactic oral thiamine to harmful or dependent workers:
        - If they are malnourished or at risk of malnourishment.
        - If they have decompensated liver disease.
        - If they are in acute withdrawal.
        - Before and during a planned medically-assisted alcohol withdrawal.
- Arrange follow-up after the initial appointment based on the person's choice and the goals set.
- Primary care follow-up under a shared care agreement may be appropriate for people who have completed a detoxification programme and who are taking acamprosate for maintenance therapy.

Prescribe information

Thiamine

Contraindications and cautions

There are no noted contraindications or cautions for thiamine.

Adverse effects

- **Adverse effects of thiamine include:**
    - Mild gastrointestinal events such as nausea, vomiting, diarrhoea, and abdominal pain.
    - Allergic and anaphylactic reactions, with symptoms of pruritus, urticaria, itching, hives, angioedema, abdominal pain, respiratory distress, tachycardia, palpitations, and shock (rare).

Drug interactions

- There are no clinically significant drug interactions with thiamine.

Dose

- For people with:
    - Mild deficiency — prescribe 50-100 mg daily.
    - Severe deficiency — prescribe 200-300 mg daily in divided doses.

Acamprosate

Contraindications and cautions

Do not prescribe acamprosate to:

- People with renal impairment if serum-creatinine is greater than 120 micromol/ L.
- Pregnant women (unless the benefit outweighs the potential risk).
- Women who are breastfeeding.

Adverse effects

- Common or very common adverse effects of acamprosate include:
  - Abdominal pain, diarrhoea, flatulence, nausea, and vomiting.
  - Sexual dysfunction.
  - Skin reactions.

Drug interactions

- There are no clinically significant drug interactions with acamprosate.

Dose and monitoring

Dose

- Acamprosate treatment should be started as soon as possible after assisted withdrawal and prescribed for up to 6 months, or longer for those benefiting from the drug who want to continue with it.
- The dose is usually 666 mg three times daily (1998 mg daily).
  - For people who weigh less than 60 kg, the dose is 666 mg once daily at breakfast, and 333 mg twice daily at midday and night (1332 mg maximum daily dose).

Monitoring

- People taking acamprosate should be supervised at least monthly for 6 months, and at reduced but regular intervals if the drug is continued after 6 months.
- Stop acamprosate if drinking persists for 4-6 weeks after starting the drug.
- Blood tests are not required routinely, but may be considered to monitor recovery of liver function and may aid motivation.

Psychological intervention

- Psychological interventions focused on alcohol-related problems include:
  - Cognitive behavioural therapies and behavioural therapies — these should usually consist of one 60-minute session per week, for 12 weeks.
  - Social network and environment-based therapies — these should usually consist of eight 50-minute sessions over 12 weeks.
  - Behavioural couples therapy — this should also focus on the impact on relationships. It should aim for abstinence, or a level of drinking predetermined and agreed by the therapist and the service user to be reasonable and safe. It should usually consist of one 60-minute session per week, for 12 weeks.

TOPIC: Opioid dependence

Definition

- **An opioid is** either a natural derivative of opium (for example, heroin) or a synthetic substance (for example, methadone, or buprenorphine) with agonist, partial agonist, or mixed agonist and antagonist activity at opioid receptors.
- **An opiate is** either a natural derivative or a semi-synthetic constituent of opium.
- **Opioid dependence develops after a period of regular use of opioids**, with the time required varying according to the quantity, frequency and route of administration, factors of individual vulnerability and the context in which drug use occurs. Opioid dependence is not just a heavy use of opioids but a complex health connotation that has social, psychological and biological determinants and consequences, including changes in the brain. It is not a weakness of character or will.
    - Physical and psychological dependence can develop within a relatively short period of continuous use (2–10 days).
- **The key elements of opioid dependence are:**
    - A strong desire or sense of compulsion to take opioids.
    - Difficulty in controlling use.
    - A physiological withdrawal state when opioid use has ceased or been reduced.
    - Evidence of tolerance.
    - Progressive neglect of alternative pleasures and interests.
    - Persistence with opioid use despite clear evidence of overtly harmful consequences.
- Opioid dependence should be viewed as a chronic relapsing-remitting disorder which has physical, social, and psychological dimensions.

Prognosis

- Opioid dependence is a chronic disease, with relapse rates greater than 90% in untreated people. With effective treatment, relapse rates are similar to those in other chronic medical illnesses, such as diabetes, hypertension, and asthma.
- The mortality risk of people dependent on illicit diamorphine is around 12 times that of the general population.
- People with other coexisting conditions have a poorer prognosis, for example:
    - Mental health problems.
    - Cognitive impairment.

Complications

- **Complications of opioid dependency include:**
    - **Death**
    - **Overdose.**
    - **Infection — people who are dependent on opioids have a greater risk of infection, including:**
        - ***Staphylococcus aureus* and Group A *streptococci***
        - **HIV**
        - **Hepatitis B**
        - **Hepatitis C**
        - **Clostridium infections** — such infections are associated with injecting into the skin or muscle (a technique known as 'popping').
        - **Bacillus anthracis (anthrax)**
        - **Tuberculosis**

- **Deep vein thrombosis and pulmonary emboli**
    - **Superficial thrombophlebitis:**
        - Dependent users are susceptible to superficial thrombophlebitis because of repetitive trauma, unsterilized techniques, and irritation caused by the drug mixtures.
        - Street drugs are often cut with substances that promote thrombosis (for example, lactose, sucrose, dextrose).
        - Cocaine, in particular, seems to promote thrombosis.
    - **Venous and arterial thrombosis** can result from poor injecting technique, especially in people injecting into the groin.
- **Poor nutrition and dental disease** — people who have a history of substance use problems are more likely to have poorer oral and dental health, and may suffer from poor nutrition.
- **Social problems, including:**
    - Crime — it is estimated that half of all recorded crime is drug related.
    - Imprisonment.
    - Effect on partner and children including child protection issues.
    - Deprivation and social exclusion.
    - Involvement in the sex trade.
    - Homelessness.
    - Loss of driving licence.
- **Psychological problems, including:**
    - Craving and fear of withdrawal.
    - Guilt.
    - Anxiety.
    - Loss of memory or cognitive skills.

Diagnosis

Suspecting opioid dependency

- **Drug misuse may become apparent while a person is consulting with another medical problem (that may or may not be drug-related), or present in a variety of ways:**
    - **An active request for help with:**
        - Drug dependency — the user has realized that there are problems associated with taking drugs, or has become motivated to change behaviour, or a crisis has developed (perhaps as a result of an impending court case, health or relationship problems).
        - Alcohol dependency — always ask about drug misuse.
    - **A** complication **of drug use.**
    - **Clinical features of opioid intoxication** — symptoms and signs include:
        - Constriction of pupils, itching and scratching, sedation and somnolence, lower blood pressure, slower pulse, hypoventilation.
    - **Acute withdrawal syndrome** — symptoms and signs include:
        - Watering eyes, rhinorrhoea, yawning, sneezing, cool and clammy skin, dilated pupils, cough.
        - Abdominal cramps, nausea, vomiting, diarrhoea.
        - Tremor, sleep disorder, restlessness, anxiety, irritability, hypertension.
    - **Certain features:**
        - Psychiatric history — overdoses, depression, psychosis.
        - Forensic history — past custodial sentences, probation, community service.

        - Social history — family problems, unemployment, accommodation issues, financial problems.
    - On physical examination, there may be evidence of poor nutrition, dental caries, other signs of neglect, needle tracks, skin abscess, and signs of drug intoxication or withdrawal.
    - On mental health assessment, there may be indications of abnormal general behaviour, disorders of mood (particularly anxiety or low mood), delusions or hallucinations, confusion.

Initial investigations

- **Analysis of urine for drug screen:**
    - Different opioids persist in the urine for different lengths of time (for example, up to 48 hours for heroin, up to 7–9 days for methadone metabolite).
- **Mouth swab tests** (saliva and oral mucosal transudate) — this is easier to collect and harder to switch or adulterate samples. However, drugs are present in lower concentrations and there is a shorter detection window (for example, heroin can be detected only up to 24–48 hours later with mouth swabs), so only very recent drug use can be detected.
- Be aware that false-positive and false-negative results can occur. Interpret results in conjunction with clinical signs such as visible injection sites and signs of withdrawal. If necessary repeat the test.
    - Interpret inconsistent or unexpected results in conjunction with other clinical information.

Table 1. Duration of detectability of drugs in urine.

Drug or its metabolite(s)	**Duration of detectability***
Amphetamines, including methylamphetamine and MDMA	2–4 days
Buprenorphine (sublingual)	1–2 weeks
Cocaine metabolites	2–4 days
Methadone	3–4 days
Opiates (codeine, morphine)	2–3 days (metabolites 3–6 days)
Oxycodone	1–2 days
Benzodiazepines	
Short-acting	1–3 days
Long-acting	1–2 weeks (up to 6 weeks)
Cannabis	
Occasional	1–3 weeks
Chronic heavy use	4–6 weeks (may be up to 12 weeks)

Management
Assessment
How should I assess the person?

- **The purpose of the initial assessment is to:**
  - Identify and respond to any emergency or acute problem.
  - Confirm the person is taking psychoactive substances (based on history, examination and drug testing, and through accessing any relevant additional information from clinical records).
  - Identify the degree of problem use or dependence.
  - Identify physical and mental health problems.
  - Identify social problems.
  - Assess the family history for substance use and dependence and relevant medical, psychiatric or psychosocial factors.
  - Determine the person's understanding of treatment options and motivation for change.
  - Explore and identify strengths the person can use to achieve their treatment and recovery goals.
  - Determine the need for substitute medication or other prescribing for dependence.
  - Assess risk behaviours, including those associated with injecting.
- Comprehensive assessment may need to be conducted over several sessions and is then ongoing. It may be appropriate for concerned friends, relatives, carers or other professionals already involved to attend.
- **Take a history and ask about:**
  - Degree of dependence (past and current) — types of psychoactive drugs used (including prescribed and over-the-counter medicines and tobacco), quantity and frequency of use, pattern of use, routes of administration (including any injecting), sources of drugs obtained and evidence for harmful use or dependence (including any experience of withdrawal syndromes).
  - Motivation — assess the person's understanding of treatment options and motivation for change.
  - Risk behaviours, including those associated with injecting:
    - Supply of needles and syringes; sharing habits; knowledge of how to inject safely; correct disposal of used equipment; knowledge of HIV and hepatitis A, B, and C; issues of transmission and safer sex.
    - For drug-misusing parents or other adults with dependent children, obtain information on the children and any drug-related risks to which they may be exposed.
  - Medical history — current or previous physical complications of drug use such as infection with blood-borne viruses or continuing related risky behaviours, liver disease, abscesses, overdoses, enduring severe physical disabilities and sexual health problems, last menstrual period (in order to consider pregnancy and contraception), last cervical smear.
  - Psychiatric history — current or previous psychological problems, such as personality problems and disorders, self-harm, history of abuse or trauma, depression, anxiety and severe psychiatric comorbidity (details of contact with mental health services should be recorded).
  - Family history of substance use and dependence and relevant medical, psychiatric or psychosocial factors.
  - Social problems:
    - Problems in personal relationships (including with partners) and of social integration, including domestic violence and abuse, family, housing and living arrangements, education, employment, benefits and financial problems.

        - Childcare issues, including parenting, pregnancy and child protection.
        - Criminal involvement, offending and other legal issues, including arrests, fines, outstanding charges and warrants, probation, imprisonment, violent offences and criminal activity, and involvement with workers in the criminal justice system.
        - Consider asking sensitively, where relevant, how drug use is funded (for example, sex working, diversion of family income).
    - Any drug-related risks to which the person's own children or other children in the home may be exposed to. Any important current risks should be clarified and reflected in a risk management plan. The assessment should take into consideration the:
        - Effect of drug misuse on functioning, for example, intoxication, agitation.
        - Effect of drug-seeking behaviours, for example, leaving children unsupervised, contact with unsuitable characters.
        - Impact of parent's physical and mental health on parenting.
        - Emotional availability to children.
        - Effects on family routines, for example, getting children to school on time.
        - Other people supporting the children, for example, family, and their fitness to provide that support.
        - Ability to access professional support.
        - Storage of illicit drugs, prescribed medication and drug-using paraphernalia.
    - Past contact with treatment services — previous efforts to reduce or stop taking drugs; past contact with treatment and rehabilitation, social, medical, and community services.
    - Strengths, including:
        - Personal, family, social and other strengths and positive networks they can use to achieve their treatment and recovery goals.
        - Past successes and difficulties in achieving stability or in making improvements (for example, what helped in the past and what didn't help or what caused any relapse).
- **Examine the person in order to:**
    - Confirm evidence of drug misuse (for example, needle tracks, signs of drug intoxication or withdrawal).
        - Injecting into the groin or neck are indications of high-risk drug use which may benefit from specialist advice or referral.
    - Check for the presence of complications.
    - Assess mental health.
    - Assess respiratory, cardiovascular and other body systems.
- **Consider investigations to exclude complications** — the choice of investigations is guided by clinical judgement, but may include:
    - Tests for hepatitis B, and C, and HIV.
    - Liver, renal or thyroid function tests.
    - Full blood count (to exclude anaemia, signs of infection).
    - Neurological examination.

Assessing suitability for substitution therapy

- **Substitution therapy (maintenance or detoxification) should normally only be considered if:**
    - Opioids are being taken on a regular basis – usually daily.
    - There is convincing evidence of current dependence.
    - The initial assessment clearly substantiates the diagnosis and the need for treatment.

    - The clinician is satisfied that the person may be able to comply with the prescribing regimen.
    - The person is not receiving an opioid prescription for management of dependence from another clinician.
- **If possible, check liver function tests (LFTs) at initial assessment** — methadone and buprenorphine both undergo extensive metabolism in the liver.
    - If the person is well, do not delay starting methadone or buprenorphine whilst waiting for the results.
- **If liver function tests are normal, explain that they should be monitored every 6–9 months whilst taking methadone or buprenorphine.**
- Seek specialist advice if there is:
    - Impaired hepatic function.
    - A marked deterioration in liver function.
- In people with advanced liver disease the risks from methadone or buprenorphine are insignificant compared to the risks from street drugs.
    - People with end-stage liver disease should only be started on methadone or buprenorphine with extreme care and should be referred to a specialist.

Information and advice

- **Offer information and advice on:**
    - Immunisation against hepatitis B (and possibly hepatitis A). Ideally, vaccinate all drug users against hepatitis A and hepatitis B.
        - Use the accelerated schedule for hepatitis B (given on days 0, 7, and 21) with a booster at 12 months.
        - Advise that partners and children should be vaccinated against hepatitis B.
    - Immunisation against tetanus — ensure that the person is immunized against tetanus, as people who inject drugs are at increased risk of tetanus. Give booster doses if there is any doubt about the person's immunization status.
    - Blood-borne viruses and other infections — provide safer injecting advice and ensure that the person has access to clean injecting equipment, needles and syringes.
    - Inform the person about the signs and symptoms of overdose and advise them to ensure that family and friends are also aware of these signs and to seek immediate medical help if they occur.
    - Overdose prevention training and take-home naloxone.
    - Contraception and safer sex — refer to a sexual health service if appropriate.
    - Local NHS dental services or direct referral to special care dental services if appropriate.
- **Explain the benefits the person can expect** from reducing the medicine and aim to reach agreement using a shared decision-making approach.
- **Understand that the person might be reluctant or anxious about discussing problems** associated with dependence. Reassure them that dependence is an expected effect of these medicines and that problems associated with dependence sometimes develop. Be sensitive to the use of terminology that may apportion blame to the person or be perceived adversely.
- **Acknowledge and discuss with the person any differences between their views and your own** about the risks and benefits of the medicine.
- **Be prepared for queries about prescribing decisions made previously.** Explain that our understanding of the balance of risks and benefits of a medicine can change over time. If sufficient clinical detail is available, discuss the possibility that past prescribing was done in the person's best interests using the knowledge available at the time.
- **Offer carers specific information and advice about:**

    - The risks from blood-borne viruses and overdose and, if appropriate, offer vaccination or referral.
    - Overdose and naloxone availability.
    - Safe storage of medicines.
- **Inform anyone who is opioid dependent or who is persistently using opioids that they need to inform the Driver and Vehicle Licensing Agency (DVLA) and explain that persistent use of, or dependence on, heroin, morphine, buprenorphine, or methadone will lead to refusal or revocation of a driving licence.**
    - Some individuals may refuse to do this, and in certain circumstances confidentiality may need to be overridden. Whether the GP should breach confidence and inform the DVLA without the person's consent is a complex ethical issue. GPs may need to consult their defence union.
    - A licence will be considered only if the person has been free of drug use for at least 1 year for Group 1 entitlement (car and motorcycle), and at least 3 years for Group 2 entitlement (bus and lorry). An exception may be made for an individual complying fully with a consultant-supervised methadone or buprenorphine maintenance programme.

Management of withdrawal symptoms

- **Do not be pressurized into prescribing opioids in an unplanned manner in order to abolish withdrawal symptoms** — all people require an adequate assessment. Find a balance that allows for the necessary information to be obtained efficiently without adding delay and without placing the person at risk of dropping out of treatment. The extent of the initial assessment will depend on the individual.
    - A more comprehensive assessment can take place over the following weeks during treatment optimisation.
    - Seek specialist advice if required.
- **Warn the person about the dangers of overdose if street drugs are used to alleviate symptoms, especially if injected** — explain that loss of tolerance may lead to an unintentional overdose and death.
- **Consider using lofexidine, which is licensed for the management of symptoms of opioid withdrawal** — the treatment course is between 7–10 days with doses starting at 800 micrograms daily and rising to a maximum of 2.4 mg in divided doses. The dose is then reduced over subsequent days.
- **Consider prescribing symptomatic treatment to help with the symptoms of withdrawal:**
    - Diarrhoea — loperamide 4 mg immediately followed by 2 mg after each loose stool for up to five days. The usual dose is 6-8 mg daily, maximum 16 mg daily.
    - Nausea and or vomiting — metoclopramide 10 mg eight-hourly (for a maximum of five days to minimise neurological and other adverse reactions) or prochlorperazine 5 mg three times a day or 12.5 mg intramuscularly 12-hourly.
    - Stomach cramps — mebeverine 135 mg three times daily.
    - General aches and pains (for example, headache, muscular pains)— paracetamol, ibuprofen or other nonsteroidal anti-inflammatory drugs or topical rubefacients.
    - Agitation and anxiety, sleeplessness — diazepam (oral) up to 5–10 mg three times daily when required (or zopiclone 7.5 mg at bedtime for people who have been dependent on benzodiazepines).
        - In severe cases of anxiety and agitation, obtain suitable psychiatric advice from an addiction psychiatrist or the on-call duty psychiatrist.

TOPIC: Bipolar Disorder

Definition

- Bipolar disorder (also known as bipolar affective disorder or manic depressive disorder) is a serious long-term mental illness, which is usually characterized by episodic depressed and elated moods and increased activity (hypomania or mania).
    - A manic episode is a distinct period during which there is abnormally and persistently elevated, expansive, or irritable mood lasting at least 1 week, accompanied by at least three additional symptoms, and which:
        - Is severe enough to cause marked impairment in social or occupational functioning or necessitate hospitalization, or
        - Includes psychotic features such as delusions or hallucinations.
    - A hypomanic episode is similar to a manic episode except that a diagnosis only requires that symptoms have lasted for 4 days, the episode is not severe enough to cause marked impairment in social or occupational functioning or necessitate hospitalization, and there are no psychotic features.
    - A depressive episode is a period of at least 2 weeks during which there is either depressed mood or loss of interest or pleasure in nearly all activities (or irritability in children and adolescents), accompanied by at least four additional symptoms.
    - A mixed episode is:
        - A mixture or rapid alternation of manic and depressive symptoms, or
        - A period of time (at least 1 week) in which the criteria are met for either a manic or hypomanic episode and at least three symptoms of depression are present during the majority of the days of the current or most recent episode of mania or hypomania, or
        - A period of time (at least 2 weeks) in which the criteria for a major depressive episode are met and at least three manic or hypomanic symptoms are present during the majority of the days of the current or most recent episode of depression.
- Rapid-cycling bipolar disorder is defined as the experience of at least four depressive, manic, hypomanic, or mixed episodes within a 12-month period.

Causes

- Bipolar disorder is thought to be caused by an interplay between genes and environment:
    - Bipolar disorder is known to be one of the most heritable psychiatric disorders.
    - First-degree relatives of a person with bipolar disorder have a lifetime risk of developing the illness that is at least five times greater than that of a person in the general population.
- Environmental factors/triggers that have been associated with the onset, or with relapses of bipolar disorder include:
    - Early life stress, maternal death before a child reaches five years of age, childhood trauma, childhood abuse, emotional neglect/abuse.
    - Toxoplasma gondii exposure.
    - Cannabis use, cocaine exposure.

Diagnosis

- The Diagnostic and statistical manual of mental disorders (DSM-5) makes a distinction between bipolar I disorder and bipolar II disorder:

- Bipolar I disorder is characterized by at least one manic episode with or without a history of major depressive episodes.
- Bipolar II disorder is characterized by one or more major depressive episodes and by at least one hypomanic episode, but no evidence of mania.

Management in Primary care

- Refer all people with suspected bipolar disorder to a specialist mental health service to confirm the diagnosis, treat the acute episode, and establish a care plan. Clinical judgement should be employed to determine urgency of admission/referral, based on the results of an assessment of the risk of harm to the person and/or others:
  - Determine the risks of harm to others by assessing:
    - The risk of neglect of people dependent on them for care, in particular family, children, and any other dependents. Follow local safeguarding procedures if appropriate.
    - Any risk to the public, especially if there is a risk of aggression or previous history of violence.

Follow up and review

- When carrying out a physical health review:
  - Ask about symptoms that could be due to adverse effects of medication.
  - Ensure that there are arrangements in place for monitoring of lithium, valproate, and antipsychotics (if applicable).
  - Ask about alcohol intake and substance misuse, and encourage people who smoke to stop. Note that smoking enhances the metabolism of olanzapine and clozapine.
  - Ask about the person's diet, and level of physical activity, check the person's weight, and measure their waist circumference.
  - Measure the person's pulse and blood pressure, and assess and manage the person's cardiovascular risk. .
  - In addition, assess for respiratory conditions such as chronic obstructive pulmonary disorder (COPD) if appropriate.
  - Perform the following blood tests:
    - Fasting glucose, $HbA1_c$.
    - Lipid profile.
    - Urea and electrolytes.
    - Full blood count.
    - Liver function tests.
    - Thyroid function and calcium levels if the person is taking long-term lithium.

Prescription information
SSRI (mentioned in depression)
Antipsychotics
General information

- The most commonly prescribed antipsychotics to treat bipolar disorder are the second generation anipsychotics olanzapine, quetiapine, and risperidone.
- An antipsychotic may be used in monotherapy or prescribed concurrently with lithium or valproate.
- CKS recommends that people requiring a change in the dose of their antipsychotic or a change of drug should be referred to specialist mental health services, or that specialist advice should be sought.

- When discontinuing antipsychotics, the dose should be reduced gradually over at least 4 weeks if the person is continuing with other antimanic drugs. If the person is not continuing with other antimanic drugs or if there is a history of manic relapse, a withdrawal period of up to 3 months should be considered.

How should I monitor someone taking antipsychotics?

- **When antipsychotics are initiated, baseline measurements should be taken in secondary care. Many adults with bipolar disorder will remain under the responsibility of the secondary care team for the first 12 months,** or until their condition has stabilized (whichever is longer).
- **Regular monitoring may subsequently be done in primary care on specialist advice or depending on the person's care plan.** This may include:
    - Body weight, or body mass index (BMI) - weekly for the first 6 weeks, then at 3 months. Thereafter every 12 months, or more often if the person is gaining weight rapidly.
    - Serum electrolytes and urea including creatinine and estimated glomerular filtration rate - every 12 months.
    - Full blood count - every 12 months.
    - Blood lipids - 3 months after starting treatment, then every 12 months.
    - Plasma glucose or HbA1c - at 3 months after starting treatment, then every 12 months. Additionally for clozapine and olanzapine repeat after the first month of treatment. Ask about symptoms of hyperglycaemia (such as polydipsia, polyuria, and increased appetite).
        - In some cases both plasma glucose and HbA1c may be monitored.
    - Pulse and blood pressure - during dose titration and at each dose change.
        - Not required for amisulpride, aripiprazole, trifluoperazine, and sulpiride.
    - Electrocardiography (ECG) - after dose changes. Ideally, also annually.
        - Mandatory for haloperidol, pimozide, and sertindole; not required for antipsychotics with no effect, or a low-to-moderate effect on the QT interval and where there are no other risk factors for arrhythmia.
    - Prolactin - 6 months after starting treatment, then every 12 months. Also ask about symptoms of raised prolactin (these include low libido, sexual dysfunction, menstrual abnormalities, gynaecomastia, and galactorrhoea).
        - Not required for aripiprazole, clozapine, quetiapine, or olanzapine (less than 20 mg daily).
    - Liver function tests - every 12 months.
    - Creatinine kinase if neuroleptic malignant syndrome is suspected.
    - Monitoring for the emergence of movement disorders.
- **Tests which need to be done every 12 months may be carried out at the annual physical review.**
- **Clozapine**
    - People taking clozapine are managed exclusively in secondary care. Clozapine can cause neutropenia or agranulocytosis, and frequent monitoring of the full blood count is required. This is carried out by the clozapine monitoring service.
    - Clozapine has been associated with varying degrees of impairment of intestinal peristalsis, ranging from constipation, which is very common, to very rare intestinal obstruction, faecal impaction, and paralytic ileus. People taking clozapine and their carers should be advised to seek immediate medical advice before taking the next dose of clozapine if constipation develops.
    - Note: following fatal cases involving toxicity of clozapine and other antipsychotic medicines, the MHRA advises that monitoring blood concentration of amisulpride, aripiprazole, clozapine, olanzapine, quetiapine, risperidone, and sulpiride may be

helpful in certain circumstances, such as patients presenting symptoms suggestive of toxicity, or when concomitant medicines may interact to increase blood concentration of these medicines.

What are the contraindications and cautions?

Intramuscular use of olanzapine is contraindicated in people with:

- Acute myocardial infarction.
- Bradycardia.
- Recent heart surgery.
- Severe hypotension.
- Sick sinus syndrome.
- Unstable angina.

Antipsychotics should be used with caution in people with:

- Blood dyscrasias.
- Cardiovascular disease.
- Conditions predisposing to seizures; epilepsy.
- Depression.
- Diabetes (may raise blood glucose).
- History of jaundice.
- Myasthenia gravis.
- Parkinson's disease (may be exacerbated).
- Photosensitisation (may occur with higher dosages).
- Prostatic hypertrophy.
- Severe respiratory disease.
- Susceptibility to angle-closure glaucoma.

What adverse effects of antipsychotics should I be aware of?

- **Antipsychotics can cause a wide range of adverse effects. The risk varies with the type of antipsychotic (first-generation or second-generation) and the individual drug. Adverse effects include:**
    - **Extrapyramidal symptoms** - more common with first-generation antipsychotics. They include:
        - Dystonic reactions (abnormal movements of the face and body), and pseudoparkinsonism (tremor, bradykinesia, and rigidity) - these can be alleviated by antimuscarinic drugs, such as procyclidine (should not be prescribed routinely).
        - Akathisia (motor restlessness) - can often be relieved by reducing the dose of the antipsychotic.
        - Tardive dyskinesia - late-onset movement disorder that can occur with prolonged use of antipsychotics. It is characterized by rhythmical, involuntary movements, usually lip-smacking and tongue rotating, although it can affect the limbs and trunk. It may be persistent and can sometimes worsen on treatment withdrawal. The drug should be discontinued on appearance of early signs.
        - Oculogyric crisis has been reported as an adverse effect of aripiprazole.
    - **Weight gain -** common with all antipsychotics, but more frequent with second-generation antipsychotics. In general, clozapine and olanzapine have the greatest

potential to cause weight gain, followed by chlorpromazine, quetiapine, and risperidone.

- **Dyslipidaemia -** phenothiazines (such as chlorpromazine), clozapine, olanzapine, quetiapine, and risperidone all increase lipid levels. Offer dietary advice and consider treatment with a statin according to national guidelines.
- **Hyperprolactinaemia -** most antipsychotics can cause hyperprolactinaemia that may lead to galactorrhoea, amenorrhoea, gynaecomastia, hypogonadism, sexual dysfunction, and an increased risk of osteoporosis. Clozapine, olanzapine, quetiapine, and aripiprazole do not increase prolactin above the normal range in standard doses.
- **Sedation -** chlorpromazine, clozapine, promazine, and zotepine cause the highest incidence of sedation. Amisulpride, aripiprazole, sertindole, and sulpiride are associated with a very low incidence of sedation. Performance of skilled tasks (such as driving) may be affected; effects of alcohol are enhanced. Tolerance to sedation usually develops.
- **Anticholinergic effects** (such as dry mouth, blurred vision, urinary retention, constipation, and cutaneous flushing) — chlorpromazine and clozapine have potent anticholinergic effects. Tolerance may develop, but it is very variable, and these adverse effects are often poorly tolerated.
- **Postural hypotension -** commonly associated with clozapine, chlorpromazine, aripiprazole, quetiapine, and risperidone.
- **Hypertension -** commonly reported with clozapine but there are also reports with aripiprazole, olanzapine, quetiapine, and risperidone. Hypertension can occur as:
    - A small, steady increase in blood pressure over time (may be associated with weight gain), *or*
    - An unpredictable, sharp increase in blood pressure on starting a new drug.
- **Reduced seizure threshold -** seizures are a recognized adverse effect of antipsychotics (the higher the dose, the greater the risk). Clozapine carries the greatest risk.
- **Impaired glucose tolerance -** has been associated with both first-generation and second-generation antipsychotic drugs. Hyperglycaemia, and sometimes diabetes (including ketoacidosis and coma) can occur in people taking clozapine, olanzapine, risperidone, and quetiapine.
- **Cardiomyopathy, myocarditis, and cutaneous vasculitis** – these have been reported with quetiapine, but a causal relationship has not been established. Consider discontinuing treatment if cardiomyopathy or myocarditis are suspected.
- **QT interval prolongation -** the most widely reported cardiac conduction defect caused by antipsychotics and considered to be a class effect. However aripiprazole and paliperidone are thought to have no effect on the QT interval.
    - Avoid co-prescribing other drugs that are known to prolong the QT interval (for example tricyclic antidepressants, erythromycin, or antiarrhythmics), and monitor potassium levels at least annually as there is an increased risk of arrhythmia if the person has hypokalaemia.
    - People taking antipsychotics who experience palpitations or any other symptoms that suggest cardiac disease should undergo electrocardiography.
- **Stroke risk -** olanzapine and risperidone have been associated with an increased risk of stroke in elderly people with dementia. The Committee on Safety of Medicines advised that:
    - For acute psychotic conditions in elderly people with dementia, risperidone should be limited to short-term use under specialist advice; olanzapine is not licensed for acute psychosis.
    - The possibility of cerebrovascular events should be considered carefully before treating people with a history of stroke or transient ischaemic attack;

risk factors for cerebrovascular disease (for example hypertension, diabetes, smoking, and atrial fibrillation) should also be considered.

- **Venous thromboembolism (VTE) -** antipsychotic use may be associated with an increased risk of VTE.
  - Data are insufficient to determine any difference in risk between second-generation and first-generation antipsychotics, or between individual drugs.
  - All possible risk factors for VTE should be identified before and during antipsychotic treatment and preventive measures undertaken.
- **Neuroleptic malignant syndrome (NMS)-** is a rare but potentially fatal adverse effect of all antipsychotics. Signs and symptoms of NMS include fever, increased sweating, rigidity, confusion, fluctuating consciousness, fluctuating blood pressure, tachycardia, raised creatine kinase, leucocytosis, and raised liver function tests.
- **Pneumonia -** recent evidence from observational studies suggest that all antipsychotics are associated with an increased risk of pneumonia. The mechanism by which this occurs is unclear.
- **Neutropenia -** stop the suspected drug if neutrophils fall below 1.5 x 10^9/L and seek urgent advice from a secondary care specialist.
- **Abnormal liver function tests (LFTs) -** stop the suspected drug if LFTs suggest hepatitis (transaminases rise to three times normal) or prothrombin time or albumin are abnormal. Seek advice from a secondary care specialist.
- **Photosensitivity -** is common with chlorpromazine. Adequate use of sunscreen should prevent sunburn in affected people. Some high-factor sunscreens (sun protection factor 30 or above) are available on prescription. The prescription should be endorsed with ACBS.
- **Skin and subcutaneous tissue disorders -** olanzapine has been associated with Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), frequency unknown.
- **Diplopia -** is an uncommon side effect with aripiprazole.
- **Misuse and abuse -** cases of misuse and abuse have been reported with quetiapine. Quetiapine should be prescribed with caution in people who have a history of drug or alcohol abuse.
- **Salivary hypersecretion.**

What are the drug interactions of antipsychotics?

- **The following interactions are common to all antipsychotics:**
  - **Drugs with a sedative action** (such as alcohol, analgesics, tricyclic antidepressants, and sedating antihistamines) will enhance the sedative effects of antipsychotics.
  - **Drugs with a hypotensive effect** (for example antihypertensives) will enhance the hypotensive effect of antipsychotics.
  - **Drugs that prolong the QT interval,** such as anti-arrhythmics, macrolides (for example erythromycin), and tricyclic antidepressants, may have a synergistic effect on the QT interval. Avoid co-prescribing drugs that prolong the QT interval. These drugs are contraindicated with haloperidol, sertindole, or pimozide therapy.
  - **Diuretics** may cause hypokalaemia, which may increase the risk of arrhythmias; monitor potassium levels in people taking diuretics.
- **Azole antifungals**
  - Azole antifungals may increase levels of some antipsychotics, for example:
    - Aripiprazole levels are predicted to be increased by itraconazole.
    - Haloperidol levels are increased by 30% or more by itraconazole.
  - If azole antifungals are given concurrently with antipsychotics, monitor for signs of adverse effects of antipsychotics and consider reducing the dose as necessary.

    - The manufacturer of aripiprazole recommends halving the dose of aripiprazole if itraconazole is given concomitantly.
- **Carbamazepine**
    - Carbamazepine reduces plasma levels of clozapine, haloperidol, and risperidone by half. Carbamazepine also reduces levels of aripiprazole, fluphenazine, olanzapine, quetiapine, and sertindole.
        - Monitor the person's symptoms to ensure that antipsychotics remain effective.
    - Carbamazepine levels are increased by haloperidol, quetiapine, or risperidone, or chlorpromazine with amoxapine.
        - Monitor carbamazepine levels if these drugs are given together.
- **Grapefruit juice**
    - Advise the person not to drink grapefruit juice if they are taking pimozide. Grapefruit juice increases the levels of pimozide, possibly leading to torsades de pointes potentially fatal arrhythmias.
- **Selective serotonin reuptake inhibitors (SSRIs)**
    - SSRIs increase levels of some antipsychotics, for example:
        - Haloperidol levels are increased by 20–30% by fluoxetine and by 20–60% by fluvoxamine.
        - Risperidone levels are increased by fluvoxamine, fluoxetine, and paroxetine.
        - Sertindole levels are increased two- to three-fold by fluoxetine and paroxetine.
        - Clozapine and olanzapine levels are increased by fluoxetine, fluvoxamine, paroxetine, sertraline, and possibly citalopram.
    - Where antipsychotic drug levels are increased, the person should be monitored and the dose adjusted accordingly.
- **Stopping smoking**
    - Smoking induces the metabolism of olanzapine and clozapine. If the person stops smoking, monitor for increased adverse effects and seek advice about dose adjustment if necessary.

TOPIC: Schizophrenia and psychosis

Definition

- **The term 'psychosis' encompasses a number of symptoms associated with significant alternations to a person's perception, thoughts, mood, and behaviour.**
- **Individuals experiencing psychosis will have a different combination of symptoms, which may include:**
    - **Positive symptoms, such as:**
        - **Hallucinations (perceptions in the absence of stimulus).** The most common are auditory hallucinations, in which voices are heard and may provide a running commentary on actions, argue with the person, command, or echo the person's thoughts. Hallucinations which are visual, smell, taste, or tactile in nature occur less commonly.
        - **Delusions (fixed or falsely-held beliefs)** can include delusions of reference (the belief that ordinary events, objects, or the behaviour of others has a meaning specifically for the person, for example, that people on the radio are talking to, or about, them), delusions of control (the belief that the person's thoughts, feelings, or behaviour are being controlled by others. They include thought insertion, thought withdrawal, and thought broadcasting), and delusions of persecution (the belief that other people are plotting against the person).
        - **Disorganised behaviour, speech, and/or thoughts (thought disturbance).**
    - **Negative symptoms, such as:**
        - Emotional blunting.
        - Reduced speech.
        - Loss of motivation.
        - Self-neglect.
        - Social withdrawal.
- **Schizophrenia is the most common psychotic disorder. The diagnostic criteria for schizophrenia (using the International Classification of Diseases-10 [ICD-10] criteria) require the following symptoms to be present most of the time for 1 month or more:**
    - **One or more of the following features:**
        - Hallucinatory voices giving a running commentary on the person's behaviour, or discussing the person among themselves, or other types of hallucinatory voices coming from some part of the body.
        - Thought echo, thought insertion or withdrawal, and thought broadcasting.
        - Delusions of control, influence, or passivity, clearly referred to body or limb movements or specific thoughts, actions, or sensations.
        - Persistent delusions of other kinds that are culturally inappropriate and completely impossible, such as religious or political identity, or superhuman powers and abilities (for example being able to control the weather, or being in communication with aliens from another world).
    - **Or any two of the following criteria:**
        - Persistent hallucinations in any form, when accompanied by fleeting or half-formed delusions without clear affective content, or by persistent over-valued ideas (similar to preoccupations), or when occurring every day for weeks or months on end.
        - Breaks or interpolations in the train of thought, resulting in incoherence or irrelevant speech, or neologisms (invented words).
        - Catatonic behaviour, such as excitement, posturing, or waxy flexibility; negativism; mutism; and stupor.

        - Negative symptoms, such as marked apathy, reduced speech, and blunting or incongruity of emotional responses, usually resulting in social withdrawal and lowering of social performance; it must be clear that these are not due to depression or to antipsychotic medication.
        - A significant and consistent change in the overall quality of some aspects of personal behaviour, manifest as loss of interest, aimlessness, idleness, a self-absorbed attitude, and social withdrawal.
- **Other psychotic disorders include:**
    - Schizoaffective disorder — where symptoms of schizophrenia and a mood disorder (depressed or manic) are equally prominent.
    - Drug-induced psychosis — substance-induced and usually remits within a month of cessation of use.
    - Persistent delusional disorder — where the most pervasive symptom is delusion.

Causes and risk factor

- **The precise cause of psychotic disorders is unknown, but interactions between a number of genetic, social, and environmental risk factors appear to be involved.**
- **Psychotic disorders exhibit a clear heritability which twin studies estimate to be around 85%.**
- **Identified social and environmental risk factors include:**
    - **Stressful life events (such as bereavement, job loss, eviction, and relationship breakdown)**
    - **Childhood adversity (such as abuse, bullying, parental loss or separation)**
    - **Family**
    - **Migration (especially from a developing country)**
    - **Urban living**
    - **Cannabis use**
    - **Other substance use** — use of amphetamines, cocaine, ketamine, LSD, or inhaled substances such as toluene and certain types of glue, can cause acute psychosis. Heavy use may also lead to long-term psychotic symptoms that may persist for years after the last exposure.
    - **Medication use** — high-dose corticosteroid use can precipitate psychosis.
    - **Early life factors** — such as exposures *in utero* to medication, maternal stress, nutritional deficiency, and infection; intrauterine growth restriction, birth and postnatal trauma.
    - **Parental age** — a paternal age of > 40 years and parental age of < 20 years have both been associated with an increased risk of schizophrenia.
    - **Exposure to the protozoan parasite *Toxoplasma gondii*** — associated with a two-fold increase in risk of schizophrenia.

Prognosis

- **After pharmacological, psychological, and other interventions, many people experience regression or resolution of symptoms, although some negative symptoms may remain:**
    - It is estimated that four out of every five people show some response to treatment within the first year.
    - One in every five people will have no further psychotic episodes within the next five years.
    - The most common course is initial improvement of symptoms with ongoing recurrent acute psychotic episodes or relapses over many years.

- It has been reported that around 15% of people experience persistent psychotic symptoms that are unresponsive to treatment two years after the acute episode. These people may require rehabilitation and support including help with activities of daily living.

- **Factors associated with a poor prognosis include:**
  - Longer duration of untreated psychosis.
  - Early or insidious onset of schizophrenia.
  - Male sex.
  - Negative symptoms.
  - Family history of schizophrenia.
  - Low IQ, low socioeconomic status, or social isolation.
  - Significant psychiatric history.
  - Continued substance misuse.

Complications

- **Premature death** Various factors contribute to this including:
  - **Increased risk of suicide**
    - Risk factors strongly associated with suicide include younger age, male sex, having a higher level of education, prior suicide attempts, a family history of suicide, depression, active hallucinations and delusions, the presence of insight (i.e. recognition that the person's experiences are caused by a mental illness), and comorbid substance misuse.
  - **Increased risk of certain physical disorders, including:**
    - **Cardiovascular disease**
    - **Type 2 diabetes** — also thought to be caused by a combination of lifestyle factors, such as poor diet and lack of exercise, and treatment with antipsychotic drugs. However, research has shown that impaired glucose homeostasis is more commonly present at disease onset in people with schizophrenia and is therefore not solely explained by drug treatment.
    - **Smoking-related illness**
    - **Cancer** — studies have shown this to be the cause of approximately 9% of the excess deaths in people with psychotic disorders. Increased risk of early death is thought to be caused by both late recognition and under-treatment. Women with schizophrenia have been found to have a 31% increased risk of developing breast cancer.
    - **Infections such as HIV, hepatitis C and tuberculosis**.
- **Social exclusion** — negative symptoms, such as emotional blunting, reduced speech, loss of motivation, self-neglect, and social withdrawal, are often the major cause of social exclusion by impairing the person's ability to learn, work, and maintain relationships
- **Substance misuse** — up to one-third of people with schizophrenia misuse drugs, which as well as posing direct risks to the person's health, may be associated with an increased risk of blood-borne viruses.

Diagnosis

- **Suspect frank psychosis in a person with positive symptoms such as hallucinations and delusions, and negative symptoms such as affective flattening (lack of spontaneity or reactivity of mood), avolition (lack of drive), anhedonia (lack of pleasure), attention deficit, or impoverishment of speech and language.**

- **Be aware that psychosis may be preceded by a prodromal period that can last from a few days to around 18 months.** Identifying the prodromal stage offers a critical treatment window to delay or prevent the person's transition to frank psychosis.
    - **The prodromal period is characterised by increasing distress and a decline in personal and social functioning. Specific symptoms may include:**
        - **Transient, low-intensity psychotic symptoms** — intermittent, self-limiting episodes, typically lasting less than a week. May include hallucinations/unusual perceptual experiences, unusual thoughts (including new preoccupation with mystical or religious themes, concerns about being under surveillance, etc) and may manifest as unusual or uncharacteristic behaviour.
        - **Reduced interest in daily activities** — may manifest as poor personal hygiene and/or reduced performance at school or work.
        - **Problems with mood, sleep, memory, concentration, communication, affect, and motivation.**
        - **Anxiety, irritability, or depressive features.**
        - **Incoherent or illogical speech** — suggestive of thought disturbance.
    - **A positive family history** (particularly a first-degree relative with psychosis or schizophrenia) should also raise clinical suspicion in a person exhibiting deterioration in personal and social functioning.
    - **Be aware of other risk factors for psychosis** and in particular, consider asking the person about any recent or past stressful or traumatic experiences and whether they have used cannabis.
    - **Positive symptoms may not be readily disclosed by the person.** It may, therefore, be necessary to directly ask them whether they are experiencing hallucinations ('feeling, seeing, or hearing things that other people cannot') or experiencing delusional thoughts ('feeling that you are being talked about, watched, or given a hard time for no reason, suspecting you are being externally controlled, or having thoughts inserted or removed from your mind, or others being aware of your thoughts').
    - **Where symptoms are suggestive of the prodromal period of psychosis, consider conducting a medical assessment to rule-out an alternative underlying cause:**
        - Review the person's history to rule out use of prescribed drugs that can cause psychosis, for example, anticonvulsants, high-dose corticosteroids, levodopa and dopamine agonists, or opioids.
        - Carry out a urine drug screen if use of recreational drugs that can cause psychosis, such as amphetamines or cocaine, is suspected.
        - Use clinical judgement to determine whether to screen for HIV and/or syphilis as both infections can cause psychiatric symptoms.
        - Carry out a full blood count if anaemia is a potential cause of negative symptoms.
        - If a neurological condition such as temporal lobe epilepsy or cerebrovascular disease is suspected.

Differential diagnosis

- **Severe affective (mood) disorders associated with psychotic symptoms,** including severe depression or bipolar disorder.
    - These conditions are distinguished from schizophrenia by the predominance of affective symptoms that may at times trigger psychosis. In contrast, schizophrenia is dominated by psychotic symptoms that may at times be associated with affective symptoms.

- **Drug-induced psychosis** caused by cannabis, corticosteroids, opioids, cocaine, or amphetamines.
- **Sepsis** — consider in people without a prior history of psychotic symptoms, particularly where there are features suggestive of specific infection, such as dysuria or productive cough.
- **Another underlying medical condition,** such as cerebrovascular disease, cerebral tumours, or temporal lobe epilepsy.
- **Post-traumatic stress disorder (PTSD)** may include flashbacks that have a hallucinatory quality and hyper-vigilance that may reach paranoid proportions. PTSD is distinguished from psychotic disorders by the existence of a traumatic event and characteristic features such as reliving or re-enacting the event.
- **Obsessive compulsive disorder (OCD),** where strong irrational beliefs are held, but related to specific fears, and for which the person has developed rituals.
- **Autism spectrum disorder or communication disorders** – people with these disorders may display symptoms that resemble a psychotic episode. They may be distinguished from psychotic disorders by their deficits in social interaction with repetitive and restricted behaviours.

Management

How should I manage someone at risk of psychosis, with a first episode of psychosis, or with a confirmed diagnosis?

- **For all people at risk of developing a psychotic disorder and people with psychotic symptoms, undertake an assessment of the risk of harm to the person and/or others:**
  - **Determine the level of risk to the person by enquiring about/considering:**
    - History of self-harm.
    - Suicidal ideation and plans, any previous attempts.
    - Feelings of hopelessness.
    - Misuse of recreational drugs and/or alcohol.
    - The likelihood of accidental or non-accidental injury.
    - Whether the person is experiencing 'command' hallucinations and whether they feel compelled to act upon them.
    - Level of family/social support.
    - Timing — be aware that the highest risk of suicide tends to be around the time of a psychotic episode and shortly after hospital discharge.
  - **Assess the person's risk of unintentional harm to themselves** caused by disorganized behaviour or poor judgement of risk due to their absorption with psychotic experiences and beliefs.
  - **Determine the risk of harm to others by enquiring about/considering:**
    - The potential for neglect of individuals dependent on the person for care, in particular family, children, and any other dependents. Follow local safeguarding procedures if necessary.
    - Any risk to the public, especially if there is a previous history of confrontation with others, violence, carrying weapons, and use of stimulant drugs.
    - Whether the person is experiencing delusions focused on a particular individual.
  - **Assess risk from others including any adult safeguarding issues,** for example vulnerability to traumatic injuries, accidents, assault, or exploitation.
- **For people judged to be at high risk of harm to themselves or others, arrange same-day specialist mental health assessment by the early intervention in psychosis service (if available).**
- **If the person is not thought to present a high risk of harm to themselves or others, refer for specialist assessment:**

    - **For people experiencing psychosis** refer without delay to the early intervention in psychosis service. If the service cannot provide urgent intervention, refer the person to a crisis resolution and home treatment team (with support from early intervention in psychosis services).
    - **For people suspected to be in the prodromal phase of psychosis**, refer without delay to an early intervention in psychosis service (if available) or a specialist mental health service. A consultant psychiatrist or a trained specialist with experience in at-risk mental states should carry out the assessment.
- **Do not start antipsychotic drug treatment while awaiting specialist assessment unless under advice from a consultant psychiatrist**.
- The secondary care team should maintain responsibility for monitoring the person's physical health and the effects of any antipsychotic medication for at least the first 12 months of treatment, or for longer until the person's condition has stabilised. After this, the responsibility for monitoring may be transferred to primary care, depending on locally agreed shared care guidelines or the person's care plan.
- **Ongoing primary care management may include:**
    - Prescribing medication initiated in secondary care and undertaking any required monitoring.
    - Monitoring the person's symptoms, and if necessary, following crisis plans developed in secondary care and liaising with secondary care specialists.
    - Reviewing the person's physical health, mental health, and medication at least annually and more often if the person, carer or healthcare professional has any concerns.
    - Asking the person about the support they receive from family and carers, and considering referral to the community mental health service for family intervention to manage any conflict or difficulties that could trigger a relapse.
    - Ensuring that people with psychotic disorders and their family and/or carers are informed about support available to help them back into work or education, their housing options, entitlement to benefits, and entitlement to drive.
    - Ensuring that the person's family or carers:
        - Have ideally been offered an initial assessment of their own needs (done by mental health services) and that a copy of the carer's care plan has been received by primary care.
        - Have ideally been offered a focused education and support programme.
        - Are aware of their right to a formal carer's assessment by social care services.
        - Know where to obtain help during a crisis.
- **If a person with a psychotic disorder is being managed solely in primary care, re-refer them to secondary care if:**
    - There is a poor or partial response to treatment or treatment adherence is poor.
    - The person's functioning declines significantly.
    - They develop intolerable or medically important adverse effects from medication.
    - Comorbid alcohol or drug misuse is suspected.
    - There is potential risk to the person or others.
    - A woman with a psychotic disorder is pregnant or planning a pregnancy.

How should I manage a woman with psychosis or schizophrenia who is planning a pregnancy or presents with unplanned pregnancy?

- **Refer women who are pregnant or planning a pregnancy to a specialist perinatal mental health service if available, or to the community mental health service for an assessment and discussion of drug treatment.**

- **Do not alter or stop drug treatment (and advise the woman not to alter her own medication) without seeking specialist advice.**
    - Explain to the woman that following discussion with a specialist of the risks to the mother of under-treatment (relapse), the potential benefits of a particular treatment, and the (largely hypothetical) risks to the fetus that may be associated with any drug treatment, she may be offered the following options:
        - Remaining on current drug treatment throughout conception, pregnancy, and birth — commonly advised when a woman's condition is well-controlled.
        - Switching to another drug treatment.
        - Stopping or reducing the dose of her medication.
    - Discuss the likelihood that sleep and routine will be disturbed after the baby is born. Participate in multi-agency plans for practical support after the baby is born (from the woman's family, health visitor, organizations such as Sure Start, and social care) so that the woman gets enough rest and sleep.
- **For women who are planning a pregnancy, give general pre-conception advice (such as cessation of smoking and alcohol consumption) and prescribe folic acid. Note that a high dose (5 mg/day) is indicated in women taking certain medications and who are obese.**
- **Be aware that women in the postnatal period are at increased risk of relapse of psychotic symptoms.**

What issues should I consider if a woman is breastfeeding?

- **Support a woman with a psychotic disorder in the choice of feeding method that best suits her and her family.** Breastfeeding may be appropriate unless the woman is taking clozapine. Ensure support for women who choose not to breastfeed.
- **Infants exposed via breastmilk to antipsychotics should, as a precaution, be monitored for adverse effects**, particularly if at high risk (e.g. preterm or low birth weight infants). These may include drowsiness, irritability, motor abnormalities and poor feeding.
- **If more advice on breastfeeding and medication is required,** liaise with specialist mental health services or seek specialist advice

Prescription information
Antipsychotics

Table 1. Antipsychotics available to prescribe.

Oral first-generation (typical)	Oral second-generation (atypical)	Antipsychotic depot injections*
Benperidol	Amisulpride	Aripiprazole
Chlorpromazine	Aripiprazole	Flupentixol decanoate
Flupentixol	Clozapine	Fluphenazine decanoate
Haloperidol	Olanzapine	Haloperidol
Levomepromazine	Paliperidone	Olanzapine embonate

Oral first-generation (typical)	Oral second-generation (atypical)	Antipsychotic depot injections*
Pericyazine	Quetiapine	Paliperidone
Perphenazine	Risperidone	Pipotiazine palmitate
Pimozide		Risperidone
Prochlorperazine		Zuclopenthixol decanoate
Promazine		
Sulpiride		
Trifluoperazine		
Zuclopenthixol		
* Antipsychotic depot injections are used for maintenance therapy when adherence to oral treatment is unreliable. They are administered every 1–4 weeks.		

TOPIC: personality disorders

Cluster A: 'Odd or Eccentric'

Paranoid

- Hypersensitivity and an unforgiving attitude when insulted
- Unwarranted tendency to questions the loyalty of friends
- Reluctance to confide in others
- Preoccupation with conspiratorial beliefs and hidden meaning
- Unwarranted tendency to perceive attacks on their character

Schizoid

- Indifference to praise and criticism
- Preference for solitary activities
- Lack of interest in sexual interactions
- Lack of desire for companionship
- Emotional coldness
- Few interests
- Few friends or confidants other than family

Schizotypal

- Ideas of reference (differ from delusions in that some insight is retained)
- Odd beliefs and magical thinking
- Unusual perceptual disturbances
- Paranoid ideation and suspiciousness
- Odd, eccentric behaviour
- Lack of close friends other than family members
- Inappropriate affect
- Odd speech without being incoherent

Cluster B: 'Dramatic, Emotional, or Erratic'

Antisocial

- Failure to conform to social norms with respect to lawful behaviours as indicated by repeatedly performing acts that are grounds for arrest;
- More common in men;
- Deception, as indicated by repeatedly lying, use of aliases, or conning others for personal profit or pleasure;
- Impulsiveness or failure to plan ahead;
- Irritability and aggressiveness, as indicated by repeated physical fights or assaults;
- Reckless disregard for the safety of self or others;
- Consistent irresponsibility; repeated failure to sustain consistent work behaviour or honour financial obligations;

- Lack of remorse, as indicated by being indifferent to or rationalizing having hurt, mistreated, or stolen from another

Borderline

- Efforts to avoid real or imagined abandonment
- Unstable interpersonal relationships which alternate between idealization and devaluation
- Unstable self-image
- Impulsivity in potentially self-damaging area (e.g. Spending, sex, substance abuse)
- Recurrent suicidal behaviour
- Affective instability
- Chronic feelings of emptiness
- Difficulty controlling temper
- Quasi psychotic thoughts

Histrionic

- Inappropriate sexual seductiveness
- Need to be the centre of attention
- Rapidly shifting and shallow expression of emotions
- Suggestibility
- Physical appearance used for attention seeking purposes
- Impressionistic speech lacking detail
- Self-dramatization
- Relationships considered to be more intimate than they are

Narcissistic

- Grandiose sense of self importance
- Preoccupation with fantasies of unlimited success, power, or beauty
- Sense of entitlement
- Taking advantage of others to achieve own needs
- Lack of empathy
- Excessive need for admiration
- Chronic envy
- Arrogant and haughty attitude

Cluster C: 'Anxious and Fearful'

Obsessive-compulsive

- Is occupied with details, rules, lists, order, organization, or agenda to the point that the key part of the activity is gone
- Demonstrates perfectionism that hampers with completing tasks
- Is extremely dedicated to work and efficiency to the elimination of spare time activities
- Is meticulous, scrupulous, and rigid about etiquettes of morality, ethics, or values
- Is not capable of disposing worn out or insignificant things even when they have no sentimental meaning

- Is unwilling to pass on tasks or work with others except if they surrender to exactly their way of doing things
- Takes on a stingy spending style towards self and others; and shows stiffness and stubbornness

Avoidant

- Avoidance of occupational activities which involve significant interpersonal contact due to fears of criticism, or rejection.
- Unwillingness to be involved unless certain of being liked
- Preoccupied with ideas that they are being criticised or rejected in social situations
- Restraint in intimate relationships due to the fear of being ridiculed
- Reluctance to take personal risks due to fears of embarrassment
- Views self as inept and inferior to others
- Social isolation accompanied by a craving for social contact

Dependent

- Difficulty making everyday decisions without excessive reassurance from others
- Need for others to assume responsibility for major areas of their life
- Difficulty in expressing disagreement with others due to fears of losing support
- Lack of initiative
- Unrealistic fears of being left to care for themselves
- Urgent search for another relationship as a source of care and support when a close relationship ends
- Extensive efforts to obtain support from others
- Unrealistic feelings that they cannot care for themselves

TOPIC: Depression in Adults

Definition

- Depression is characterised by the absence of a positive affect (a loss of interest and enjoyment in ordinary things and experiences), low mood, and a range of associated emotional, cognitive, physical, and behavioural symptoms.
    - It is defined in the fifth edition of the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM-5) by the presence of at least five out of a possible eight defining symptoms, during the same two-week period, where at least one of the symptoms is depressed mood or loss of interest or pleasure.
- The severity of depression depends on the intensity and frequency of symptoms, their duration, and impact on personal and social functioning. The National Institute for Health and Care Excellence (NICE) guideline classifies new episodes of depression according to severity on the PHQ-9 scale
    - 'Less severe depression' — this encompasses subthreshold and mild depression, defined as depression scoring less than 16 on the PHQ-9 scale.
    - 'More severe depression' — this encompasses moderate and severe depression, defined as depression scoring 16 or more on the PHQ-9 scale.
- Chronic depressive symptoms are defined by the NICE guideline in people with:
    - 'Symptoms which continually meet the criteria for the diagnosis of a major depressive episode for at least two years, or have persistent subthreshold symptoms for at least two years, or who have persistent low mood with or without concurrent episodes of major depression for at least two years'.
- Seasonal affective disorder describes episodes of depression which recur annually at the same time each year with remission in between (usually appearing in winter and remitting in spring).

Causes

The underlying cause of depression is unknown, but is likely to result from a complex interaction of genetic, environmental, biological, cultural, and psychological factors.

- Risk factors for depression include:
    - Female sex.
    - Older age.
    - Past history of depression.
    - Personal, social, or environmental factors, such as relationship issues or breakdown, bereavement, stress, poverty, unemployment, homelessness, social isolation, or past history of child maltreatment.
    - Postpartum period.
    - Past history of depression.
    - Family history of depressive illness (first-degree relatives of a person with a 'major' depressive episode have a three-fold increased risk of depression) or suicide.
    - History of other mental health conditions and/or substance misuse.
    - Other chronic physical health conditions associated with functional impairment (such as diabetes mellitus, chronic obstructive pulmonary disease, cardiovascular disease, chronic pain syndromes, epilepsy, stroke disease).
- Risk factors for relapse of depression include:
    - Older age of onset.
    - History of recurrent episodes of depression, particularly if frequent or within the past two years.
    - Incomplete response to previous treatment, including residual symptoms.

- Unhelpful coping styles or behaviours, such as avoidance or rumination.
- History of severe depression (including severe functional impairment).
- Other chronic physical or mental health conditions, especially in the elderly.
- Ongoing personal, social, or environmental factors

Diagnosis

Consider a diagnosis of depression if a person has possible risk factors, non-specific symptoms, or any of the symptoms of depression listed below.

- Be aware that a person with depression may present with non-specific symptoms rather than low mood.
  - In particular, older people may only present with physical symptoms or a deterioration in cognitive functioning.
- Ask the two 'depression identification questions':
  - During the last month, have you often been bothered by feeling down, depressed, or hopeless?
  - During the last month, have you often been bothered by having little interest or pleasure in doing things?
- If the person answers 'yes' to one of the questions and symptoms have been present most days, most of the time, for at least 2 weeks, ask about associated symptoms of depression:
  - Disturbed sleep (decreased or increased compared to usual).
  - Decreased or increased appetite and/or weight.
  - Fatigue or loss of energy.
  - Agitation or slowing down of movements and thoughts.
  - Poor concentration or indecisiveness.
  - Feelings of worthlessness or excessive or inappropriate guilt.
  - Recurrent thoughts of death, recurrent suicidal ideas, or a suicide attempt or specific plan.

Management

Initial Management

If a person has been diagnosed with depression in primary care following initial assessment:

- Provide advice on the nature and course of depression, recovery, and sources of information and support including self-help materials, support groups, and peer support.
- Provide advice on activities to improve sense of wellbeing, such as:
  - Physical activitity including walking, jogging, swimming, dance, gardening, particularly if outdoors.
  - Maintaining a healthy lifestyle through diet, alcohol intake, and sleep.
- Offer social support including for any family and/or carers who are supporting the person.
- Arrange urgent referral to specialist mental health services if the person has:
  - More severe depression and is at signficant risk of self-harm or suicide, harm to others, or self-neglect — crisis resolution and home treatment (CRHT) team input may be needed.
  - Pyschotic symptoms — CRHT team or specialist psychiatry input may be needed.
- Follow local safeguarding procedures if:
  - The person is a carer for a child or vulnerable adult(s) and there are safeguarding concerns.
- Discuss the options for treatment of depression and develop a treatment plan, depending on the person's wishes, any physical or mental health comorbidities, experiences with

previous treatment(s), hopes and expectations of treatment, involving family/carers in shared decision-making where appropriate.

- If a person has a new episode of less severe depression and does not want treatment, or feels that symptoms are improving:
    - Offer active monitoring, with the option to consider treatment at any time if needed.
    - Ensure the person has adequate social support and is aware of sources of help if symptoms worsen.
    - Arrange an initial review, usually within 2–4 weeks, and ensure follow-up if the person does not attend.
- If a person has a new episode of less severe depression and wishes to consider treatment, match treatment to their clinical needs and wishes:
    - Consider offering guided self-help first-line.
    - Do not routinely offer an antidepressant as first-line treatment. If the person wishes to start drug treatment, offer a selective serotonin reuptake inhibitor (SSRI) first-line.
    - Offer other treatment options, depending on the person's preferences, their previous experience of treatment(s), and local referral pathways and service provision.
        - First line treatment options, which include cognitive behavioural therapy (CBT), group exercise, group mindfulness and meditation, and counselling.
- If a person has a new episode of more severe depression, match treatment to their clinical needs and wishes:
    - Offer any treatment option first-line, depending on the person's preferences, their previous experience of treatment(s), and local referral pathways and service provision.
        - Arrange an initial review usually 2–4 weeks after starting treatment, and ensure follow-up if the person does not attend.
        - If a person wishes to start an antidepressant, offer an SSRI or serotonin noradrenaline reuptake inhibitor (SNRI) first-line.
    - Advise the person they must not drive and must notify the Driver and Vehicle Licensing Agency (DVLA) if they have significant memory or concentration problems, agitation, behavioural disturbance, or suicidal thoughts.
- Do not recommend taking the over-the-counter preparation St John's wort for management of depression.
- Arrange appropriate management of any associated mental and physical health condition(s).
- Arrange regular monitoring and follow-up, depending on clinical judgement.
    - Consider using a validated depression questionnaire to monitor response to treatment, such as PHQ-9.

What should I advise to prevent relapse of depression?

During follow-up of a person with depression, advise that continuation of treatment (antidepressant or psychological intervention) after full or partial remission of symptoms, may help reduce the risk of relapse.

- Discuss with the person the relative risks and benefits of continuing or stopping treatment, taking into account:
    - Possible risk factors for relapse.
    - Possible risks of continuing antidepressants long-term, such as adverse effects and withdrawal symptoms on stopping treatment.
        - Advise a person taking an antidepressant that treatment should be taken for at least six months following remission of symptoms, to prevent relapse.

- If a person has been taking an antidepressant alone and is in remission but at higher risk of relapse, discuss options to prevent relapse, such as:
    - Continuing antidepressant medication, maintaining the dose that led to full or partial remission of symptoms, or
    - Engaging with a psychological intervention, such as group cognitive behavioural therapy (CBT) or mindfulness-based cognitive therapy (MBCT), or
    - Continuing antidepressant medication and engaging with a psychological intervention (CBT or MBCT).
- If a person has been taking an antidepressant in combination with a psychological intervention and is in remission but at higher risk of relapse, discuss options to prevent relapse, such as:
    - Continuing with one or both treatments.
- If a person has engaged with a psychological intervention alone and is in remission but at higher risk of relapse, discuss options to prevent relapse, such as:
    - Continuing to engage with psychological intervention with a focus on relapse prevention skills.
- If a person is taking antidepressant medication longterm to prevent relapse:
    - Arrange review at least every six months, to monitor symptoms, check concordance with medication, assess for adverse effects, risk factors for relapse, any suicidal ideas, and their wishes for ongoing treatment.
        - Consider using a validated depression questionnaire to monitor response to treatment, such as PHQ-9.
    - Ensure the person has adequate social support and is aware of sources of help if symptoms worsen.
- If a person is continuing with a psychological intervention to prevent relapse:
    - Arrange for review when they are finishing treatment to monitor symptoms, assess risk factors for relapse, and arrange ongoing follow-up and treatment depending on clinical judgement.

Ongoing management

- Ask about and manage:
    - Any ongoing symptoms of depression, impact on daily functioning including work, relationships, and any carer role for a child or vulnerable adult(s).
    - Response of symptoms to any psychosocial intervention(s) or antidepressant drug treatment, including adherence to treatment, and any adverse effects.
        - Consider using a validated depression questionnaire to assess for depression, severity of symptoms, and response to treatment, such as PHQ-9.
    - Any thoughts, plans, or intent to self-harm or commit suicide and any other risks to others or risk of self-neglect.
        - Ensure the person has a crisis plan identifying potential triggers and strategies to help
    - Any new symptoms of other mental health disorders, including anxiety, eating disorders, bipolar disorder, or psychosis.
    - Any new or ongoing personal, social, or environmental factors that may impact on symptoms and recovery.
- Offer support to the family and/or carers who are supporting a person with depression, particularly if there are severe or chronic symptoms.
- If there is no improvement in symptoms at follow-up after 4 weeks of antidepressant medication at a recognized therapeutic dose, or after 4–6 weeks of psychological therapy or combined psychological therapy with antidepressant medication:
    - Assess and manage any modifiable risk factors for depression.

    - Assess and manage any issues affecting concordance with treatment (including stopping or reducing the dose of antidepressant due to adverse effects; or explore reasons for missed sessions with psychological therapy).
    - Consider whether an alternative diagnosis or comorbid condition may be limiting the person's response to or engagement with treatment.
- Discuss the options for further-line treatment, depending on the person's wishes, experiences with previous treatment(s), hopes and expectations of treatment, involving family/carers in shared decision-making where appropriate.
- If there is limited or no improvement with psychological interventions alone, options include:
    - Switching to an alternative psychological therapy.
    - Adding in a selective serotonin reuptake inhibitor (SSRI) antidepressant.
    - Switching to an SSRI alone.
- If there is limited or no improvement with antidepressant medication alone, options include:
    - Augmenting with group exercise.
    - Switching to a psychological intervention.
    - Increasing the antidepressant dose or switching the antidepressant to a drug in the same class or different class.
        - Advise that antidepressant dose increases may not be more effective and may increase adverse effects. Arrange regular monitoring and follow-up.
        - Advise that options to switch antidepressant include to an SSRI or SNRI.
    - Changing to a combination of psychological intervention and antidepressant medication.
- If there is limited or no improvement with a combination of psychological intervention and antidepressant medication, options include:
    - Switching to an alternative psychological therapy.
    - Increasing the antidepressant dose or switching the antidepressant to a drug in the same class or different class.
    - Seeking specialist advice or arranging referral for consideration of an additional antidepressant drug or other specialist augmentation of treatment, depending on clinical judgement.
        - Advise that adding in another antidepressant may not be more effective and may increase adverse effects.

How should I advise to switch an antidepressant?

If a person is taking an antidepressant, ideally at a recognized therapeutic dose, but wishes to switch treatment due to lack of response, adverse effects, or personal preference:

- If treatment is needed for a recurrent episode of depression:
    - Consider prescribing an antidepressant that the person has previously found helpful or prefers.
    - Try to avoid prescribing an antidepressant that the person has previously failed to respond to or could not tolerate due to adverse effects.
    - Advise that symptoms of mood changes, anxiety, agitation, hopelessness, or suicidal thoughts or ideas may increase when changing treatment. Advise to seek urgent review if symptoms worsen significantly or the person/carers are concerned.
        - Arrange to review the person's treatment, depending on clinical judgement.
- When switching to a different treatment, match the antidepressant drug choice to the person's needs, taking into account the following factors:

    - The person's preferences for treatment.
    - Potential toxicity in overdose. If the person is at risk of suicide:

        - Avoid tricyclic antidepressants ([TCAs] except for lofepramine) and venlafaxine (a serotonin noradrenaline reuptake inhibitor [SNRI]) due to their risk of death from overdose.
        - Consider limiting the amount of medication available to the person, to reduce the risk of death or medical complications if taken in overdose, depending on clinical judgement.
    - Tolerability and potential adverse effects and withdrawal symptoms.
        - Advise on the risk of serotonin syndrome when combinations of serotonergic antidepressants are prescribed.
    - Any drug treatment for associated mental health or physical health conditions which may affect prescribing.
    - Any current drug treatment(s) or substances that may potentially interact with an antidepressant.
        - This includes over-the-counter medication, herbal remedies, alcohol, or recreational drugs.
- Advise on how to switch to a new antidepressant safely, depending on the drug class(es), potential drug interactions, and recommended washout period.
    - Explain that cross-tapering usually involves gradually reducing the dose of the current antidepressant while starting the new antidepressant at a low dose, and gradually increasing the new drug dose as the previous drug is withdrawn.
        - Note: it may be possible to switch abruptly by withdrawing the current antidepressant and starting the new drug the next day, for example when switching from one selective serotonin reuptake inhibitor (SSRI) to another SSRI or an SNRI — the exception to this is fluoxetine, due to its long half-life, where a 'washout' period is needed.

How to advise stopping an antidepressant?

Do not stop antidepressants suddenly unless there are exceptional medical circumstances, such as the occurrence of serious side effects (for example upper gastrointestinal bleeding). In these circumstances, consider scheduling more frequent reviews.

If a person is taking an antidepressant and wishes to stop treatment due to recovery, lack of response, adverse effects, or personal preference:

- Advise that treatment can be stopped at any stage, but in general, the dose should be reduced gradually in stages over time ('tapering') with regular monitoring for withdrawal symptoms and signs of relapse, the frequency of follow-up depending on clinical judgement.
    - When agreeing a dose reduction schedule with the person:
        - Explain the risk of abrupt discontinuation and that the rate of safe withdrawal varies between people and can vary over time for the same person.
        - Balance the risk of adverse events from continued exposure to the medicine with minimising the risk of withdrawal symptoms by slow dose reduction and withdrawal.
        - Ensure that the planned rate of reduction is acceptable to the person.
        - Explain that although withdrawal symptoms are to be expected, the reduction schedule can be modified to allow intolerable withdrawal symptoms to improve before making the next reduction.
        - Consider giving the person additional control over the process of dose reduction (for example by issuing their usual daily dose in a form that allows them to reduce the amount in small decrements at a pace of their choosing, rather than issuing successive prescriptions for reduced daily doses).

        - Agree on regular intervals for reviewing and adjusting the reduction schedule as needed.
        - Ensure the person knows who to contact if problems occur.
    - Advise a slow step-wise taper of the drug dose proportionate to the existing dose over time, for example, by prescribing 50% of the previous dose. Consider prescribing smaller reductions (such as 25% of the previous dose) as the dose is lowered towards zero.
        - Ensure that any withdrawal symptoms have resolved or are tolerated before prescribing a further dose reduction.
        - Advise that withdrawal may take weeks or months to complete.
- Consider arranging a more rapid drug withdrawal if the person:
    - Has serious or intolerable adverse effects, such as hyponatraemia, cardiac arrhythmias, or upper gastrointestinal bleeding.
    - Wishes to switch antidepressant drugs.
    - Is taking fluoxetine (due to its long half-life) — if taking fluoxetine 20 mg daily, a period of alternate day dosing may be suitable for drug withdrawal.
        - Note: a more gradual withdrawal schedule is needed for people taking higher doses of fluoxetine.

Management of chronic depressive symptoms

If a person presents with chronic depressive symptoms and significant impairment of personal and social functioning, discuss options for management, depending on the person's wishes, hopes and expectations of treatment, involving family/carers in shared decision-making where appropriate.

- If the person has not received previous treatment for depression,discuss one of the following options for treatment:
    - Cognitive behavioural therapy (CBT) — this should focus on managing factors that may maintain symptoms, such as unhelpful coping styles or behaviours including avoidance or rumination, or interpersonal difficulties.
    - A selective serotonin reuptake inhibitor (SSRI) — if a specific SSRI is not tolerated, consider treatment with an alternative SSRI.
    - A serotonin noradrenaline reuptake inhibitor (SNRI).
    - A tricyclic antidepressant (TCA).
    - Combination therapy with CBT and either an SSRI or TCA.
- If the person has had, or is still receiving treatment for depression,discuss options for further-line treatments.
- If there is no improvement in symptoms at follow-up after a trial of SSRIs or SNRIs:
    - Assess and manage any modifiable risk factors for depression.
    - Assess and manage any issues affecting concordance with treatment (including stopping or reducing the dose of antidepressant due to adverse effects; or explore reasons for missed sessions with psychological therapy).
    - Consider whether an alternative diagnosis or comorbid condition may be limiting the person's response to or engagement with treatment.
    - Ensure the person has adequate social support and is aware of sources of help.
    - Offer referral to a specialist mental health service or seek specialist advice.

Prescription information
Antidepressants dosing and titration

- The Maudsley prescribing guidelines advise to prescribe the starting dose of an antidepressant and titrate up to the recognized minimum effective dose.

- See Table 1 for recommended starting doses, minimum effective doses, and maximum doses of different antidepressants.
- For selective serotonin reuptake inhibitors (SSRIs), mirtazapine, reboxetine, and venlafaxine, starting doses are often effective and up-titration may not be necessary.
- For people who show a clear clinical response to a low-dose TCA, this dose may be maintained without further dose increases, providing the person maintains symptom improvement.

Table 1. Recommended starting doses, minimum effective doses, and maximum doses of antidepressants.

Antidepressant	Starting dose (per day)	Usual minimum effective dose (per day)	Max dose in depression (per day)
Serotonin reuptake inhibitors			
Citalopram	20 mg	20 mg	40 mg (20 mg in the elderly and in those with reduced hepatic function)
Escitalopram	10 mg (5 mg in the elderly and in those with reduced hepatic function)	10 mg	20 mg (10 mg in the elderly and in those with reduced hepatic function)
Fluoxetine	20 mg	20 mg	60 mg
Fluvoxamine	50 mg	50 mg	300 mg
Paroxetine	20 mg	20 mg	50 mg* (40 mg in the elderly)
Sertraline	50 mg	50 mg	200 mg
Tricyclic antidepressants			
Imipramine	10–75 mg	At least 75 mg to 100 mg† (possibly 125 mg)	150–200 mg
Lofepramine	140 mg	140 mg	210 mg
Others			
Duloxetine	60 mg	60 mg	60 mg
Mirtazapine	15–30 mg	30 mg	45 mg
Reboxetine	8 mg	8 mg	12 mg
Trazodone	150 mg	150 mg	300 mg
Venlafaxine	75 mg	75 mg	375 mg
Vortioxetine	10 mg (5 mg in the elderly)	10 mg	20 mg
†Lower doses may be effective.			

Antidepressants toxicity and overdose

- **Tricyclic antidepressants (TCAs) and monoamine oxidase inhibitors (MAOIs) have the highest toxicity in overdose.**
  - Among TCAs, ddosulepin has particularly high toxicity, while lofepramine has relatively low toxicity.
  - TCAs have potentially fatal cardiovascular effects (tachycardia, postural hypotension, slowed cardiac conduction) when taken in overdose. Overdose can also cause sedation, coma, and seizures.

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Side effects of antidepressant medications[1-7]

Drug	Anticholinergic	Drowsiness	Insomnia/ agitation	Orthostatic hypotension	QTc prolongation*	Gastrointestinal toxicity	Weight gain	Sexual dysfunction
Selective serotonin reuptake inhibitors¶								
Citalopram	0	0	1+	1+	3+Δ	1+¶	1+	3+
Escitalopram	0	0	1+	1+	2+	1+¶	1+	3+
Fluoxetine	0	0	2+	1+	1+	1+¶	0	3+
Fluvoxamine	0	1+	1+	1+	1+	1+¶	1+	3+
Paroxetine	1+	1+	1+	2+	0 to 1+	1+¶	2+	4+
Sertraline	0	0	2+	1+	1 to 2+	2+¶◇	1+	3+
Atypical agents								
Agomelatine§ (not available in United States)	0	1+	1+	0	0	1+	0	0 to 1+
Bupropion	0	0	2+ (immediate release) 1+ (sustained release)	0	1+	1+	0	0
Mirtazapine	1+	4+	0	0	1+	0	4+	1+
Serotonin-norepinephrine reuptake inhibitors¶¥								
Desvenlafaxine‡	0	0	1+	0	0	2+	Unknown	1+
Duloxetine	0	0	1+	0	0	2+¶	0 to 1+	1+
Levomilnacipran‡	0†	0	0 to 1+	0 to 1+	0	2+¶	0	1+
Milnacipran‡	0	1+	0	0	0	2+¶	0	1+
Venlafaxine‡	0	1+	1+	0	1 to 2+	2+	0 to 1+	3+
Serotonin modulators								
Nefazodone**	1+	2+	0	1+	0	2+	0	0
Trazodone	0	4+	0	1+ (hypnotic dose) 3+ (antidepressant dose)	1 to 2+	1+ (hypnotic dose) 3+ (antidepressant dose)	0 (hypnotic dose) 1+ (antidepressant dose)	1+¶¶
Vilazodone	0	0	2+	0	0	4+ΔΔ	0	2+
Vortioxetine	0	0	0	0	0	3+	0	1+
Tricyclic and tetracyclic antidepressants								
Amitriptyline	4+	4+	0	3+	1 to 2+	1+◇◇	4+	3 to 4+
Amoxapine	2+	2+	2+	2+	1+	0◇◇	2+	ND
Clomipramine	4+	4+	1+	2+	3+	1+◇◇	4+	4+
Desipramine	1+	2+	1+	2+	1 to 2+	0◇◇	1+	ND
Doxepin	3+	3+	0	2+	3+	0◇◇	4+	3+
Imipramine	3+	3+	1+	4+	3+	1+◇◇	4+	3+
Maprotiline	2+	3+	0	2+	1+	0◇◇	2+	ND
Nortriptyline	2+	2+	0	1+	1 to 2+	0◇◇	1+	ND
Protriptyline	2+	1+	1+	2+	1+	1+◇◇	1+	3 to 4+
Trimipramine	4+	4+	1+	3+	1+	0◇◇	4+	ND
Monoamine oxidase inhibitors								
Isocarboxazid	1+	1+	2+	2+	0	1+	1+	4+
Phenelzine	1+	2+	1+	3+	0	1+	2+	4+
Selegiline	1+	0	1+	1+	0	0	0	0
Tranylcypromine	1+	1+	2+	2+	0	1+	1+	4+

Scale: 0 = none; 1+ = slight; 2+ = low; 3+ = moderate; 4+ = high; ND = inadequate data.

SSRI: selective serotonin reuptake inhibitors; SNRI: serotonin-norepinephrine reuptake inhibitor.

* Relative mean QTc prolongation at therapeutic doses; arrhythmogenic potential can be significantly increased in overdose (eg, for cyclic antidepressants, bupropion, citalopram, duloxetine, venlafaxine, and some others). QTc prolongation classifications are based upon US Food and Drug Administration guidance.[6] The use of other classification criteria may lead to some agents being classified differently by other sources. Refer to UpToDate topics on acquired long QT syndrome and acute antidepressant poisonings.

¶ All SSRIs and SNRIs can cause transient nausea and gastrointestinal discomfort when starting therapy or increasing dose.

Δ Based upon reports of dose-related QTc prolongation and arrhythmia, the maximum recommended dose of citalopram is 40 mg/day in most patients; for patients at increased risk of elevated serum concentrations (eg, age >60 years, significant hepatic impairment, receiving interacting medications), the maximum daily dose is 20 mg.

◇ Sertraline is associated with higher rates of diarrhea.

§ Agomelatine may be hepatotoxic and is contraindicated in any degree of liver impairment. Transaminase monitoring is required.

¥ SNRIs do not have significant anticholinergic effects. However, SNRIs can produce anticholinergic-like effects (which appear to be mediated by noradrenergic stimulation) such as dry mouth and constipation, and they should be used with caution in narrow angle glaucoma. Levomilnacipran is associated with urinary hesitancy.

‡ May cause persistent dose-related increases in blood pressure (primarily diastolic) and heart rate. Monitor blood pressure regularly.

† Levomilnacipran can cause dose-dependent urinary hesitancy.

    - Moclobemide (an MAOI) prolongs the QT interval and toxic effects in overdose may include hypertensive crisis, serotonin and noradrenaline toxicity, agitation, aggressiveness, and behavioural changes.
- **Venlafaxine and mirtazapine are less toxic than TCAs, but have a higher toxicity compared with selective serotonin reuptake inhibitors (SSRIs).**
    - Arrhythmia is rare even after a massive overdose of venlafaxine. Venlafaxine commonly causes vomiting, sedation, tachycardia, hypertension and seizures in overdose.
    - Duloxetine has low toxicity in overdose. It can cause somnolence, coma, serotonin syndrome, seizures, vomiting, and tachycardia in overdose.
    - SSRIs have a low toxicity when taken in overdose, but have been reported to prolong the QTc interval. Citalopram and escitalopram have a higher risk of cardiac toxicity than other SSRIs in overdose.
    - There is a risk of serotonin syndrome in overdose of serotonergic drugs such as SSRIs, which may present with insomnia, anxiety, nausea, diarrhoea, hypertension, tachycardia, hyper-reflexia, agitation, myoclonus, tremor, flushing, low-grade fever, hyperthermia, confusion, rigidity, respiratory failure, coma, or death.

SNRI

Contraindications and cautions

- **Do not prescribe serotonin noradrenaline reuptake inhibitors (SNRIs) to people with:**
    - Uncontrolled hypertension.
    - Hepatic impairment (duloxetine).
    - Severe renal impairment – creatinine clearance less than 30 mL/min (duloxetine).
- **Prescribe SNRIs with caution to people with:**
    - A history of bleeding disorders.
    - A history of epilepsy.
    - A history or family history of mania or bipolar disorder.
    - Cardiac disease — monitor blood pressure.
    - Conditions associated with high risk of cardiac arrhythmia.
    - Hypertension.
    - Diabetes mellitus — blood sugar control may be affected.
    - Hepatic impairment — manufacturer advises consider dose reduction of 50% in mild to moderate impairment and of more than 50% in severe impairment (venlafaxine).
    - Renal impairment (venlafaxine).
    - Susceptibility to angle-closure glaucoma (duloxetine).

Adverse effects

Possible adverse effects associated with serotonin noradrenalin reuptake inhibitors (SNRIs) include:

- Cardiac — palpitations (common); tachycardia, hypertension, supraventricular arrhythmia (uncommon); torsade de pointes, QT interval prolongation (rare), hypotension, syncope.
- Gastrointestinal — dry mouth, altered taste, decreased appetite, nausea (very common); constipation, vomiting, diarrhoea, dyspepsia, flatulence, abdominal pain (common); gastrointestinal haemorrhage, weight changes, hepatitis (rare), pancreatitis (rare).
- Central nervous system — headache, somnolence (very common); dizziness, drowsiness, akathisia, lethargy, tremor, paraesthesia (common); akathisia, movement disorders, sleep disorders, tinnitus, vision problems, seizure (rare).

- Psychiatric — insomnia, confusion, anxiety, abnormal dreams, agitation (common); suicidal thoughts, depersonalisation.
- Skin — sweat changes, rash (common), photosensitivity reaction (uncommon), alopecia, angio-oedema.
- Other — dyspnoea, flushing, menstrual cycle irregularities, sexual dysfunction (may persist after treatment stopped); urinary symptoms, hyponatraemia (rare), neuroleptic malignant syndrome (rare), serotonin syndrome (rare), neutropenia (rare), syndrome of inappropriate antidiuretic hormone secretion (SIADH, rare), thrombocytopenia (rare).

Drug interactions

Possible drug interactions associated with use of serotonin noradrenaline reuptake inhibitors (SNRIs) include:

- Antifungals (fluconazole, itraconazole, ketoconazole) — plasma concentration of venlafaxine increased. Manufacturer advises that concomitant use with fluconazole should only be undertaken with caution; for other antifungals monitor for adverse effects and adjust venlafaxine dose if necessary.
- Aspirin, NSAIDs, anticoagulants, and antiplatelets — increased risk of bleeding if taken concurrently with SNRI. Consider additional INR monitoring if taking warfarin.
- Bupropion — plasma concentration of SNRI increased. Monitor for adverse effects and adjust SNRI dose if necessary.
- Carvedilol, metoprolol — levels may be increased by duloxetine. Monitor blood pressure.
- Ciprofloxacin — avoid concomitant use (metabolism of duloxetine is inhibited by ciprofloxacin).
- Clarithromycin, erythromycin — levels of venlafaxine may be increased. Possible increased risk of QT interval prolongation.
- Diltiazem, verapamil — plasma concentration of venlafaxine may be increased. Monitor for adverse effects and adjust venlafaxine dose if necessary.
- HIV protease inhibitors (for example ritonavir, lopinavir, atazanavir) — levels of venlafaxine may be increased. Monitor for adverse effects and adjust dose of venlafaxine if necessary.
- Lithium — concurrent use with SNRI may cause serotonin syndrome. In addition, lithium has been associated with QT interval prolongation. Monitor for adverse effects.
- Monoamine oxidase inhibitors (MAOIs) — concurrent use is contraindicated. Fatal reactions may occur (serotonin syndrome).
- SSRIs (paroxetine, sertraline) — increased risk of serotonin syndrome when given concurrently with an SNRI. Monitor for symptoms (such as fever, tremors, diarrhoea, agitation).
    - Venlafaxine may also increase the risk of QT interval prolongation if taken concurrently with an SSRI. Consider ECG monitoring.
- Other sedative drugs (alcohol, barbiturates, benzodiazepines) — SSRIs are sedating and co-administration with other sedating drugs may have a synergistic effect.
- Other drugs, which if taken with an SNRI may increase the risk of serotonin syndrome include mirtazapine, opioids, St John's wort (avoid concurrent use), triptans, vortioxetine.
- QT interval prolongation — possible increased risk of QT interval prolongation if venlafaxine is taken with other drugs known to cause QT interval prolongation, including antiarrhythmics (amiodarone, dronedarone, quinidine); antipsychotics (phenothiazine derivatives, pimozide, haloperidol); tricyclic antidepressants (TCAs, clomipramine, imipramine); sildenafil.

Reboxetine

Contraindications and cautions

- **Prescribe reboxetine with caution to people with:**
    - A history of bipolar disorder.
    - A history of cardiovascular disease.
    - A history of epilepsy.
    - Hepatic impairment (risk of increased exposure).
    - Prostatic hypertrophy.
    - Renal impairment (needs initial dose reduction).
    - Susceptibility to angle-closure glaucoma.
    - Urinary retention.

Adverse effects

Possible adverse effects associated with reboxetine include:

- Cardiac — palpitations, tachycardia, (common), hypertension, hypotension.
- Vision — accommodation disorder (common); mydriasis (uncommon).
- Gastrointestinal — dry mouth, decreased appetite, altered taste, nausea, constipation (very common); vomiting (common).
- Central nervous system — dizziness (very common); insomnia, headache, akathisia, paraesthesia, vertigo.
- Psychiatric — insomnia (very common); anxiety, agitation (common); sleep disorder.
- Skin — hyperhidrosis (very common); skin reactions.
- Other — sexual dysfunction, urinary disorders, chills.

Referral

Arrange referral to specialist mental health services for co-ordinated multidisciplinary care, or seek specialist advice if:

- Considering adding in an additional antidepressant medication from a different drug class, if there is a limited response or no response to treatment in primary care.
    - Options include adding trazodone or mirtazapine to a selective serotonin reuptake inhibitor (SSRI).
- Considering other specialist combination drug treatment or physical therapy (such as electroconvulsive therapy [ECT]).
    - Specialist drug options include combining an antidepressant with a second-generation antipsychotic (such as aripiprazole, olanzapine, quetiapine, risperidone), or augmentation with lithium, lamotrigine, or triiodothyronine.

Complications

Possible complications of depression include:

- Impact on personal and social functioning including work, college, relationships, and looking after dependents, and reduced quality of life.
- Potential increased risk of medical complications, functional impairment, and poorer prognosis of any coexisting physical and mental health conditions.
- Increased risk of associated alcohol and substance misuse.
- Increased risk of morbidity and mortality due to self-harm, self-neglect, and suicide.
    - Depression is a major risk factor for attempted and completed suicide.
    - There is a 10-fold increased risk of suicide in people receiving treatment for a mental health condition compared with the general population.

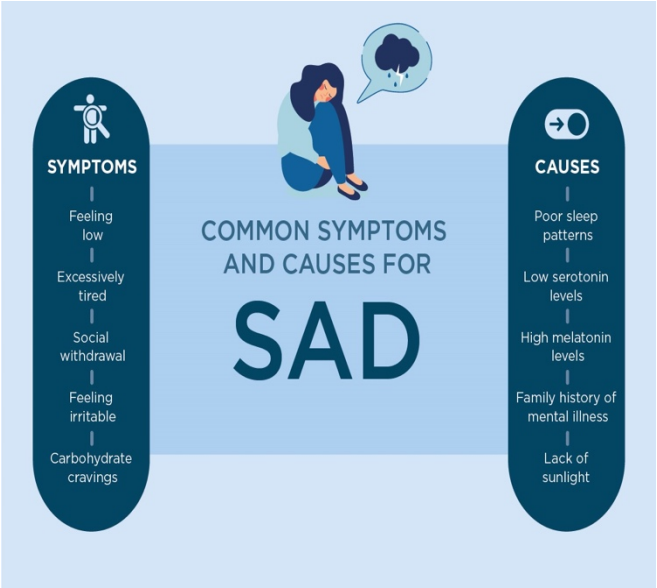
TOPIC: Seasonal affective disorder

Seasonal affective disorder (SAD) describes depression which occurs predominately around the winter months. SAD should be treated the same way as depression, therefore as per the NICE guidelines for mild depression, you would begin with psychological therapies and follow up with the patient in 2 weeks to ensure that there has been no deterioration. Following this an SSRI can be given if needed. In seasonal affective disorder, you should not give the patient sleeping tablets as this can make the symptoms worse. Finally, the evidence for light therapy is limited and as such it is not routinely recommended.

Definition:
is as a variant of depression, characterised by depressive episodes that recur annually at the same time each year, usually during the winter months. It can be very disabling for patients, some of whom may require hospitalisation for SAD at some point

Classification and seasonal affective disorder variants :
SAD with autumn- or winter-onset depression, with symptoms remitting or hypomanic or manic symptoms during the spring or summer, is the most common.
Sub-syndromal SAD is characterised by seasonal mood disruptions with milder functional impairments where the mood disruption and impairments may still be significant.
SAD with summer onset of depressive symptoms, with remission or hypomanic or manic symptoms during the winter months (less common than autumn- or winter-onset depression).

Presentation:
Patients may develop the following symptoms in a step-wise fashion, beginning in September and culminating with depression and anxiety in December. SAD symptoms persist until springtime in April.



- Difficulty waking Decreased energy.
- Carbohydrate craving, lethargy
- Increased appetite, wt gain, increased sleep
- Difficulty concentrating.
- Decreased libido.
- Withdrawal from family and friends.
- Depression/anxiety/irritability.
- Other symptoms include:
- Family problems.

Tearfulness Physical symptoms - eg, headache, palpitations and generalised aches and pains

Differential diagnosis
The patient should be assessed for other psychological conditions, including:

- Major depressive disorder.
- Bipolar I and II disorders (all patients with autumn- or winter-onset depression should be screened for spring or summer hypomania or mania symptoms. An estimated 20% of people with SAD may present with a bipolar disorder).
- Dysthymic disorder.
- Cyclothymic disorder.
- Premenstrual dysphoric disorder (associated with bloating and breast tenderness with onset during the later part of the luteal phase of the menstrual cycle and remission of symptoms during the follicular phase).

- Chronic fatigue syndrome
- hypothyroidism
- Substance misuse.
- Alcohol misuse.

. The seasonality of symptoms may be determined by asking the patient to record symptoms in a diary or using the seasonal pattern assessment questionnaire (SPAQ).

Investigations:

TFTs and urine and blood tests for illicit substances and alcohol will help to rule out some of the above conditions.

Diagnosis:

SAD is categorised as a form of depressive disorder in the DSM-IV. The gold standard for diagnosis is a structured interview to determine whether patients fulfil the DSM-IV criteria. The SIGH-SAD clinical assessment tool is commonly used.

Using this tool, the diagnosis of SAD is based on:

- Depression cycles on a regular basis during autumn/winter.
- Full remission of symptoms in spring/summer.
- Seasonal symptoms for at least two consecutive years.
- Atypical features, which may or may not be present.
- During the psychological examination, it is also important to assess:
- Suicidal ideation.
- Abnormal mechanisms of coping - eg, social isolation, alcohol use.

Treatment:

The most current National Institute for Health and Care Excellence (NICE) guidelines for depression state that depression in SAD patients should be managed in the same way as non-seasonal depression[.

Cognitive behavioural therapy (CBT), light therapy and prescribed medications have all been found to help to induce remission of SAD symptoms during winter months. In patients with SAD, or with symptoms that are highly suspicious of it, it is important not to wait until they are showing signs of depression but to start non-drug treatment while they have early symptoms such as lethargy and carbohydrate cravings around September.

Education:

- Give information about the disorder and self-help groups.
- Simple advice should include the following:
- Spend more time out of doors.
- Work in bright conditions.
- Exercise outside regularly.
- Eat a healthy diet.

Light therapy, or phototherapy

Light therapy, or phototherapy, has been associated with a reduction in depression fatigue, sleepiness and health-related quality of life.

It may be carried out using various devices such as a light box, LED screen and a light room; however, the light box is the most commonly used.

Light therapy with a light box that meets evidence-based guidelines from a reputable vendor, is recommended for first-line treatment of SAD. Self-reported reductions in depression scores have been demonstrated after one hour of light therapy exposure.

Relative contra-indications for light therapy:

Retinal disease.

Macular degeneration.

Current use of photosensitising medicines (such as some antihypertensives, antibiotics and oncology drugs).

TOPIC: Mental health in students

Definition

- **The most common mental health problems in young people in higher education are depression and anxiety disorders. Other observed mental health problems include:**
    - Bipolar disorder.
    - Psychosis and schizophrenia.
    - Eating disorders.
    - Post-traumatic stress disorder (PTSD).
    - Drug abuse

TOPIC: Eating disorders

Definition

- Eating disorders are characterised by persistent disturbance of eating or eating-related behaviour which leads to altered intake or absorption of food and causes significant impairment to health and psychosocial functioning.
- The main types of eating disorder are:
  - **Anorexia nervosa** – restriction of food intake or persistent behaviour which interferes with weight gain and leads to low body weight. Anorexia nervosa is associated with body image disturbance and an intense fear of gaining weight.
  - **Bulimia nervosa** – recurrent (at least once per week for 3 months) episodes of uncontrolled eating of an abnormally large amount of food over a short time period (binge eating) followed by compensatory behaviour such as self-induced vomiting, laxative abuse or excessive exercise.
  - **Binge eating disorder** — recurrent episodes of binge eating in the absence of compensatory behaviours. Episodes are marked by feelings of lack of control.
  - **Atypical eating disorders** (also known as 'eating disorder not otherwise specified' (EDNOS) or 'other specified feeding or eating disorder' [OSFED]) — symptoms of an eating disorder such as anorexia nervosa, or bulimia nervosa, which do not meet the precise diagnostic criteria. For example, all of the criteria for anorexia nervosa are met, there is significant weight loss, but the person's weight is within or above normal range.
- The different types of eating disorders have many features in common and a person can move between the different diagnostic categories over time.

Causes

- The aetiology of eating disorders is not fully understood but is generally considered to involve a combination of genetic, neurobiological, and sociocultural factors.
- Factors associated with development and maintenance of an eating disorder include:
  - General factors such as female sex, adolescence and early adulthood, and being overweight as a child.
    - About 90% of people with an eating disorder are female.
  - Neurobiological factors:
    - Eating disorders have been associated with abnormal neurotransmitter activity (including serotonin and dopamine) and satiety-related hormones (including adiponectin and ghrelin).
    - It is not known if these are involved in the development of an eating disorder or are the result of physiological alterations due to malnutrition.
  - Psychosocial factors such as:
    - Anxiety, depression, perfectionism, low self-esteem, body dissatisfaction and overestimation of body size.
    - Societal idealizations about body shape including professional or recreational pressure to be slim (for example dancing, modelling, and athletics).
    - Household stress, social isolation and poor social support.
  - Family history of an eating disorder, psychiatric disorder or substance misuse.

Complications

- Psychological and emotional disturbance such as anxiety, social withdrawal, poor quality of life, low mood and suicidal ideation.

- Social difficulties such as disrupted relationships, isolation, limitations on employment prospects and economic disadvantage.
- Family/carer stress — the burden of caring for a person with anorexia nervosa is comparable to that of caring for a person with depression or schizophrenia.
- Physical abnormalities — usually secondary to compromised nutritional state and can affect many systems including.
    - Cardiovascular — arrhythmias, hypotension, valve prolapse, peripheral oedema, sudden death.
    - Musculoskeletal — loss of muscle strength, loss of bone density (may be irreversible), fractures, impairment of growth in children and teenagers.
    - Endocrine — thyroid abnormalities, incomplete development of secondary sexual characteristics, impaired temperature regulation.
    - Gastrointestinal (GI) — slowed GI motility, constipation, upper GI bleeding, abnormal liver function tests, enlargement of the parotid glands.
    - Haematological — low white blood cell count (particularly neutrophils), anaemia (rarely), thrombocytopenia.
    - Metabolic — dehydration, electrolyte disturbance in those who misuse laxatives or diuretics or induce vomiting, re-feeding syndrome (a complication of rapid re-feeding treatment).
    - Neurological — cognitive impairment, seizures (rarely).
    - Renal — kidney stones and acute or chronic kidney disease (chronic volume depletion can lead to the development of kidney disease).
    - Dental — erosion of tooth enamel from vomiting.
    - Dermatological — dry skin, alopecia, lanugo (fine, white hairs on the body) hair and chilblains.
    - Obstetric and gynaecological — infertility (may be irreversible), sexual dysfunction, risk of polycystic ovaries, miscarriage, hyperemesis, anaemia, intrauterine growth restriction, preterm birth, postpartum depression and anxiety.
- Mortality — most deaths due to anorexia nervosa are a result of the medical consequences of starvation (especially cardiac and severe infection) however around 20% are due to suicide.

Clinical features

Clinical features of anorexia nervosa

- **Restriction of energy intake resulting in low body weight.**
    - Low weight is defined as less than minimally normal in adults (typically BMI less than 18.5) or minimally expected in children and young people (typically BMI-for-age less than the 5th percentile on an appropriate growth chart).
- **Intense fear of gaining weight.**
    - Preoccupation with food and weight is often related to a pursuit of thinness, or later in the course of illness, a fear of gaining weight.
    - Changes in behaviour may be noticed for example becoming increasingly selective about food, avoiding eating with others, saying they aren't hungry, that they have already eaten, or refusing food they used to enjoy.
    - Other behaviours may include repeated weighing, measuring and checking in the mirror.
- **Behaviour that interferes with weight gain.**
    - Compensatory behaviours such as self-induced purging (by vomiting or misuse of laxatives), excessive exercise, and use of appetite suppressant medication or diuretics may be present.
- **Psychological disturbance which may include.**
    - Distortion of body image, with a dread of being overweight.

    - Low self-esteem and a drive for perfection.
    - Over-evaluation of self-worth in terms of body weight and shape.
- **Denial of seriousness of malnutrition and its impact on physical health.**
    - Many people with anorexia nervosa have difficulty acknowledging there is a problem, do not consider themselves to be ill and may be ambivalent about change.
    - In others, there may be secrecy about the symptoms, for example hiding weight loss.
- **Hormonal disturbance.**
    - Females with anorexia nervosa may present with amenorrhoea or other menstrual irregularities.
    - Hormonal disturbance is a common feature of anorexia nervosa but amenorrhea is no longer in the criteria for diagnosis in females and is only one of a number of markers of hormonal insufficiency due to low weight.
    - Males with anorexia nervosa may present with loss of libido and potency.
    - In children, onset of puberty, growth and physical development can be affected.
- **Many physical signs are associated with anorexia nervosa** including dry skin, hair loss, bradycardia, orthostatic hypotension, hypothermia, loss of muscle strength, oedema, constipation, fainting, dizziness, and fatigue.

Clinical features of bulimia nervosa

- **Recurrent episodes of binge eating** occurring on average at least once a week for 3 months.
    - A binge is defined as consuming an excessive amount of food in a discreet time period accompanied by a sense of loss of control over eating at that time.
    - In-between binges there are also typically continuing attempts to restrict eating.
- **Recurrent inappropriate compensatory behaviour to prevent weight gain (occurring on average at least once a week for 3 months).**
    - Compensatory behaviours include vomiting, purging, fasting, excessive exercise, laxative, diuretic or diet pill use.
- **Weight is often within normal limits or above weight range for age.**
    - Bulimia nervosa may be kept secret for many years, as the person's appearance is typically unremarkable and they can often eat normally in public.
    - Many people delay seeking help because of the shame associated with binge eating or because they do not know that treatment is available.
- **Psychological features**, which may include:
    - Over-evaluation of self-worth in terms of body weight and shape.
    - Fear of gaining weight, with a sharply defined weight threshold set by the person.
    - Mood disturbance and symptoms of anxiety and tension.
    - Persistent preoccupation and craving for food and feelings of guilt and shame about binge eating.
    - Self-harm, often by scratching or cutting.
- **Physical symptoms**, such as bloating, fullness, lethargy, gastro-oesophageal reflux, abdominal pain, and sore throat (from vomiting).
    - In severe cases, physical signs may include Russell's sign (knuckle calluses from inducing vomiting), dental enamel erosion, and salivary gland enlargement.

Clinical features of binge eating disorder

- **Recurrent episodes of binge eating (at least once per week for 3 months) in the absence of compensatory behaviours.**

    - A binge is defined as consuming an excessive amount of food in a discreet time period accompanied by a feeling of loss of control where the person cannot stop eating or control the amount of food they eat at that time.
    - During a binge the person may eat more rapidly than normal, eat until uncomfortably full or when not hungry and experience significant distress and feelings of guilt and shame.
- **Body weight may be maintained at normal, overweight or obese.**
    - Many people with binge eating disorder are overweight or obese.

Clinical features of atypical eating disorders

- Atypical eating disorders have features that closely resemble but do not meet the strict criteria for other diagnostic categories for example the person's weight may be just above the diagnostic threshold for anorexia nervosa or binge eating/purging may occur infrequently.
    - The majority of eating disorders are atypical.
- Over-concern with body weight and shape is generally present.
- Many people with atypical eating disorders have had anorexia or bulimia nervosa previously or may in the future develop the full diagnostic criteria for anorexia or bulimia nervosa.

Assessment

Diagnosis of an eating disorder is based on suggestive clinical features supported, where possible, by corroboration from a relative/carer or friend. Be aware that:

- **Eating disorders can be difficult to detect in primary care** as those affected may be slow to present, reluctant to disclose symptoms, or be unaware they have an eating disorder.
- **A variety of tools can be used for screening and risk assessment such as:**
    - A baseline eating disorder assessment tool available on the National Institute of Health and Care Excellence (NICE) website.
    - The SCOFF questionnaire — two or more positive answers to the following questions are suggestive of anorexia nervosa or bulimia nervosa.
        - 'Do you ever make yourself sick because you feel uncomfortably full?'
        - 'Do you worry that you have lost control over how much you eat?'
        - 'Have you recently lost more than one stone in a 3-month period?'
        - 'Do you believe yourself to be fat when others say you are too thin?'
        - 'Would you say that food dominates your life?'
- **Assessment may be normal even in medically unstable people.**
    - Severe malnutrition and purging behaviours can lead to life-threatening complications such as cardiovascular instability or severe electrolyte disturbance.
    - Primary care clinicians should have a low threshold for seeking advice – emergency admission may be required.

Take a history asking about:

- symptoms of eating disorder such as:
    - Change in weight (increase, decrease, or failure to thrive).
    - Dietary restriction or binge eating.
    - Fear of gaining weight and body image disturbance — ask about perceived ideal weight.
    - Compensatory behaviours (such as excessive exercise, purging, vomiting or use of weight loss medications) — consider insulin misuse in diabetic patients.

- Complications of eating disorders such as fatigue, constipation, reflux, hair loss, and amenorrhoea.
    - Be alert for symptoms suggestive of a serious complication requiring urgent admission such as syncope, pre-syncope or severe abdominal pain.
- **Co-morbidities and symptoms suggestive of an** alternative cause such as inflammatory bowel disease or coeliac disease.
- **Mental health**
    - Mood disorders (such as anxiety, obsessive compulsive disorder, and depression) and social withdrawal are often associated with eating disorders.
    - Assess the risk of self-harm and suicide.
        - Approximately 20% of deaths in people with anorexia nervosa are due to suicide.
- **Social history and social support** including:
        - Stressors at school, work or home (including the internet and social media) – consider possibility of bullying and abuse.
    - Alcohol and drug misuse.
- **Family history** of eating disorders, depression, or substance abuse.
- **Medication** (including over-the-counter).

Examine the person:

- **Calculate the person's body mass index (BMI) and compare with previous recordings.**
    - Use centile charts if the person is younger than 18 years of age — it is important to make an early diagnosis in children because they are at risk of irreversible growth impairment.
    - Be aware that some people may refuse to be weighed or falsify their weight by drinking large amounts of water beforehand or by hiding heavy objects in their clothes.
- **Check vital signs including:**
    - Temperature (hypothermia is a red flag).
    - Pulse (bradycardia for example <50 beats per minute or postural tachycardia are red flags).
    - Blood pressure checking for postural differences (hypotension or orthostatic hypotension are red flags).
    - Hydrations state.
    - Peripheral circulation.
- **Look for muscle wasting and assess muscle strength:**
    - Scores of 2 or less (especially if scores are falling) on the Sit up–Squat–Stand (SUSS) test are a red flag.
        - The sit up test — the person lies flat on a firm surface such as the floor and has to sit up without, if possible, using their hands.
        - The squat test — the person is asked to rise from a squatting position without, if possible, using their hands.
- **Carry out a general examination looking for** complications **of eating disorders or signs suggestive of an** alternative cause.

Consider the need for investigations:

- **Extensive investigation is not usually required in primary care.**
    - Most people with an eating disorder will have normal blood results which are a poor indicator of risk, however, some tests may be useful to rule out complications.

- **Depending on clinical situation, consider the following investigations in primary care (seeking specialist advice or admitting to hospital may be more appropriate):**
    - Full blood count — may show anaemia from malnutrition or gastrointestinal losses, or mild leucopenia or thrombocytopenia from malnutrition.
    - Erythrocyte sedimentation rate (ESR) — usually normal in people with anorexia, a raised ESR may indicate an organic cause of weight loss.
    - Urea and electrolytes — hypokalaemia is suggestive of vomiting or laxative abuse; hyponatraemia may be a result of excess water intake. Electrolytes may be elevated due to dehydration.
    - Liver function tests — may be slightly elevated from malnutrition.
    - Blood glucose.
    - Creatinine, and urinalysis — chronic hypokalaemia and chronic volume depletion can lead to the development of kidney disease.
    - Electrocardiography (ECG) — this should be considered for all people with rapid weight loss, excessive exercise, severe purging behaviours (such as laxative or diuretic use or vomiting), bradycardia, hypotension, excessive caffeine (including from energy drinks), prescribed or non-prescribed medications, muscle weakness, electrolyte imbalance, or previous abnormal heart rhythm.
- Further tests may be required in more severe cases or to assess complications (seek specialist advice):
    - Calcium, magnesium, phosphate.
    - B12, folate and ferritin.
    - Thyroid function tests.
    - Follicle stimulating hormone, luteinising hormone, oestradiol, prolactin and urinalysis (including pregnancy test) may be considered if presenting with amenorrhoea.
- Other investigations may be indicated if an alternative diagnosis is suspected (for example coeliac screening).

Differential diagnosis

- **The differential diagnosis of weight loss includes:**
    - Malabsorption for example coeliac disease, inflammatory bowel disease or peptic ulcer.
    - Malignancy.
    - Drug or alcohol misuse.
    - Infection for example tuberculosis, HIV, infectious mononucleosis.
    - Autoimmune disease including rheumatological disorders.
    - Endocrine disorder for example hyperthyroidism, diabetes mellitus, hypercortisolism, adrenal insufficiency.
- **The differential diagnosis of amenorrhoea includes:**
    - Pregnancy.
    - Primary ovarian failure.
    - Polycystic ovary syndrome.
    - Pituitary prolactinoma.
    - Hypothalamic causes.
- **The psychiatric differential diagnosis includes:**
    - Depression.
    - Anxiety.
    - Obsessive-compulsive disorder.
    - Body dysmorphic disorder.
    - Substance misuse.
    - Psychosis or schizophrenia.

Management

How should I manage a suspected eating disorder in primary care?

- **Consider the need for emergency medical or psychiatric admission for anyone at risk of serious physical complications, suicide or serious self-harm.**
- **For all other people with a suspected eating disorder:**
  - **Refer *immediately* to an age appropriate eating disorder service** for specialist assessment and management – do *not* use a watchful waiting strategy for managing eating disorders.
  - **While awaiting specialist assessment:**
    - Arrange regular review (with frequency dependant on the clinical situation [for example weekly in children]) to monitor level of physical and mental health risk and consider the need for urgent admission, further investigations, or increasing the urgency of referral.
    - Consider the possibility of complications and monitor/manage appropriately, seeking specialist advice when indicated.
    - Ensure that people with co-morbidities (such as diabetes) or pregnant women have specialist input — increased monitoring and alterations to treatment may be required.
    - When prescribing medication for people with a suspected eating disorder, take into account the impact of malnutrition and compensatory behaviour on medication effectiveness (for example the oral contraceptive pill) and the risk of side effects (in particular, cardiac side effects).

When should I admit a person with a suspected or confirmed eating disorder?

Be aware that some medically unstable people with eating disorders can appear deceptively well – a low threshold for concern is essential, seek specialist advice if unsure.

- **Consider emergency admission for anyone at risk of serious physical or psychological complications such as:**
  - **Severely compromised physical health including:**
    - BMI or body weight below a safe range — degree of dietary restriction and rate of weight loss is important, for example more than 1kg per week indicates high risk.
    - Cardiovascular instability for example bradycardia of 40 beats or less per minute), tachycardia on standing, prolonged QT interval on ECG, or hypotension (including postural).
    - Hypothermia.
    - Reduced muscle power on the Sit up–Squat–Stand (SUSS) test .
    - Concurrent infection.
    - Overall ill health or rapid deterioration.
    - Abnormal blood tests such as electrolyte imbalance or hypoglycaemia.
  - **Risk of refeeding syndrome:**
    - Risk is increased by rapid weight loss, fasting for over five days, BMI less than 16kg/m2, compensatory behaviours (such as laxative misuse or vomiting), dehydration, use of diet pills or diuretics, water loading or excessive exercise.
  - **Acute mental health risk (such as risk of suicide attempt or serious self-harm):**
    - Consider psychiatric crisis care or psychiatric inpatient care – discuss with a specialist if unsure whether admission to medical/paediatric ward is more appropriate.

    - **Lack of support at home:**
        - Consider admission if the person may not be kept from significant harm at home or where the home environment impedes recovery.
- **If unsure where to admit (for example acute psychiatric ward, specialist eating disorders unit or acute medical/paediatric ward) discuss with a specialist.**
    - The most appropriate place to admit a person with a life-threatening eating disorder is not always obvious and depends on local services and practice.
    - Children and adolescents should be admitted to age-appropriate facilities.
- **If a person's physical health is at serious risk due to their eating disorder and they can only be treated safely as an inpatient but they do not consent to admission**:
    - Consider the need for compulsory admission and seek specialist advice.

How should I manage someone with a confirmed eating disorder in primary care?

- **Ensure there is a clear agreement between primary and secondary or tertiary care about the responsibility for monitoring a person with an eating disorder.**
    - Monitoring of general medical problems usually occurs in primary care.
    - Good co-ordination of care is particularly important when young people move from child to adult services, if more than one service is involved in care, or care is needed in different places at different times (for example university students).
- **For people who are being treated in secondary care — monitor the level of risk to the person's mental and physical health as treatment progresses.**
    - Be aware that medically unstable people with eating disorders (such as anorexia nervosa and bulimia nervosa) can appear deceptively well.
    - Risk may increase, for example following weight change or at times of transition between services — consider the need for admission or increased specialist input.
    - Monitor growth and development closely in children and young people with anorexia nervosa who have not completed puberty — discuss with a specialist if unsure.
- **If the person has chronic anorexia nervosa and is not receiving ongoing treatment in secondary care,** review their physical and mental health at least annually in primary care.
    - A baseline assessment tool is available on the NICE website.
    - Review should include discussion of treatment options and:
        - Weight or BMI (using centile charts where appropriate).
        - Blood pressure.
        - Relevant blood tests.
        - Discussion of problems with daily function.
        - ECG for people with purging behaviours and/or significant weight changes.
        - Assessment of risk (both physical and mental health).
- **If the person exhibits compensatory behaviours such as:**
    - **Vomiting —** give advice on the importance of regular dental and medical review. They should:
        - Avoid brushing their teeth immediately after vomiting this can brush acid on to the teeth causing further damage.
        - Rinse with water or a non-acidic mouthwash after vomiting.
        - Reduce the acidity of the oral environment (for example by avoiding acidic foods and drinks).
    - **Misuse of laxatives or diuretics** — advise them that:
        - Laxatives and diuretics do not reduce calorie absorption and do not help with weight loss — laxative or diuretic use should be gradually reduced and stopped.
    - **Excessive exercise —** advise them to stop.

- **Put an alert in the person's prescribing record to highlight the potential risk of adverse drug effects.**
    - Adverse drug effects, such as prolonged QT interval and cardiac dysrhythmias are more likely in malnourished people and those with electrolyte abnormalities — review all drugs prescribed to people with an eating disorder.
    - Offer ECG monitoring to people with an eating disorder who are taking medication that can compromise cardiac functioning for example drugs that may cause electrolyte imbalance, bradycardia, hypokalaemia, or a prolonged QT interval.
- **Manage complications depending on their type and severity**.
    - Some people will require inpatient or specialist outpatient care while others can be managed in primary care— seek advice if unsure.
    - For suspected developmental delay or faltering growth in children and young people with an eating disorder seek specialist paediatric advice.
    - If low bone mineral density is suspected — discuss bone density scanning and treatment with the eating disorder team. Advise people with anorexia nervosa and osteoporosis/related bone disorders to avoid activities that are high-impact or significantly increase the chance of falls or fractures.
- **Manage comorbidities with input from specialists where appropriate, for example:**
    - Diabetes — insulin may be misused in people with eating disorders. Close monitoring and collaboration between the eating disorder and diabetes teams is required — have a low threshold for checking blood glucose and ketones and seeking specialist advice.
    - Associated mental health problems and substance abuse – seek specialist advice.
- **Be aware that women with an eating disorder:**
    - **Who are sexually active** should be advised of the risk of unplanned pregnancy and the need for effective contraception even if they have amenorrhoea.
        - Long-acting reversible contraception (LARC) methods (such as implant and intra-uterine contraceptives) are most effective in this group.
        - Implants should only be fitted by experienced clinicians as some woman with eating disorders may have little subcutaneous tissue.
        - Consideration of the UK Medical Eligibility Criteria for contraceptive use (UKMEC) is required.
        - Anorexia nervosa is a significant risk factor for osteoporosis — use of progesterone only injectables in anorexia requires careful informed discussion and consideration by the woman.
    - **Who are planning a pregnancy and are underweight:**
        - Should be informed that the risk of adverse pregnancy outcomes is increased when underweight and advised to delay conception until in remission. Give advice on the importance of maintaining good mental health, adequate nutrition, a healthy body weight and stopping behaviours such as binge eating, vomiting, laxatives and excessive exercise.
    - **Who are pregnant** need increased monitoring and support during pregnancy and post-partum.
        - Eating disorders increase possible health risks to mother and child and are associated with a high risk of mental health problems in the perinatal period – seek specialist advice.
- **Ensure the person and their family/carers have access to information and support. Be aware that:**
    - The family/carers of a person with an eating disorder may experience severe distress.
    - The need for support (including practical support and emergency plans if the person with the eating disorder is at high medical or psychiatric risk) should be assessed.

What specialist treatments are recommended for eating disorders?

Anorexia nervosa

- **For adults with anorexia nervosa, one of the following may be considered:**
  - Individual eating-disorder-focused cognitive behavioural therapy (CBT-ED) — typically up to 40 sessions over 40 weeks, with twice-weekly sessions in the first 2 or 3 weeks.
  - If individual CBT-ED, MANTRA, or SSCM is unacceptable, contraindicated or ineffective eating-disorder-focused focal psychodynamic therapy (FPT) may be considered — typically 40 sessions over 40 weeks.
  - Antipsychotic or SSRI can be given in high dose.
- **For anorexia nervosa in children and young people:**
  - Anorexia-nervosa-focused family therapy for children and young people (FT-AN) delivered as single-family therapy or a combination of single and multi-family therapy will be considered — typically 18–20 sessions over 1 year.
  - If FT-AN is unacceptable, contraindicated or ineffective for children or young people with anorexia nervosa, individual CBT-ED or adolescent focused psychotherapy for anorexia nervosa (AFP-AN) will be considered.
    - Individual CBT-ED for children and young people with anorexia nervosa typically consists of up to 40 sessions over 40 weeks, with individual and family sessions.
    - AFP-AN for children and young people typically consists of 32–40 individual sessions over 12–18 months, with 8–12 additional family sessions.

Bulimia nervosa

- **Psychological treatments in adults:**
  - Bulimia-nervosa-focused guided self-help programmes may be considered for example 4 to 9 sessions lasting 20 minutes each over 16 weeks, running weekly initially.
  - If bulimia-nervosa-focused guided self-help is unacceptable, contraindicated, or ineffective after 4 weeks of treatment, individual eating-disorder focused cognitive behavioural therapy (CBT-ED) may be considered. This typically consists of up to 20 sessions over 20 weeks.
  - SSRI or Antipsychotics can be given.
- **Psychological treatments in children and young people:**
  - Bulimia-nervosa-focused family therapy (FT-BN) may be considered in children and young people with bulimia nervosa — typically 18–20 sessions over 6 months.
  - If FT-BN is unacceptable, contraindicated or ineffective, individual eating-disorder-focused cognitive behavioural therapy (CBT-ED) for children and young people with bulimia nervosa may be considered — typically 18 sessions over 6 months, with more frequent sessions early in treatment and up to 4 additional sessions with parents or carers.

Binge eating disorder

- **Psychological interventions that may be offered to adults and children and young people with binge eating disorder include:**
  - Evidence-based self-help programmes with brief supportive sessions (for example 9 sessions over 16 weeks).

TOPIC: Panic disorder

Panic disorder is frequently associated with agoraphobia and/or social phobia. There is a significant association with mood disorders, particularly depression, with lifetime prevalence rates as high as 50-60%. There appears to be a higher risk of suicide attempts than in the general population.

Physical health conditions

These may co-exist but one would not generally use the term panic disorder if the symptoms arise directly from the physical illness. Cardiovascular conditions such as mitral valve prolapse, cardiomyopathy and hypertension are associated. Chronic obstructive airways disease and migraine headaches are also present in a larger proportion of sufferers than chance would suggest, as are disorders such as irritable bowel syndrome and tension-type headache. A link with joint hypermobility disorder has been elucidated.

Differential diagnosis

Mental health conditions

- agoraphobia (often co-exists).
- Social anxiety disorder (often co-exists).
- Anxiety disorders, including GAD (may co-exist).
- Adjustment disorders.
- Bipolar disorder
- depression
- dissociative disorder
- factitious disorder
- somatization disorder
- Mental symptoms arising as a result of physical illness.
- OCD
- Specific phobic disorders.
- PTSD
- Stimulant-drug misuse (including caffeine-related illness).

Physical health conditions

- HYPERTHYROIDISM
- pheochromocytoma
- Carcinoid syndrome.
- Hypoglycaemic episodes (possibly due to insulinoma in those not using insulin/oral hypoglycaemic agents).
- Paroxysmal cardiac dysrhythmia.
- Mitral valve prolapse
- MI
- Recurrent small pulmonary emboli.
- Epileptiform disorders, particularly temporal lobe epilepsy.
- Withdrawal from alcohol/sedatives/opiates.
- Paroxysmal vestibular disorders - eg, Meniere's disease.

Panic disorder is characterised by recurrent unexpected panic attacks that are not situational. It can be a severe and disabling illness and is common in primary care. It can be difficult to assess when it presents acutely, as many of its symptoms may also be experienced with physical illness. Those who experience panic disorder often present repeatedly to their GP or local emergency department with worrying episodes of multiple symptoms that the patient may ascribe to life-threatening illness. This condition often co-exists with agoraphobia - the avoidance of exposed situations for fear of panic or inability to escape. Panic disorder is often classified into panic disorder with, or without,

agoraphobia. Social anxiety disorder can often co-exist and is diagnosed where the situations avoided are predominantly social and interactive in nature.
The first panic attack may be associated with a stressful episode but gradually the attacks become dissociated and occur 'out of the blue'. It is a chronic condition which relapses and can lead to much distress and social dysfunction.

Diagnosis:

To satisfy the DSM-5 diagnostic criteria:

- Panic attacks are recurrent and unexpected.
- At least two unexpected attacks are necessary for the diagnosis.
- At least one attack must be followed by at least one month of persistent concern about additional attacks or their consequences, or by significant maladaptive behavioural changes related to the attacks.
- Panic attacks must not arise as a result of alcohol or substance misuse, medical conditions or other psychiatric disorders.

A panic attack is defined as a discrete episode of intense subjective fear or discomfort, where at least four of the characteristic symptoms, listed below, surge abruptly and peak within minutes of the onset of the attack. Attacks usually last at least 10 minutes but their duration is variable.

Symptoms

Panic disorder manifests as the sudden, spontaneous and unanticipated occurrence of panic attacks, with variable frequency, from several in a day to just a few per year:

- Palpitations, pounding heart or accelerated heart rate.
- Sweating.
- Trembling or shaking.
- Dry mouth.
- Feeling short of breath, or a sensation of smothering.
- Feeling of choking.
- Chest pain or discomfort.
- Nausea or abdominal distress.
- Feeling dizzy, unsteady, light-headed or faint.
- Derealisation or depersonalisation (feeling detached from oneself).
- Fear of losing control or 'going crazy'.
- Fear of dying.
- Numbness or tingling sensations.
- Chills or heat sensations.

Investigation

There are no specific investigations to diagnose the condition but clinicians may wish to refer the patient, or carry out tests to exclude underlying physical causes for the symptoms.

Management:

Psychological therapy, medication, and self-help have all been shown to be effective in the treatment of panic disorder.There is no high-quality, unequivocal evidence to support one psychological therapy over another.There is also no strong evidence regarding the relative effectiveness of

psychological therapies and pharmacological treatment.NICE recommends a stepped care approach.

Step 1: recognition and diagnosis

This has been dealt with in the 'Presentation', 'Differential diagnosis' and 'Investigations' sections, above.

Step 2: treatment in primary care

For people with mild to moderate panic disorder, offer or refer for one of the following low-intensity interventions:

- Individual non-facilitated self-help.
- Individual facilitated self-help.

Self-help

- Offer information about the condition, and about support groups or other help where available.
- Give the patient details of books based on CBT principles, and contact details of any available support groups. There is evidence that self-help interventions are an effective option for people with panic disorder.
- Advise on the benefits of exercise for general good health. There is some evidence that exercise reduces anxiety symptoms. A systematic review suggested that the effect is not as great as antidepressants but it could be a useful adjunct.
- Advise avoiding anxiety-producing substances - eg, caffeine.
- Patients may benefit from advice on how they can control some of their symptoms by using abdominal/diaphragmatic breathing.

It is important to exclude alcohol or drug misuse as a factor and to treat these problems if present. Reassessment after successful management of substance-related issues will reveal if this is true panic disorder. Response to pharmacological/psychological therapies is likely to be poor in the face of alcohol/drug misuse or dependence.

Monitor the patient on a regular basis, usually every 4-8 weeks, preferably using a self-completed questionnaire.

Step 3: review and offer alternative treatment if appropriate

For people with moderate to severe panic disorder (with or without agoraphobia), consider referral for:

- CBT; or
- An antidepressant if the disorder is long-standing or the person has not benefited from, or has declined, psychological intervention.

Cognitive behavioural therapy (CBT)

- Treatment focused on the recognition of factors which trigger the panic, and behavioural methods to cope with the symptoms, have been found to be very useful.
- Trained and supervised personnel should be involved in the delivery of treatment, working to empirically grounded protocols.
- 1-2 hours a week is suitable for most people, for a total of 7-14 hours to be completed within four months from starting.
- Briefer CBT of about seven hours may be appropriate for some patients, combined with self-help materials.
- More intensive CBT over a shorter period of time may be suitable for some patients.
- Overall, there is evidence of a moderate treatment effect with psychological treatments in primary care.

Antidepressant medication

General principles

- Before prescribing, consider age, previous treatment, tolerability, other medication, comorbidities, personal preference, cost and risk of self-harm (SSRIs are less dangerous than tricyclics in overdose).

- Inform the patient about possible side-effects (including a temporary increase in anxiety at the start of treatment), delay in onset of effect, possible discontinuation symptoms, the length of treatment and the need to follow dosage instructions.
- Follow MHRA safety advice on citalopram, including enquiring about concomitant cocaine use and the potential for increased bleeding as a result.
- Provide written information appropriate to the patient's needs.
- Start with a low dose to minimise side-effects.
- Some patients may need long-term treatments and a dose at the upper end of the range.
- Do not prescribe benzodiazepines, sedative antihistamines and antipsychotics for panic disorder.
- Antidepressant drugs have been shown to be effective in reducing the amplitude of panic, reducing frequency of, or eliminating, panic attacks and improving quality-of-life measures in this group of patients.
- The classes of antidepressants that have an evidence base for effectiveness are the selective serotonin reuptake inhibitors (SSRIs), serotonin-noradrenaline reuptake inhibitors (SNRIs) and tricyclic antidepressants (TCAs).
- Offer an SSRI licensed for this indication first-line unless contra-indicated.
- Consider imipramine or clomipramine if there is no improvement after 12 weeks and further medication is indicated.
- Review the patient after two weeks to check for side-effects and efficacy, and at four, six and 12 weeks.
- If there has been an improvement after 12 weeks, continue for six months after the optimum dose has been reached.
- If medication is used for longer than 12 weeks, review at 8- to 12-weekly intervals.
- Follow the summary of product characteristics of the individual drugs for other monitoring requirements.
- Use self-completed questionnaires to monitor outcomes where possible.
- At the end of treatment, withdraw the SSRI gradually, as dictated by patient preference, and monitor monthly for relapse for as long as appropriate to the individual.
- If there is no improvement after a 12-week course, offer an antidepressant from the alternative class (if appropriate) or another form of therapy.

Step 4: review and offer referral from primary care if appropriate

If two interventions (including self-help, psychological therapy, and medication) have been offered without benefit, consider referral to specialist mental health services. Specialist treatment may include management of comorbid conditions, structured problem solving, other types of medication and treatment at tertiary centres.

Prognosis

The literature is contradictory about prognosis. In one study, panic disorder had a long course lasting several years.In another study in primary care, only a quarter of patients with panic disorder and agoraphobia had remission in a three-year follow up. Yet, three quarters of those with just panic disorder improved.There is an increased risk of attempted suicide in people with panic disorder. Those who have recovered should be made aware that the condition may relapse and that they should seek early help for further treatment if panic attacks return. They can be helped to recognise and ameliorate triggers to panic.

TOPIC: Generalized anxiety disorder

Diagnosis

- Be aware that, in primary care, people with GAD often present solely with physical symptoms such as headaches, muscle tension, gastrointestinal symptoms, back pain, and insomnia, and may not readily report worry or psychological distress.
- International diagnostic criteria can be used to aid the diagnosis:
    - The DSM-5-TR diagnostic criteria for GAD include:
        - Excessive anxiety and worry (apprehensive expectation), occurring more days than not for at least 6 months, about a number of events or activities (such as work or school performance).
        - The individual finds it difficult to control the worry.
        - The anxiety and worry are associated with three (or more) of the following six symptoms (with at least some symptoms present for more days than not for the past 6 months): restlessness or feeling 'keyed up' or on edge, easily fatigued, difficulty concentrating or mind going blank, irritability, muscle tension, sleep disturbance (difficulty falling or staying asleep, or restless, unsatisfying sleep).
        - The anxiety, worry, or physical symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
        - The disturbance is not attributable to the psychological effects of a substance, and cannot be explained by another mental disorder.
- Consider using a validated assessment tool such as the GAD-2 or GAD-7 questionnaires to determine the severity of GAD.
- Assess the risk of suicide.

Differential diagnosis

The differential diagnoses of generalized anxiety disorder (GAD) include:

- Situational anxiety — suggested by controllable anxiety that is not generally associated with pathological symptoms and relates to a particular life event (such as an upcoming examination).
- Adjustment disorder — suggested by temporary anxiety that has occurred in response to a life stressor and persists for no longer than 6 months after the stressor ends.
- Depression — suggested by an inability to feel pleasure and feelings of hopelessness.
- Panic disorder — suggested by recurrent episodes of sudden onset anxiety, in the absence of multi-themed worry. During an acute attack, the person experiences at least four symptoms from the following: shortness of breath, palpitations, shakiness, nausea, hot or cold flushes, dizziness, and fear of dying. Also frequently accompanied by avoidance behaviours (of activities where escape would be difficult). Panic may exist alongside GAD.
- Social phobia — suggested by anxiety or persistent fear that is limited to social situations and fear of social scrutiny or embarrassment. Avoidance behaviour is commonly present.
- Obsessive-compulsive disorder — suggested by anxiety due to compulsions or obsessions.
- Post-traumatic stress disorder — suggested by anxiety that is caused by exposure to reminders of past trauma. The person may report feeling as if they are reliving these events through flashbacks and nightmares.
- Somatoform disorders — suggested by anxiety related to specific physical complaints, which have no medical basis.
- Anorexia nervosa — suggested by anxiety related to fear of gaining weight.
- Substance or drug-induced anxiety disorder — suggested by anxiety directly related to exposure to a substance (for example, alcohol, illicit or prescribed drugs). Consider a urine drug screen if substance misuse requires confirmation.
    - Salbutamol, theophylline, corticosteroids, antidepressants, some herbal medicines, and caffeine can be triggers.

- Central nervous system depressant withdrawal — suggested by anxiety during withdrawal of a substance (for example, alcohol, opioids, sedatives). Signs and symptoms include shakiness, tachypnoea, tachycardia, and disorientation.
- Cardiac disease — suggested by symptoms such as palpitations, sensation of rapid or skipped heartbeat, dizziness, dyspnoea and chest pain on exertion, and numbness. Examination may reveal hypertension, hypotension, tachycardia or bradycardia, or S3 or S4 gallops. Investigations such as angiogram, echocardiogram, exercise stress test, and/or ECG should be arranged to confirm the diagnosis.
- Pulmonary disease — suggested by a history of lung conditions such as asthma or chronic obstructive pulmonary disorder, the person being a current or previous smoker, and signs/symptoms such as wheezing, cough, respiratory distress, or sputum production, and a feeling of suffocation. Pulse oximetry may reveal low oxygen saturation. Pulmonary function tests should be arranged to confirm the diagnosis.
- Hyperthyroidism — suggested by weight loss, warm moist skin, heat intolerance, ophthalmopathy, or goitre. Thyroid function tests should be arranged to confirm the diagnosis.
- Infection — suggested by anxiety and additional symptoms of fever, night sweats, or cough, which resolve following successful treatment.
- Irritable bowel syndrome — suggested by alteration of bowel habit associated with pain, and abdominal discomfort, bloating, or distention.
- Phaeochromocytoma — suggested by anxiety accompanied by hypertension, headache, diaphoresis, pallor, and palpitations. There may be a family history of phaeochromocytoma. 24-hour urine for vanillylmandelic and metanephrines should be arranged to confirm the diagnosis.

Risk factors

The aetiology of anxiety disorders is multifactorial with both genetic and environmental factors playing a role. The following factors have been shown to increase a person's risk of developing generalized anxiety disorder (GAD):

- Female sex — GAD is twice as common in women than in men.
- Comorbid anxiety disorder — for example, panic disorder or social phobia.
- Family history of anxiety disorders, depression, or other psychiatric disorders.
- Childhood adversity such as:
    - Maltreatment (for example, sexual or physical abuse), neglect.
    - Maternal depression, family disruption (for example, divorce).
    - Domestic violence, parental alcoholism, or drug use.
    - Parental mental health problems.
    - Exposure to an overprotective or overly harsh parenting style.
    - Bullying or peer victimization among youths.
- History of physical, sexual, or emotional trauma, such as:
    - Physical or sexual abuse or assault.
    - Motor vehicle accident.
    - Sudden bereavement.
- Sociodemographic factors, such as:
    - Separated, widowed, divorced.
    - Unemployment.
    - Low socioeconomic status.
    - Low education levels.
- Substance dependence or exposure to organic solvents — these can exacerbate the development of anxiety disorders.
- Chronic physical condition, for example, cardiovascular disease, cancer, respiratory disease, diabetes mellitus, or arthritis.

Management

How should I manage a person with generalized anxiety disorder?

The National Institute for Health and Care Excellence (NICE) recommends a stepped approach in the management of generalized anxiety disorder (GAD).

- Step 1 — for all people with GAD:
    - Communicate the diagnosis of GAD as early as possible to help people understand the disorder and start effective treatment promptly.
    - Provide information about the nature of GAD and treatment options.
    - Arrange active monitoring of the person's symptoms and functioning at intervals based on clinical judgement.
    - And a comorbid depressive or other anxiety disorder, treat the primary disorder first (that is, the one that is more severe and in which it is more likely that treatment will improve overall functioning). Also ensure that other comorbid conditions are optimally managed.
- Step 2 — for people with GAD whose symptoms have not improved, offer one of the following low-intensity psychological interventions, guided by the person's preference:
    - Individual non-facilitated self-help — based on cognitive behavioural therapy (CBT) principles, this should include suitable written or electronic materials that the person works through systematically over a period of at least 6 weeks. It usually involves minimal therapist contact, for example an occasional short telephone call of no more than 5 minutes, if required.
    - Individual guided self-help — should include suitable written or electronic materials, and be supported by a trained practitioner who facilitates the programme and reviews progress and outcomes. This usually consists of 5–7 weekly or fortnightly face-to-face or telephone sessions, each lasting 20–30 minutes.
    - Psychoeducational groups — based on CBT principles, this should have an interactive design and encourage observational learning through presentations and self-help manuals. Conducted by trained practitioners, there should be a ratio of one therapist to 12 participants and usually consist of 6 weekly 2-hour sessions.
- Step 3 — for people with GAD and marked functional impairment, or with GAD that has not improved following step 2 interventions, offer one of the following guided by the person's preference:
    - An individual high-intensity psychological intervention such as CBT or applied relaxation.
        - These usually consist of 12–15 weekly sessions each lasting 1 hour.
        - Inform the person that the response to psychological treatment is not immediate and that a prolonged course is usually needed to maintain an initial response.
    - Drug treatment — offer a selective serotonin reuptake inhibitor (SSRI) first line.
        - If sertraline is ineffective, offer an alternative SSRI, for example, paroxetine or escitalopram, or a selective serotonin-noradrenaline reuptake inhibitor (SNRI), such as duloxetine or venlafaxine.
        - If the person cannot tolerate SSRIs or SNRIs, consider offering pregabalin.
        - Before prescribing any medication, discuss the treatment options, any concerns the person may have, explain the reason for prescribing and provide written and verbal information on the likely benefits, potential for adverse effects, and withdrawal symptoms.
        - Explain that adverse effects early in treatment with an SSRI or SNRI may include increased anxiety, agitation, and sleeping problems, and that there will be a gradual improvement in symptoms over 1 week or more before they experience the full anxiolytic effect.
        - Review the effectiveness and adverse effects of the drug every 2–4 weeks during the first 3 months of treatment and every 3 months thereafter. Dose

adjustment may be required. Modest benefit is usually seen within 6 weeks and continues to increase over time.
    - Advise people aged under 30 years that in a minority of people aged under 30 years, SSRIs and SNRIs are associated with an increased risk of suicidal thinking and self-harm. Anyone in this age group receiving an SSRI or SNRI should therefore be seen within 1 week of first prescribing, and the risk of suicidal thinking and self-harm should be monitored weekly for the first month.
    - Do not offer a benzodiazepine for the treatment of GAD in primary care, except as a short-term measure during crises.
    - Do not offer an antipsychotic for the treatment of GAD in primary care.
- If a pregnant woman with GAD requires step 3 management:
  - Ideally, offer a high-intensity psychological intervention first-line.
  - A decision on starting drug treatment should take into account the benefits and harms of treatment in pregnancy, including the risk of not treating the condition, and the risks or harms to the woman and fetus (or baby) associated with the treatment.
    - This should be discussed with the woman.
    - Data on the risk of congenital malformations (including heart defects) following use of SSRIs or SNRIs in early pregnancy are conflicting and confounded, so the teratogenic potential is unproven. The available data for pregabalin do not currently indicate that maternal pregabalin use in pregnancy is associated with an increased risk of malformations, or miscarriage, or adversely affects fetal growth.
  - If a woman becomes pregnant while on medication for GAD, discuss the option of stopping the medication and gradually switching to a psychological intervention. Take into account the risk of relapse when considering discontinuing or switching medication.
    - In cases where drug treatment is continued in pregnancy, the lowest effective dose should be used.
  - Treatment with an SSRI or SNRI after around 20 weeks of pregnancy may increase the risk of persistent pulmonary hypertension of the newborn (PPHN) and/or can lead to neonatal withdrawal.
    - Women using these medications should be informed of the potential increased risk of PPHN.
  - Seek specialist advice, preferably from a specialist in perinatal mental health, if there is any uncertainty about drug treatment (for example if starting medication, or if changes need to be made to SSRI or SNRI medication in pregnancy).
- Step 4 — for people with complex, treatment-refractory GAD and very marked functional impairment, or high risk of self-harm:
  - Offer referral for specialist assessment of needs and risks.
  - Inform people with GAD who have not been offered or have refused the interventions in steps 1–3 about the potential benefits of these interventions and offer any they have not tried.
  - Advise carers about their right to carer assessment, and assessment for respite care and other support.
- For all people with GAD managed in primary care, also consider providing self-care about:
  - Sleep hygiene — such as going to bed and waking up at the same time each day, eliminating alcohol after 6 pm, avoiding caffeine after 3 pm, and getting out of bed if unable to fall asleep to avoid negative associations with the sleep environment.
  - The benefits of regular exercise — this can improve overall health and has been shown to improve anxiety symptoms.

    - Over-the-counter medicines and preparations — explain the potential for interactions with other prescribed and over-the-counter medications and the lack of evidence to support their safe use.

How should I monitor a person with generalized anxiety disorder?

- Review people aged under 30 years receiving an SSRI or SNRI within 1 week of first prescribing, and monitor the risk of suicidal thinking and self-harm weekly for the first month.
- Review all other people after 4–6 weeks treatment (or earlier if they are considered at high risk), taking into account factors including severity, and duration of symptoms, as well as the degree of distress and functional impairment.
    - Consider using anxiety screening questionnaires such as GAD-2 or GAD-7 to compare with previous scores.
- Check adherence to any treatment and inquire about adverse effects.
    - If necessary, advise dose adjustment or a switch to a different drug, based on reported adverse effects and assessment of progress.
    - If a drug is effective, advise the person to continue taking it for at least a year to reduce the risk of relapse.
    - If there is no response to drug treatment, offer an alternative drug, or a high-intensity psychological intervention.
    - If there is no response to a high-intensity psychological intervention, consider offering a drug treatment.
    - If there is a partial response to either a drug treatment or a high-intensity psychological intervention, consider offering the other in addition.
- Reassess the need for referral or specialist advice.
- Be alert to suicidal ideation and assess suicide risk, especially if the person has comorbid depression.
- Re-evaluate the required frequency of follow up based on:
    - The person's preference.
    - Severity of anxiety.
    - Comorbid conditions.
    - Change since last review and response to interventions.
    - Symptoms during treatment changes.

Prescription information

SSRI

General information

- The SSRIs most commonly prescribed in the UK for the treatment of generalized anxiety disorder (GAD) are:
    - Sertraline.
    - Paroxetine.
    - Escitalopram.
- There is no evidence that any one is more effective than the others in the treatment of GAD.
- Escitalopram and paroxetine are licensed in the UK for the treatment of GAD, whilst the others are not.
- Sertraline is often used first-line as it is the most cost-effective, but as it is not licensed for the treatment of GAD, informed consent should be obtained and documented.
- Other known clinical advantages of sertraline include its minimal drug interactions (making it preferable to paroxetine), a lack of associated electrocardiographic changes (compared with escitalopram), and its lower risk of symptoms on discontinuation (compared with paroxetine).
- Choice of SSRI should also take into account the person's preference, and their prior experience of treatment with individual drugs (particularly adherence, effectiveness, adverse effects, and experience of withdrawal syndrome).
- Treatment should be continued for at least 1 year.

- Modest benefit is usually seen within 6 weeks and continues to increase over time.

SNRI

General information

- The SNRIs duloxetine and venlafaxine modified-release are both licensed in the UK for the treatment of generalized anxiety disorder (GAD).
- Both have been shown to be effective in the treatment of GAD.
- Choice of SNRI should also take into account the person's preference, and their prior experience of treatment with individual drugs (particularly adherence, effectiveness, adverse effects, and experience of withdrawal syndrome).
- Treatment should be continued for at least 1 year.
    - Modest benefit is usually seen within 6 weeks and continues to increase over time.

Pregabalin

Diazepam

What dose of diazepam should I prescribe?

- Prescribe the lowest possible dose for the shortest period of time and review the patient regularly. Treatment should not exceed 2–4 weeks.
- For anxiety, prescribe 2 mg three times a day.
    - If needed, the dose can be increased to 15–30 mg daily in three divided doses (half the dose should be prescribed in elderly or debilitated people).
- For insomnia associated with anxiety, prescribe 5–15 mg at bedtime.

TOPIC: Phobia

Specific phobia is an anxiety disorder characterized by clinically significant anxiety associated with anticipation of, or exposure to, a circumscribed situation or object that often leads to avoidant behavior and results in considerable distress and impairment in functioning.
Specific phobia is among the most common mental disorders, associated with high rates of comorbidity. it can be associated with significant impairment and disability but is not typically seen as the primary presenting problem in clinical settings.
There are five main types of specific phobias in the American Psychiatric Association's Diagnostic and Statistical Manual of **Mental Disorders, Fifth Edition (DSM-5) that are based on the nature of the phobic stimulus:**

- Animal (eg, spiders, insects, dogs).
- Natural environment (eg, heights, storms, water).
- Blood-injection-injury (eg, needles, invasive medical procedures).
- Situational (eg, airplanes, elevators, enclosed places).
- Other (eg, situations that may lead to choking or vomiting; in children, eg, loud sounds or costumed characters).

Agoraphobia

Agoraphobia primarily, describe a fear of open spaces but also includes related aspects, e.g. the presence of crowds or the difficulty of escaping to a safe place.

Diagnosis:

A thorough assessment should include detailed information on the focus of the anxiety and the reasons for avoidance of situations. Additionally, the range of situations that are feared, the variability of the intensity, the predictability of fear, and the safety behaviors used to reduce anxiety should be assessed.

Criteria that need to be met for a diagnosis of specific phobia in adults include:

•Marked fear or anxiety about a specific object or situation
•Immediate anxiety after exposure to the phobic object or situation
•Avoidance of the phobic object or enduring it with disproportionate fear
•The fear, anxiety, or avoidance cause clinically significant distress or impairment in social, occupational, or other areas of functioning

Management:

- Cognitive-behavioral therapy (CBT): first line
- Exposure treatment

TOPIC: Obsessive compulsive disorder

Definition

- **Obsessive-compulsive disorder (OCD) is characterized by recurrent obsessional thoughts or compulsive acts or, commonly, both, which may cause significant functional impairment and/or distress.**
    - **An obsession** is defined as an unwanted intrusive thought, image, or urge that repeatedly enters the person's mind, and that usually causes marked anxiety or distress. Common obsessions in OCD include:
        - Contamination from dirt, germs, viruses (e.g. HIV), bodily fluids or faeces, chemicals, sticky substances, and dangerous materials (e.g. asbestos).
        - Fear of harm.
        - Excessive concern with order or symmetry.
        - Superstition, fear of 'bad' numbers 'magical' thinking, religious obsessions.
        - 'Forbidden' thoughts, urges or impulses (such as being a paedophile, blasphemy, violence, sexual or criminal acts, harm to others, harming own baby).
    - **Compulsions** are repetitive behaviours or rituals, including repetitive mental acts that the person feels driven to perform by their obsession(s) or according to rules that must be applied rigidly or to achieve a sense of 'completeness'. A compulsion can either be overt and observable by others, or a covert mental act that cannot be observed. Compulsions either are not connected realistically to a feared event (e.g. arranging objects symmetrically to prevent harm to a loved one) or are clearly excessive (e.g. showering daily for hours to prevent illness.) Common compulsions in OCD include:
        - Repetitive hand washing — due to fear of contamination.
        - Checking (e.g. doors are locked, electrical items unplugged, gas taps are off) — due to fear of harm to self or others.
        - Ordering, arranging, and/or repeating — due to excessive concern with order or symmetry.
        - Mental compulsions (e.g. special words or prayers repeated in a set manner, asking for forgiveness, excessive counting) — due to religious beliefs, 'magical' thinking, and superstitions.
        - Memory checking and avoidance of triggers — due to concerns about 'forbidden' thoughts or images.
    - **In children and young people:**
        - Young children's obsessional thoughts are more likely to include 'magical' or superstitious thinking (e.g. If I don't count up to 20, my parents will die).
        - Members of the family are almost always involved in a young person's compulsive rituals.
        - Young children may not be able to articulate the aims of these behaviours or mental acts.

Risk factors

It is not known what causes OCD. Risk factors for the development of obsessive-compulsive disorder (OCD) include:

- **Family history:**
    - First-degree relatives of people with OCD are at increased risk of developing the disorder. Genetic factors seem to be important as monozygotic twins are much more likely to exhibit OCD symptoms than dizygotic twins, and evidence exists that the disorder is transmitted in an autosomal dominant fashion. However, many people

with OCD cannot identify a family history, and there are monozygotic twins in whom only one has OCD, so there appear to be other factors involved.

- **Age:**
  - A bimodal onset has been observed, with peak mean ages of onset at approximately 10 years and 21 years.
  - Onset over the age of 30 years is unusual but may occur, or sub-clinical symptoms may manifest later in life.
- **Environmental factors**. A number of risk factors have been suggested but there is currently no convincing evidence for an association with any one of them. It may be that events, environmental or developmental factors, or certain infections can trigger OCD in susceptible individuals. A subset of cases of sudden onset OCD symptoms or tics in children has been associated with streptococcal infection, possibly due to an autoimmune response (paediatric acute-onset neuropsychiatric syndrome, PANS).
- **Pregnancy and the postnatal period**
  - Pregnant and postpartum women are more likely to experience OCD compared to the general population.
  - Common obsessions in these periods are worries about harming or abusing the baby and/or not being careful enough e.g. with sterilizing feeding equipment. Compulsions include avoidance behaviour, repeatedly seeking approval, and checking the baby is still breathing.

Complications

- **Complications of obsessive-compulsive disorder (OCD) include:**
  - Reduced quality of life — adverse effects on daily life, personal relationships, and ability to work and/or study. Fear of contamination can prevent access of appropriate health care. Childhood or adolescent onset may prevent the person from socialising with peers and may eventually cause difficulties with independent living.
  - Dermatitis — due to excessive handwashing.
  - Self-harm and suicide — people with OCD, particularly those who also have depression, may be at increased risk of self-harm and/or suicide. There is an increased risk of suicide-related behaviour in people with OCD independently of depressive symptoms and mood instability.

Prognosis

- **If OCD is untreated, the course is usually chronic, often with waxing and waning symptoms**. Remission can occur without treatment, with studies showing remission rates among adults between 20% and around 40%, however, this is often after many years of illness.
- **Psychological therapies can be effective in the treatment of OCD:**
  - Trials of cognitive behavioural therapy (CBT) and/or exposure and response prevention (ERP) have consistently found that these interventions are effective when compared to control groups.
- **Pharmacotherapy with clomipramine or a selective serotonin reuptake inhibitor (SSRI) has also been shown to be effective in the treatment of OCD:**
- For adults seeking treatment, relapse is common, with long-term remission occurring in fewer than 40%.
- Even after adequate initial treatment, 40 to 60% of patients have residual impairing symptoms.
- Poorer prognosis is associated with the following features:
  - Longer duration of illness prior to diagnosis or treatment.

    - More severe symptoms.
    - Psychiatric comorbidity.
    - Male gender.
- In contrast, an improved prognosis may be associated with the following factors:
    - Onset in early or middle childhood (if recognised and treated).
    - Less severe symptoms.
    - Shorter duration of illness.
    - Good response to initial treatment and full remission on treatment.
    - Longer term or more prolonged treatment.

Management

For all people diagnosed with obsessive-compulsive disorder (OCD)

- **Those who should be referred for specialist assessment include:**
    - People assessed as having severe OCD.
    - Those considered at risk of suicide or there is otherwise a risk to life due to OCD symptoms. Refer urgently (same day) to the crisis team if the person is at high risk of suicide. For those in whom admission is thought to be necessary but refuse.
    - Those with severe self-neglect.
    - Those with a significant comorbidity such as substance misuse, severe depression, anorexia nervosa, or schizophrenia.
    - Where healthcare professionals are uncertain about the risk associated with intrusive sexual, aggressive, or death-related thoughts reported by people with OCD. These are common themes in people with OCD of any age and risk can be misinterpreted by those without specific expertise, and where uncertain, the clinician should consult with or refer to a specialist.
    - Where initial treatment or treatments have not led to an adequate response.
    - Those under the age of 18.
- Follow local policy when referring to specialist mental health services, with urgency as dictated by the clinical situation.
- **For people being managed in primary care, provide written material about the nature of OCD and its treatment options.** Printable leaflets on OCD.

Management in adults

- **For adults with mild functional impairment:**
    - Recommend a psychological intervention. Following assessment, low-intensity cognitive-behavioural therapy (CBT), including exposure and response prevention (ERP) may be offered. The format for low-intensity CBT should be up to 10 therapist-hours per person, of one of the following:
        - Brief individual CBT (including ERP) with structured self-help materials.
        - Brief individual CBT (including ERP) by phone.
        - Group CBT (including ERP) which may be for more than 10 hours.
    - If the person has been unable to engage in low-intensity CBT (including ERP) or the response is inadequate, treat as for moderate functional impairment.
- **For adults with moderate functional impairment**:
    - Offer the choice of intensive CBT including ERP , or a selective serotonin reuptake inhibitor (SSRI).
    - Consider prescribing clomipramine (as an alternative first-line drug treatment to an SSRI) if the person prefers clomipramine or has had a previous good response to it, or if an SSRI is contraindicated.

    - If you are *not* confident of your assessment of moderate functional impairment *or* there is an inadequate response to initial treatment, refer to the secondary care mental health team.
- **For adults with severe functional impairment:**
    - Refer to the secondary care mental health team for assessment.
    - Whilst awaiting assessment:
        - Consider offering combined treatment with an SSRI and CBT (including ERP).
        - Consider prescribing clomipramine (as an alternative first-line drug treatment to an SSRI) if the person prefers clomipramine or has had a previous good response to it, or if an SSRI is contraindicated.
- **When prescribing an SSRI:**
    - Escitalopram, fluoxetine, fluvoxamine, paroxetine, and sertraline are all licensed for the treatment of OCD in adults. Citalopram can also be prescribed as a treatment for OCD, but this is an unlicensed use.
    - Discuss the potential for adverse effects and withdrawal symptoms before drug treatment is initiated. Explain that adverse effects early in treatment with an SSRI may include increased anxiety, agitation, and sleeping problems.
    - Advise the person that treatment with an SSRI usually requires a higher dose and a longer duration of treatment (at least 12 weeks) for an initial response.
    - **Be aware that in a minority of people under 30 years of age, SSRIs are associated with an increased risk of suicidal thinking and self-harm.**
    - When adults with comorbid depression are assessed to be at a high risk of suicide, the use of additional support such as more frequent direct contact with primary care staff or telephone contacts should be considered, particularly during the first weeks of treatment. For people at high risk of suicide, a limited quantity of medication should be prescribed.
    - Adults started on SSRIs who are not considered to be at increased risk of suicide or self-harm should be monitored closely — the effectiveness and adverse effects of the drug should ideally be reviewed every 2 to 4 weeks during the first 3 months of treatment and every 3 months thereafter. Dose adjustment may be required. Bear in mind that an absence of clinical benefit within four weeks suggests that a response to unchanged treatment is unlikely, but that the full efficacy of the drug may take up to 12 weeks to be realised.
- **When prescribing clomipramine,** be aware of the need to monitor patients for the emergence of other psychiatric symptoms, as well as the emergence of suicidality. Advise the person to report such symptoms immediately.
    - Prescribe small quantities at a time for people considered at significant risk of suicide due to its toxicity in overdose. Monitor regularly until the risk of suicide has subsided.
    - For those considered at risk of suicide and adults below the age of 25, review within a week of initiating, and then as often as needed, but no later than 4 weeks after initiating clomipramine.

Management in pregnancy

- **If a pregnant woman requires treatment for OCD:**
    - When considering the need for drug treatment for OCD in pregnancy, consider seeking the opinion of a specialist in perinatal mental health.
    - It is necessary to balance the risk of harm to the foetus from medication against the risk to both the mother and foetus from untreated illness.
    - Data is limited but current advice includes:
        - There is no robust evidence to suggest one SSRI is safer than another.

        - There may be a small risk of foetal cardiovascular malformation associated with SSRI treatment in pregnancy, but this is not confirmed.
        - There may be a risk of neonatal withdrawal syndrome if the mother is taking SSRIs in the weeks prior to delivery. (Infants should be delivered in hospital and monitored.)
        - There may be a small risk of persistent pulmonary hypertension in the newborn where there is SSRI exposure beyond 20 weeks of gestation.
        - There may be an increased risk of postpartum haemorrhage where SSRIs are used in the month prior to delivery.
        - Where treatment with an SSRI is necessary, the lowest effective dose should be used.
        - The manufacturer of clomipramine does not recommend its use in pregnancy or in women of childbearing potential not using contraception, although data is limited.

Management in children and young people

- **For children and young people (under the age of 18 years) diagnosed with OCD with any level of functional impairment:**
    - **Refer to Child and Adolescent Mental Health Services (CAMHS).**
        - Those with mild OCD will be offered guided self-help (a self-help programme with support, guidance and encouragement from a mental health professional).
        - Those with moderate to severe functional impairment and those with mild functional impairment for whom guided self-help has been ineffective or refused, will be offered CBT (including ERP) that involves the family and carers and is age-appropriate. If psychological treatment is declined or ineffective, or they are unable to engage with it, an SSRI may be considered.
- **Note: a selective serotonin reuptake inhibitor (SSRI) should only be prescribed to people under 18 years of age following assessment and diagnosis by a child and adolescent psychiatrist.** The specialist should also be involved in decisions about dose changes and discontinuation and is likely to be key in arranging active monitoring of the person's symptoms, functioning, and response to treatment (if applicable) at intervals determined by clinical judgment.

How should I monitor a person with obsessive-compulsive disorder?

During each review:

- **Be alert to any suicidal ideation or intention and assess suicide risk, especially if the person has comorbid depression.**
- **Monitor progress, taking into account factors including severity, and duration of symptoms, as well as the degree of distress and functional impairment.**
- **For adults, check adherence to any treatment and inquire about adverse drug effects (if applicable).**
    - For people in the initial stages of SSRI treatment, ask about signs of akathisia or restlessness, suicidal ideation, and increased anxiety and agitation. If the person reports prolonged akathisia, restlessness or agitation, consider a switch to a different SSRI.
    - If there has not been an adequate response to a standard dose of an SSRI, and there are no significant side effects after 4–6 weeks, a gradual increase in dose

should be considered. Monitor the person around the time of dose changes for any new symptoms or worsening of their condition.
    - For adults with an inadequate response to initial treatment (a full 12 weeks of treatment with an SSRI, or more than 10 therapist hours of CBT [including ERP]), other treatment options include combined treatment with CBT (including ERP) and an SSRI, *and/or* switch to a different SSRI, *or* a switch to clomipramine. If there has been no response to a full trial of at least one SSRI alone, a full trial of combined treatment with CBT (including ERP) and an SSRI, and a full trial of clomipramine alone, the person should be referred to a multidisciplinary team with specific expertise in the treatment of OCD for assessment and further treatment planning.
    - **Note:** It is advisable to monitor cardiac function with ECG during long-term therapy with clomipramine, at intervals determined by clinical judgement and other factors including the length of treatment, the person's age, and cardiac risk factors. Advise the person to report cardiac or neurological symptoms such as palpitations, vertigo, syncope, or seizures.
- **If a drug is effective, advise the person to continue medication for a minimum of 12 months.**
- **Re-evaluate the required frequency of follow-up based on:**
    - The person's preference.
    - Severity of symptoms.
    - Comorbid conditions.
    - Change since the last review and response to interventions.
    - Symptoms during treatment changes.
- **When an adult with OCD has taken an SSRI or clomipramine for 12 months after remission (symptoms are not clinically significant and the person is fully functioning for at least 12 weeks), review the need for continued treatment.**
    - Consider the severity and duration of the initial illness, the number of previous episodes, the presence of residual symptoms, and concurrent psychosocial difficulties.
    - If treatment is continued beyond 12 months after remission, the need for continuation should be reviewed at intervals depending on clinical judgement.
    - When reducing or stopping SSRIs or clomipramine, the dose should be tapered gradually over at least several weeks, according to the person's need. The rate of reduction should take into account the starting dose, the drug half-life and particular profiles of adverse effects. Advise the person to seek advice if they experience significant discontinuation/withdrawal symptoms.

Prescription information
SSRI

Clomipramine
Dosing information

- Initially, prescribe 25 mg each day (or 10 mg daily in the elderly).
- If there are no significant adverse effects:
    - Increase gradually over 2 weeks to 100–150 mg daily (increase more slowly in the elderly).
    - At 4–6 weeks, if there is an inadequate treatment response, increase gradually up to a maximum dose of 250 mg daily.

TOPC: Insomnia

Definition

- **Insomnia** is difficulty in getting to sleep, difficulty maintaining sleep, early wakening, or non-restorative sleep which occurs despite adequate opportunity for sleep and results in impaired daytime functioning.
    - Daytime symptoms typically include poor concentration, mood disturbance, and fatigue.
    - Sleep disturbance in the absence of daytime impairment is not considered to be insomnia disorder.
- **Short-term insomnia**:
    - Insomnia symptoms occurring for less than 3 months duration (typically a few days or weeks).
- **Chronic insomnia**:
    - Insomnia symptoms occurring on at least 3 nights per week for three months or more.
- The terms primary insomnia (insomnia not attributable to an underlying cause) and secondary insomnia (insomnia due to a comorbid condition) are no longer included in classification systems as:
    - Insomnia is considered to be a disorder in its own right — the diagnosis can be made independent of and in addition to comorbidities.
    - Associations between insomnia and co-existing medical or psychiatric disorders are often bidirectional or interactive.

Causes

- **Short-term insomnia** — transient insomnia is common and typically lasts a few days or weeks. It can occur in association with:
    - Stressful events such as bereavement, illness, changes in employment, exams, pending deadlines or financial difficulties.
    - Changes in sleeping patterns due to the birth of a child or environmental disturbance such as excess noise or light or extremes of temperature.

- **Chronic insomnia** — factors involved in the maintenance of insomnia are not fully understood but are thought to include maladaptive behaviours and cognitive processes. Persistence of the initial stressor may also contribute:
    - Chronic insomnia commonly co-exists with other conditions including:
        - Psychiatric disorders such as anxiety and depression.
        - Medical disorders such as COPD, heart failure, neurodegenerative diseases, malignancy, musculoskeletal conditions and chronic pain.
        - Substance misuse such as alcohol and illicit drugs.
    - Bidirectional or interactive effects often exist between chronic insomnia and co-morbid conditions (for example anxiety, depression and substance abuse).

Prognosis

- **Short-term insomnia**
    - Acute insomnia is very common and typically lasts a few days or weeks resolving once the triggering factor is removed.

- **Chronic insomnia**
  - A longitudinal study in a UK population (n=870 with insomnia) found that 69% of people reporting insomnia at baseline had insomnia at 12-month follow-up — persistence of insomnia was significantly associated with older age.
  - A Canadian population-based longitudinal study (n=388 with insomnia) found that 74% of people had insomnia persisting for at least 1 year and 46% had insomnia persisting for 3 years — persistence of insomnia was more likely in those with severe symptoms at baseline, in women and in older adults.

Complications

Insomnia is associated with:

- Cognitive difficulties such as impaired memory, attention, and concentration.
- Decreased quality of life and function including:
  - Impaired work performance and increased work absenteeism.
  - Increased risk of motor vehicle accidents.
  - Increased risk of falls in older people.
- Psychiatric complications including anxiety, depression and substance misuse.
- Medical complications.
  - Studies have shown associations between persistent insomnia and increased risk of cardiovascular disease and type 2 diabetes, in particular with objectively measured short sleep duration (less than 6 hours per 24 hour period).
- A possible increase in all-cause mortality — findings from studies vary.

Diagnosis

Clinical features

- Suspect insomnia if (despite adequate time and opportunity to sleep) the person has:
  - Difficulty in getting to sleep, difficulty maintaining sleep, early wakening, or non-restorative sleep which results in impaired daytime functioning or wellbeing.
    - Daytime symptoms may include fatigue, decreased mood or irritability, malaise, and cognitive impairment.
- Sleep difficulties without functional impairment do not meet the diagnostic criteria for insomnia.

Assessment

Take a history and ask about:

- Symptoms such as difficulty in getting to sleep, difficulty maintaining sleep, early wakening, non-restorative sleep and daytime impairment**.**
- **Duration and frequency of symptoms**.
  - Unsatisfactory sleep without reported functional impairment does not meet the diagnostic criteria for insomnia — ask about beliefs on normal sleep.
- **Sleep schedule** including bedtimes, time taken to get to sleep, awakenings, and rise time.
- **Sleep environment**.
- **Possible triggers** such as stress, shift work and jet lag.
- **Behaviours during sleep** such as snoring, witnessed apnoea, restless legs, or sleep walking which may indicate an alternative cause.
  - A collateral history from a partner/family member can be helpful.
- **Past medical history** including:
  - Previous sleep problems and treatment.

    - Comorbidities such as chronic pain, medical disorders (such as chronic obstructive pulmonary disease, heart failure or gastro-oesophageal reflux disease) and psychological disorders (such as stress, anxiety, or depression).
- **Impact of insomnia** on quality of life, ability to drive, employment, relationships, and mood.
- **Medications and substance use** including caffeine, alcohol, nicotine and illicit drugs.

Carry out an examination where appropriate:

- Depending on the specific clinical situation examination may help identify comorbid conditions.

Consider the need for investigations:

- A sleep diary can help assess sleep difficulties and daytime impairment over time; it should be kept for 2 weeks and record:
    - The time of going to bed and getting up.
    - The time taken to get to sleep and the number and duration of episodes of waking through the night.
    - Episodes of daytime tiredness and naps.
    - Times of meals, alcohol consumption, caffeine consumption, and significant events during the day, such as exercise or stress.
    - Rating of sleep quality (ask the person to rate the quality of their sleep each night, from 1 to 5, where 1 is very poor and 5 is very good).
- Depending on the clinical situation, other investigations may be appropriate for example where a comorbid condition (such as hyperthyroidism, respiratory or cardiovascular disease) is suspected.

Normal sleep patterns

- Sleep is an important factor in many processes including neural development, learning, memory, regulation of emotion, and cardiovascular and metabolic functioning.
- There is no standard definition of normal sleep — the amount of sleep needed to ensure good health varies from person to person and with ageing.
    - With ageing the number of awakenings increases and total sleep time decreases.
- The American Thoracic Society suggests that the optimal sleep duration in adults is 7–9 hours:
    - Short sleep duration (6 hours or less per 24 hours) is associated with adverse outcomes including mortality.
    - Long sleep duration (more than 9–10 hours per 24 hours) may be normal but has also been associated with adverse outcomes — however, long sleep duration may be a marker of ill-health and physical inactivity rather than a cause of ill-health.
- The normal time taken to fall asleep (sleep-onset latency) is usually considered to be less than 30 minutes.

Differential diagnosis

Consider other sleep disorders such as:

- Obstructive sleep apnoea.
- Circadian rhythm disorders.
- Restless legs syndrome.
- Narcolepsy — may present with falling asleep in the daytime without warning and collapse or muscle weakness triggered by emotion.

- Parasomnias — may present with unusual or unpleasant experiences or behaviours associated with sleep that are troublesome or dangerous.

Management

How should I manage a person with short-term insomnia (less than 3 months duration)?

- **Consider the need for** referral to a sleep clinic or neurology if symptoms of another sleep disorder are present.
- **Address any circumstances/stressors associated with onset of insomnia.**
    - Ensure comorbidities (for example anxiety or depression) are optimally managed.
- **Offer advice on sleep hygiene.**
- **Advise the person not to drive if they feel sleepy.**
- **If sleep hygiene measures fail, daytime impairment is severe causing significant distress, and insomnia is likely to resolve soon (for example due to a short term stressor):**
    - Consider a short course (3-7 days) of a non-benzodiazepine hypnotic medication (z-drug).
        - Do not prescribe hypnotics routinely — use only for short courses if acutely distressed.
        - Do not prescribe hypnotics to older people or women who are pregnant or breastfeeding.
- **If sleep hygiene measures fail, daytime impairment is severe causing significant distress, and insomnia is not likely to resolve soon:**
    - Offer cognitive behavioural therapy for insomnia (CBT-I).
        - CBT-I typically includes behavioural interventions (such as stimulus control and sleep restriction), cognitive therapy and relaxation training and can be provided face-to-face or digitally.
    - Consider the need for adjunctive treatment with a short-term hypnotic medication (a z-drug or prolonged released melatonin if over 55 years of age).
- **If a hypnotic is prescribed:**
    - Consider the duration of action (short-acting is usually most appropriate), adverse effects, interactions and potential for dependency and abuse.
    - Use the lowest effective dose for the shortest period possible — do not continue treatment for longer than 2 weeks (preferably less than one week).
    - Inform the person that further prescriptions for hypnotics will not usually be given, ensure that the reasons for this are understood, and document this in the person's notes.
    - Do not issue further prescriptions without seeing the person again.
    - If there has been no response to the first hypnotic, do not prescribe another.
- **Arrange follow up for review** (for example 2–4 weeks, dependant on the clinical situation).
    - If symptoms have not improved, reassess the person — consider CBT-I (if not already offered), alternative diagnoses and the need for referral.
- **Do not recommend over-the-counter treatments for insomnia.**

What advice should I give regarding good sleep hygiene?

Sleep hygiene aims to increase awareness of behavioural, environmental, and temporal factors that may be detrimental or beneficial to sleep.

- Advise the person on:
    - Normal sleep and changes in sleep patterns with age.
    - Sleep environment.

        - A comfortable sleeping environment should be maintained: not too hot, cold, noisy, or bright.
        - The bedroom should only be used for sleep and intimacy.
        - Checking or watching the clock throughout the night should be avoided.
        - Bright light (including all technology) should be minimized. 'Blue light' displays on electronic devices and televisions suppress melatonin production — avoid using devices for two hours before bed.
    - Regular sleep schedules including:
        - Going to bed when sleepy — avoid going to bed too early.
        - Waking up and getting out of bed at the same time every morning including weekends and after a poor night's sleep — increase exposure to bright light in the morning.
        - Avoiding napping during the day.
    - Relaxation before going to bed (for example reading a book or having a bath).
        - Relaxation exercises in the early evening may help reduce hyper-arousal.
        - Visual imagery in bed can help with racing thoughts.
    - Limiting/avoidance of caffeine, nicotine and alcohol.
        - Caffeine should be avoided after midday and nicotine, alcohol, and large meals within 2 hours of bedtime.
        - Alcohol intake is a common maladaptive self-treatment strategy in insomnia which can contribute to poor sleep-maintenance.
    - Exercise.
        - Exercise should be avoided within 4 hours of bedtime but is beneficial earlier in the day

How should I manage someone with long-term insomnia (more than 3 months duration)?

- Consider the need for referral to a sleep clinic or neurology if symptoms of another sleep disorder are present.
- Address any triggers or factors associated with maintenance of insomnia (for example illness or other stressors).
    - Ensure comorbidities (such as anxiety and depression) are optimally managed.
- Offer advice on sleep hygiene.
- Advise the person not to drive if they feel sleepy.
- Offer cognitive behavioural therapy for insomnia (CBT-I) as the first-line treatment for chronic insomnia in adults of any age.
    - CBT-I typically includes behavioural interventions (such as stimulus control and sleep restriction), cognitive therapy and relaxation training. It can be provided face-to-face or digitally.
- Pharmacological therapy should be avoided in the long-term management of insomnia, however:
    - For some people with severe symptoms or an acute exacerbation a short course of a hypnotic drug (preferably less than 1 week) may be considered as a temporary adjunct to behavioural and cognitive treatment.
        - Do not prescribe long-term hypnotic treatment
    - For people over 55 years of age with persistent insomnia, treatment with a modified-release melatonin may be considered.
        - The recommended initial duration of treatment is 3 weeks. If there is a response to treatment, continue for a further 10 weeks only.
        - Discuss the risks (similar to those of other hypnotics including falls, and fractures) associated with melatonin treatment in the elderly.
- Arrange follow up for review (2–4 weeks, dependant on the clinical situation).

        - If symptoms have not improved, reassess the person — consider alternative diagnoses and the need for referral.
- Do not recommend over-the-counter treatments for insomnia.

When should I refer to secondary care?

- **Refer the person to a sleep clinic, specialist with expertise in sleep medicine or neurology if:**
    - Another sleep disorder (such as a parasomnia, narcolepsy, or obstructive sleep apnoea) is suspected.
    - There is doubt regarding the diagnosis.
- **Seek specialist advice/consider referral to secondary care if:**
    - Treatment in primary care has failed.
    - Insomnia occurs in a person from an occupational at-risk group, for example professional drivers.
- **Do not routinely refer adults with insomnia, jerks on falling asleep or isolated brief episodes of sleep paralysis.**

Prescription information

Z drugs

The National Institute for Health and Care Excellence (NICE) technology appraisal advises that hypnotics should be prescribed for short-term (up to two weeks) use only when non-drug measures have failed and insomnia is severe, disabling or causing extreme distress.

- They many risks associated with hypnotics such as falls, cognitive impairment, dependence and withdrawal symptoms, are well recognised.
- The lowest dose that controls symptoms for the shortest period of time should be used.

Z-drugs licensed for the short-term management of insomnia include:

- Zopiclone.
    - In adults — 7.5 mg once daily at bedtime.
    - In the elderly (avoid if possible due to increased risk of adverse effects) — if essential reduce dose; initially 3.75 mg once daily at bedtime.
    - DO NOT re-administer during the same night.
- Zolpidem.
    - In adults — 10 mg once daily at bedtime; for debilitated patients, use elderly dose.
    - In the elderly (avoid if possible due to increased risk of adverse effects) — if essential reduce dose; 5 mg maximum once daily at bedtime.
    - DO NOT re-administer during the same night.

Contraindications and cautions

- Z-drugs are contraindicated in:
    - Marked neuromuscular respiratory weakness.
    - Respiratory failure.
    - Myasthenia gravis.
    - Obstructive sleep apnoea.
    - Pregnancy and breastfeeding — risk of neonatal respiratory depression and withdrawal symptoms.
    - Severe hepatic impairment.
    - Psychotic illness.

- Z-drugs should be prescribed with caution in:
    - Chronic pulmonary insufficiency — increased risk of respiratory depression.
    - The elderly — avoid if possible due to increased risk of adverse effects including falls and fractures.
    - History of drug or alcohol abuse — manufacturer recommends extreme caution.
    - Muscle weakness.
    - Psychiatric illness including depression.
    - Hepatic and renal impairment — reduce dose, avoid if severe.

Adverse effects

- The adverse effects of Z-drugs include:
    - Gastrointestinal effects such as dry mouth, abdominal pain, nausea, vomiting and diarrhoea.
    - Psychiatric effects such as anterograde amnesia, impaired cognition and concentration, depression, suicidal ideation and attempt, anxiety, psychosis and behavioural abnormalities, restlessness, agitation, irritability, aggression, delusions, anger, nightmares, hallucinations, and inappropriate behaviour.
    - Neurological effects such as confusion, dizziness, hallucination, headache, sleep disorders, ataxia, falls, muscle weakness, respiratory depression and decreased level of consciousness.
- Drowsiness may persist the following day and affect performance of skilled tasks such as driving.
- Tolerance, dependence and withdrawal syndrome may develop.

Drug interactions

- **Drug interactions of Z-drugs include**:
    - Alcohol and opioids — avoid concomitant use as this can:
        - Induce sedation, respiratory depression, coma, and death by potentiating effects of hypnotics.
        - Affect ability to perform skilled tasks.
    - Centrally acting drugs — possible enhanced central depressive effect if co-administered with centrally acting drugs such as neuroleptics, antipsychotics, antidepressants, anaesthetics and sedative antihistamines.
    - Drugs that inhibit cytochrome P450 enzyme (for example ciprofloxacin, azole antifungals, and oestrogens).
        - Avoid concomitant use — can result in increased serum levels of Z-drugs leading to increased effect.
    - Drugs that induce cytochrome P450 enzyme (for example St John's Wort and rifampicin).
        - Cytochrome P450 inducers can accelerate hepatic elimination of Z-drugs and decrease their action.
    - Phenytoin — concomitant use may increase or decrease serum concentrations of phenytoin; monitor phenytoin concentrations.

Melatonin

Indications and dose

- The recommended oral dose of modified-release melatonin for insomnia (short-term use only) for an adult aged 55 years and over is 2 mg once daily (1–2 hours before bedtime and after food) for up to 13 weeks.

- The recommended initial duration of treatment is 3 weeks continued if there is a response to treatment for a further 10 weeks only.

Contraindications and cautions

- Melatonin should be avoided in:
  - Hepatic impairment.
  - Autoimmune disease — no clinical data available.
  - Pregnancy, breastfeeding and in women planning a pregnancy.
- Use melatonin with caution in:
  - Renal impairment — no information available.

Adverse effects

- The adverse effects of melatonin include:
  - Arthralgia, headache, increased risk of infection, abdominal pain, dyspepsia, dry mouth, mouth ulceration, nausea, weight gain, hypertension, chest pain, malaise, dizziness, restlessness, nervousness, irritability, anxiety, abnormal dreams, proteinuria, glycosuria, pruritus, rash, and dry skin.
  - Drowsiness, sleepiness, disorientation — moderate influence on ability to drive and use machines. Melatonin may cause drowsiness and may decrease alertness for several hours, therefore it is not recommended prior to driving and using machines.

Drug interactions

- Drug interactions of modified-release melatonin include:
  - Alcohol, opioids and other drugs with CNS depressant effects —can affect ability to perform skilled tasks.
  - Fluvoxamine — fluvoxamine markedly increases melatonin levels by inhibiting its metabolism by hepatic cytochrome P450; avoid concomitant use.
  - Quinolones — quinolones are thought to increase melatonin levels; manufacturer advises caution.
  - Combined hormonal contraceptives are thought to increase exposure to melatonin; manufacturer advises caution.
  - Warfarin — case reports suggest that melatonin may either increase or decrease the international normalised ratio (INR). If used concurrently consider increasing frequency of INR monitoring.

TOPIC: Acute stress disorder

Acute stress disorder is defined as an acute stress reaction that occurs in the first 4 weeks after a person has been exposed to a traumatic event (threatened death, serious injury e.g. road traffic accident, sexual assault etc). This is in contrast to post-traumatic stress disorder (PTSD) which is diagnosed after 4 weeks.

Features include:

- Intrusive thoughts e.g. flashbacks, nightmares
- Dissociation e.g. 'being in a daze', time slowing
- Negative mood
- Avoidance
- Arousal e.g. hypervigilance, sleep disturbance

Management:

- Trauma-focused cognitive-behavioural therapy (CBT) is usually used first-line
- Benzodiazepines:
    - Sometimes, used for acute symptoms e.g. agitation, sleep disturbance
    - Should only be used with caution due to addictive potential and concerns that they may be detrimental to adaptation

TOPIC: Grief reaction

It is normal for people to feel sadness and grief following the death of a loved one and this does not necessarily need to be medicalised. However, having some understanding of the potential stages a person may go through whilst grieving can help determine whether a patient is having a 'normal' grief reaction or is developing a more significant problem.

One of the most popular models of grief divides it into 5 stages:

- Denial: this may include a feeling of numbness and also pseudohallucinations of the deceased, both auditory and visual. Occasionally people may focus on physical objects that remind them of their loved one or even prepare meals for them
- Anger: this is commonly directed against other family members and medical professionals
- Bargaining
- Depression
- Acceptance
  It should be noted that many patients will not go through all 5 stages.
  Abnormal, or atypical, grief reactions are more likely to occur in women and if the death is sudden and unexpected. Other risk factors include a problematic relationship before death or if the patient has not much social support.

Features of atypical grief reactions include:

- Delayed grief: sometimes said to occur when more than 2 weeks passes before grieving begins
- prolonged grief: difficult to define. Normal grief reactions may take up to and beyond 12 months.

Typical acute grief:

The hallmark of acute grief is an intense focus on thoughts and memories of the deceased person, accompanied by sadness and yearning.

Grief can occur in response to other meaningful (nonbereavement) losses, including an interpersonal loss (eg, separation from a loved one through divorce) or loss of a pet, job, property, health, security, or community.

Presentation — Mourners focus their attention, emotions, thoughts, and behavior upon the deceased person and what has been lost. However, the painful feelings and memories are commonly intermingled with periods of respite and positive feelings, thoughts, and reminiscing. These positive experiences during bereavement reflect resilience and foretell better outcomes.

Symptom:

The symptoms of acute grief are typically related to either separation from the deceased or to stress and trauma:

Symptoms of separation distress:

1. Yearning for and seeking proximity to the deceased
2. Loneliness
3. Sadness, crying, guilt, shame, anxiety, and anger when confronted with reminders of the loss
4. Disrupted appetite and sleep, as well as somatic symptoms such as heart palpitations, stomach sensations (“butterflies”), and dizziness

5. Insistent thoughts and memories of the lost person, sometimes including hallucinations
6. Feeling drawn to things associated with the deceased
7. Social withdrawal and disinterest in other people and activities not associated with the deceased
8. Confusion about one's identity and feeling lost or uncertain without the deceased

Symptoms of trauma/stress reaction:

1. Disbelief and difficulty accepting the loss
2. Shock
3. Numbness
4. Impaired attention, concentration, or memory

Differential diagnosis:

- Prolonged grief disorder
- Major depression
- PTSD

Management:

Assessing the bereaved — Following the death of a loved one, grieving individuals may present seeking relief from symptoms such as intense sadness or disrupted sleep. The clinician should first pause, make eye contact, listen, and then offer condolences and validate grief symptoms. In addition, clinicians should assess patients to rule out psychiatric conditions that may be triggered or exacerbated by bereavement, including:

- Anxiety disorders, insomnia disorder, prolonged grief disorder, PTSD
- Unipolar major depression, suicidal ideation and behavior

Monitoring — We recommend monitoring bereaved individuals until they demonstrate that they are adapting to the loss and grief is diminishing in overall intensity. The frequency of monitoring (eg, every month to four months) depends upon the intensity of grief. Suggested questions include:

- How are you doing with the death?
- How are you experiencing your grief?
- Is life getting more bearable? How so?

Support — Support from family, friends, clergy, and clinicians is usually sufficient to attenuate the pain of acute grief and promote adaptation to the loss.

Interventions — Acute grief generally does **not** require treatment such as grief counseling or other psychotherapies or pharmacotherapy.

Intense grief — Although acute grief generally does not require treatment, grief therapy or other psychotherapies can help bereaved individuals who request help during acute grief or those who have persistent intense grief that does not meet criteria for prolonged grief disorder. Difficult grief reactions are more likely to occur with especially painful losses, such as death of a life partner or one's child, or deaths by violent or stigmatized means. Elements of psychotherapy that may be beneficial include education, stress management, identifying and accepting grief and emotional pain, talking about the deceased person and their death, learning ways to live with reminders of the loss, and decreasing avoidance behaviors, as well as helping patients to focus upon experiences intended to restore one's life, such as scheduling new activities and re-engaging with others Bereaved individuals, especially those who have lost children, often find it difficult to speak about the loss.

Pharmacotherapy may be indicated for grief-induced insomnia.

Psychiatric disorders — Bereavement may precipitate or exacerbate suicidal ideation and behavior as well as psychiatric disorders such as major depression, anxiety disorders, and PTSD. In addition, some people may develop prolonged grief disorder.

Insomnia (as mentioned before)

General medical illnesses — Bereavement is associated with an increased risk of general medical. One of the illnesses that occur more often in the bereaved than the nonbereaved is cardiovascular disease.

TOPIC: Post-traumatic stress syndrome

Definition

- **Post-traumatic stress disorder (PTSD) is a relatively common mental health condition which can affect people of any age and develops following exposure to a major traumatic event, such as:**
    - A serious/life-threatening accident.
    - Physical or sexual assault.
    - Abuse, including childhood or domestic abuse.
    - Work-related exposure to trauma, including remote exposure.
    - Trauma related to serious health problems or childbirth experiences (for example, intensive care admission or neonatal death).
    - War and conflict.
    - Torture.
- **According to the DSM-5 diagnostic criteria, the person may have:**
    - Directly experienced the traumatic event(s).
    - Witnessed an event happening to someone else.
    - Learned of a traumatic event occurring to a close family member or close friend, such as actual or threatened death that was violent or accidental.
    - Experienced repeated or extreme exposure to aversive details of a traumatic event, for example, first responders viewing human remains; police officers repeatedly exposed to details of child abuse.
- **PTSD may present with a range of symptoms that persist for at least one month after the traumatic event, including:**
    - Re-experiencing the event(s) — this may occur in the daytime when the person is awake (flashbacks, or intrusive images or thoughts) or as nightmares when asleep. This is the most characteristic symptom.
    - Avoidance of external reminders, and/or of thoughts and memories of the event.
    - Hyperarousal (including hypervigilance, anger, and irritability).
    - Negative alterations in mood and thinking.
    - Emotional numbing.
    - Dissociation.
    - Emotional dysregulation.
    - Interpersonal difficulties or problems in relationships
    - Negative self-perception (including feeling diminished, defeated or worthless).
- **Children with PTSD may exhibit symptoms such as:**
    - Dreams of the trauma, which may then change into nightmares of monsters.
    - Re-living the trauma in their play. For example, a child involved in a road traffic accident might re-enact the crash with toy cars.
    - Losing interest in things that they previously enjoyed.
    - Expressing the belief that they will not live long enough to grow up.
    - Stomach aches and headaches.
- **Complex post-traumatic stress disorder** develops in a subset of people with PTSD, usually after exposure to an event or series of events of an extremely threatening or horrific nature, most commonly prolonged or repetitive events from which escape is difficult or impossible. The disorder is characterised by the core symptoms of PTSD, as well as:
    - Severe and pervasive problems in affect regulation— that is, the ability to regulate emotional state and behaviour.
    - Persistent beliefs about oneself as diminished, defeated or worthless, accompanied by deep and pervasive feelings of shame, guilt or failure related to the traumatic event.

- Persistent difficulties in sustaining relationships and in feeling close to others.

Risk factors

- **Risk factors for post-traumatic stress disorder (PTSD) include:**
  - **Exposure to a** traumatic event — the type of event affects the risk of PTSD, with higher rates reported following rape and physical assault than following an accident. People who may be at greater risk of experiencing trauma include:
    - Members of the armed forces (including combat veterans and ex-service personnel).
    - Members of the ambulance, police, prison, and fire services, other emergency personnel (including those no longer in service).
    - Nursing and medical professionals (especially in front-line situations, such as casualty).
    - Journalists.
    - Refugees and asylum seekers (especially those from conflict zones).
  - **Severity of the incident** — in general, the more severe the traumatic event (and the greater the perceived threat to life) the higher the chance of PTSD.
  - **Female sex** — overall, women experience higher rates of PTSD than men, however, this varies depending on the type of trauma and specific age group being assessed.
  - **Younger age** — among people exposed to disaster or trauma, older adults have a lower risk of developing PTSD than younger adults.
  - **Previous experience of trauma.**
  - **Presence of multiple major life stressors.**
  - **Low social support, social disadvantage**
  - **History of a mental health disorder.**
- **In children exposed to trauma, the risk of PTSD is reduced by good family support and when there is less parental distress.**

Complications

Complications that have been associated with post-traumatic stress disorder (PTSD) include:

- **Mental health disorders** — including depression, anxiety disorders, substance use disorders, somatic symptom disorder, psychoses, suicidal ideation.
  - Children with PTSD may also be more likely to develop attention-deficit hyperactivity disorder (ADHD), oppositional defiant disorder, and/or a conduct disorder.
- **Increased rates of physical health disorders** — including cancer, chronic pain, diabetes, cardiovascular disease, autoimmune diseases, hypertension, dementia, serious disability, and premature death.
- **Adverse social consequences** — including unemployment, problems with school performance, poverty, relationship problems including with partners, peers, and family members, and engagement in problematic health behaviours such as smoking.

Diagnosis

- **Post-traumatic stress disorder (PTSD) is a potential diagnosis in anyone who has experienced or witnessed a traumatic event** — be aware that this encompasses repeated or multiple events that may have occurred recently, or many month or years before.
- **PTSD should also be considered in people repeatedly presenting with unexplained physical symptoms, and in people with depression, anxiety disorders, and substance abuse.**

    - Physical symptoms can include headaches, gastrointestinal problems, rheumatic pains, and skin disorders. Children with PTSD may report stomach aches and headaches.
- **Be aware of occupations and lifestyle factors that may be risk factors for exposure to traumatic events/the development of PTSD.**
- **Where PTSD is a potential diagnosis based on reported symptoms and/or presentation, enquire about, and give examples of additional symptoms (as described above) and ask whether the person has experienced a traumatic event, providing examples of such events.** Be aware that some people with PTSD may be anxious about engaging in assessment and treatment, and may be reluctant to discuss their trauma history and symptoms.
- **If a child or young person is being assessed for PTSD,** where it is developmentally appropriate, directly question them about the presence of PTSD symptoms. If possible, do not rely solely on the parent or carer for information.
- **Consider the** differential diagnosis **of PTSD.** However, be aware that PTSD may be co-morbid with some of these conditions, particularly depression and anxiety disorders.
- **In order to confirm a diagnosis of PTSD:**
    - For adults, occasionally, a primary care physician with appropriate training and experience may be able to confirm the diagnosis. However, in most cases, referral to a mental health specialist with expertise in managing post-traumatic stress disorder is required. The diagnosis is usually made using diagnostic criteria from the fifth edition of the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders (DSM-5) or the International Classification of Diseases (ICD-10).
    - In children, the diagnosis is confirmed by referral to a specialist mental health service with expertise in managing post-traumatic stress disorder.

Screening questionnaires

- **In primary care, the Trauma Screening-Questionnaire (TSQ) may be helpful to identify people with post-traumatic stress disorder. It:**
    - Consists of 10 questions which measure re-experiencing and arousal symptoms.
    - Takes only a few minutes to administer.
    - Is designed for use three weeks or more following exposure to a traumatic event to identify people who are likely to be currently suffering from post-traumatic stress disorder.
    - Asks the person to indicate which symptoms they have experienced at least twice in the past week.
        - If the person has six or more positive responses, they are at risk of having post-traumatic stress disorder and should be referred for further assessment (using a structured interview for post-traumatic stress disorder).

Differential diagnosis

- **Depression** — suggested by low mood, lack of energy, loss of interest, suicidal ideation.
- **Generalized anxiety disorder**
- **Panic disorder** — suggested by recurrent, unexpected panic attacks not triggered by stimuli that recall a specific trauma.
- **Specific phobias** — suggested by fear and avoidance restricted to certain situations.
- **Adjustment disorder**
- **Dissociative disorders** — suggested by persistent and recurrent feelings of detachment and estrangement from oneself (depersonalisation disorder), and/or gaps in recall, often

related to traumatic events (dissociative amnesia). There is an absence of the re-experiencing and hyper-arousal symptoms of PTSD.

- **Psychosis** — suggested by hallucinations, delusions. Flashbacks and vivid intrusive images are accompanied by perceptual and cognitive disorganisation.

Management

How do I manage someone with post-traumatic stress disorder?

Note: The role of a general practitioner (GP) in the initial management of a person with symptoms of PTSD is to determine the need for emergency physical and mental health assessment, and to co-ordinate care. Where a referral is being made to confirm a diagnosis of PTSD, it is likely that psychological therapy or drug treatment will be arranged directly by the specialist involved. However, whilst awaiting referral, the person should be assessed as described below. Where confirmation of the diagnosis has taken place in primary care, specialist referral is required for any further management. Once any interventions are established by a specialist, a GP may again become involved a person's care via a shared-care arrangement. For all people with symptoms of PTSD, ask about the effects on work or school, relationships, social life, and quality of life, and assess their degree of distress and functional impairment as mild, moderate, or severe:

  - **Mild** — distress caused by the symptoms is manageable, and the person's social and occupational functioning are not significantly impaired.
  - **Moderate** — distress and impact on functioning lie somewhere between mild and severe and there is not considered to be a significant risk of suicide, harm to self, or harm to others.
  - **Severe** — distress caused by the symptoms is felt to be unmanageable, and/or there is significant impairment in social and/or occupational functioning, and/or there is considered to be significant risk of suicide, harm to self, or harm to others.
- **Note : Clinically important symptoms of PTSD can be defined as those causing at least moderate functional impairment and/or those scoring above a clinical threshold on a validated scale (where this has been assessed). Where symptoms are deemed to be 'moderate', use clinical judgement to determine the need for intervention.**
- **Be aware that PTSD commonly co-exists with other mental health disorders**including depression, anxiety disorders, and alcohol or substance misuse.
- **If the person is exhibiting severe distress and/or functional impairment, co-morbid depression or another mental health disorder, or there are other concerns, assess their risk of suicide and self-harm.**
- **For people not requiring same-day referral:**
  - **If the person has co-morbid depression,** be aware that treating the PTSD first is often associated with an improvement in depression. However, if depression is severe enough to make psychological treatment of the PTSD difficult, the depression should be treated first.
  - **If there are concerns relating to the person's physical health,** arrange or carry out an emergency physical health assessment, using clinical judgement to determine the appropriate investigations.
  - **Assess the person's social needs** (including asking about employment or accommodation problems). Make any appropriate social care referrals.
- **For all people with subclinical PTSD symptoms, consider active monitoring and arrange regular follow-up (within one month if the person has reported symptoms in the first month following the trauma).**
- **For people with clinically important PTSD symptoms where the traumatic event occurred within the last month:**

    - For adults, arrange specialist referral for psychological therapy or drug treatment.
    - For children and young adults aged 18 years and under, use clinical judgement to determine whether active monitoring or specialist referral are required. Seek specialist advice where there is any doubt.
- **For all people with clinically important PTSD symptoms that have continued for more than one month after a traumatic event, offer referral to a specialist mental health service for** psychological therapy or drug treatment.
    - Armed forces veterans with service-related post-traumatic stress disorder can be referred to secondary care more rapidly than civilians under the veterans' priority scheme.
    - Availability and arrangements for accessing trauma-focused psychological treatments may vary locally, and there may sometimes be a significant delay before treatment begins.
- **Consider treatment with an antidepressant such as venlafaxine or an SSRI in an adult, if:**
    - The person expresses a preference for drug treatment.
    - The person declines referral for psychological therapies.
    - Referral is significantly delayed.
        - Discuss the potential for adverse effects and withdrawal symptoms before drug treatment is initiated. Explain that adverse effects early in treatment with an SSRI or SNRI may include increased anxiety, agitation, and sleeping problems.
        - Review the effectiveness and adverse effects of the drug every 2 to 4 weeks during the first 3 months of treatment and every 3 months thereafter. Dose adjustment may be required. Bear in mind that the full efficacy of the drug may take several weeks to be realised.
        - **Be aware that in a minority of people aged under 30 years of age, SSRIs and SNRIs are associated with an increased risk of suicidal thinking and self-harm.** Anyone in this age group receiving an SSRI or SNRI should therefore be seen within 1 week of first prescribing, and the risk of suicide and self-harm should be monitored weekly for the first month.
    - Note: Do *not* offer drug treatments for PTSD in children and young people aged under 18 years.
- **For people with problems sleeping:**
    - Advise on sleep hygiene and consider prescribing a hypnotic for short-term use.
- **Enquire about any other family members who have been affected by the trauma to determine if they require assessment and treatment.**
- **Arrange follow-up for monitoring of the person's symptoms, functioning, and response to treatment at intervals determined by clinical judgement and as part of any shared care arrangement (if applicable).**

Psychological therapies and drug treatment

A specialist may offer a person with PTSD:

- **Trauma-focused psychological treatments** (usually considered first-line).
    - **Trauma-focused cognitive behavioural therapy (CBT)** — up to 12 sessions are typically offered. Trauma-focused CBT consists of a combination of exposure therapy and trauma-focused cognitive therapy. More complex presentations are likely to require longer treatment. If a child/young person is being treated, trauma-focused CBT should be adapted to their age and development, and involve parents or carers as appropriate.

        - **Exposure therapy** — the person confronts traumatic memories (usually by recounting the event) and is repeatedly exposed to situations which they have been avoiding that elicit fear.
        - **Trauma-focused cognitive therapy** — this identifies and modifies misrepresentations of the trauma and its aftermath that lead the person to overestimate the threat. For example, rape victims may blame themselves, war veterans may feel that it was their fault a friend was killed, and people who survive accidents may feel that they are in danger of having another accident.
- **Eye movement desensitization and reprocessing (EMDR)** — up to 12 sessions of EMDR are typically offered. More complex presentations are likely to require longer treatment. May be offered to children from the age of seven years.
    - EMDR uses bilateral stimulation (eye movements, taps, and tones) while the person focuses on memories and associations. This is thought to help the brain process flashbacks and to make sense of the traumatic experience.
- **Antidepressants — such as venlafaxine or an SSRI (also suitable for use in primary care), as well as antipsychotics such as risperidone (initiated and monitored under specialist mental health supervision) — may be considered if the an adult:**
    - Expresses a preference for drug treatment.
    - Does not want to/cannot engage in psychological treatments.
    - Has significant comorbid depression or severe hyperarousal, which prevents them from gaining benefit from psychological treatments. Drug treatment may be used as an adjunct to psychological treatments.
    - Has had a poor response to psychological treatments.

Prescription information

SSRI

Choice of SSRI

Note: Only paroxetine and sertraline are licensed in the UK for the treatment of post-traumatic stress disorder (PTSD). Use of other SSRIs for this indication therefore constitutes off-licence use.

- **For paroxetine:** prescribe a starting dose of 20 mg a day to be taken in the morning. The maximum licensed dose for the treatment of PTSD is 50 mg/day in adults with 40 mg/day recommended in the elderly.
- **For sertraline:** prescribe a starting dose of 25 mg daily for one week, then increase to 50 mg daily. Dose may be further increased if required/tolerated in steps of 50 mg, at intervals of at least one week. The maximum licensed dose for the treatment of PTSD is 200 mg a day.

Venlafaxine

Dosage

Note: venlafaxine is not licensed for the treatment of post-traumatic stress disorder and use for this indication, therefore, constitutes an off-licence use.

- Modified release products are recommended for anxiety disorders.
- The recommended starting dose is 75 mg once daily, increased if necessary up to 225 mg once daily. Dose to be increased at intervals of at least 2 weeks; maximum 225 mg per day.

TOPIC: Suicide

Suicide can be described as a fatal act of self-harm initiated with the intention of ending one's own life. Although often seen as impulsive, it may be associated with years of suicidal behaviour including suicidal ideation or acts of deliberate self-harm.

Risk factors for suicide	Protective factors
• Previous suicide attempt or previous self-harm. • A history of self-harm is present in up to three quarters of women who die from suicide, and more than that in those under the age of 25. • Male gender. • Unemployment. • Physical health problems such as disabling or painful illness, including chronic pain. • Living alone. • Being unmarried. • Alcohol and/or drug dependence. • Active mental illness (specifically affective disorder, schizophrenia, personality disorder. • Risk factors for suicide are more common in the prison population;	Evidence is weak but protective factors *may* include: • A strong religious faith. • Family support to find alternative solutions to their problems. • Having children at home. • A sense of responsibility for others. • Problem-solving skills.

Mental disorders and risk of suicide:

- Affective disorders (45%) particularly depression.
- schizophrenia (17%). Studies suggest that patients with schizophrenia have an 8.5-fold greater risk of suicide than the general population[16].
- Personality disorder (8% where it is the primary diagnosis).
- History of alcohol abuse (45%).
- History of drug misuse (33%).

TOPIC: Unexplained Symptoms

In all parts of healthcare system, we meet patients presenting with physical symptoms that lack an obvious organic basis. The terms used to describe such symptoms - medically unexplained symptoms (MUS) or functional somatic symptoms - are purely descriptive and do not imply psychogenesis. Many, of the affected patients do not receive a correct diagnosis and undergo numerous fruitless investigations and attempts at treatment. The narrow focus on the somatic aspects of a complex problem may reinforce their concerns about having a physical disease, make them less satisfied with the healthcare system, contribute to the development of chronic disablement and cause healthcare costs to become excessive.

Causes:

The causative factors for MUS are similar to those for anxiety and depression:

- Stress at home or work, history of childhood or family illness and high deprivation index may all play a part.
- A history of past or recent abuse is often associated - particularly in the case of chronic pelvic pain, in which around a third of patients will have some history of abuse

Presenting features:

The term 'MUS' comprises a wide spectrum of complaints ranging from mild transitory illness to chronic disorders with severe disability.

MUS are more likely if there is a past or current history of depression or anxiety

The exclusion of relevant physical disease may not in itself cure the patient. He or she may still feel ill and seek medical care.

General advice:

- The whole primary care health team should be aware of the diagnosis and management plan. This will make the approach to management consistent across the practice.
- Physical exercise seems to be of benefit
- The importance of pleasurable private time should be emphasized. This may include yoga classes or meditation, bowling or nature walks, which, under the general title of 'stress management', can be presented as necessary medical treatments.

Management:

Medical care of MUS should include improvements in three interrelated elements: diagnosis, specific treatment strategies and communication.

1. **Making a diagnosis**

Diagnosis is not merely the exclusion of serious physical problems but also the combined consideration of MUS and classic psychiatric disorders

A thorough physical examination and diagnostic tests are performed to rule out physical causes - which tests are done are determined by the symptoms present.

A psychological evaluation should also be performed to rule out related disorders. However, finding evidence of a psychiatric condition does not rule somatization in or out. It can be a clue to the diagnosis.

There is considerable evidence that patients with common psychiatric conditions such as depression and anxiety disorders may present to primary care physicians with nonspecific somatic symptoms, including fatigue, aches and pains, palpitations, dizziness and nausea.

Unfortunately, many MUS patients seek care to find an organic disease they fear, but do not have. Doctors then may test for and even treat (non-existent) organic disease. This produces high use of services, unnecessary laboratory testing and consultation, increased costs and high iatrogenic complication rates – ex. ill-advised tests, drug addiction and trial treatments for presumed but absent organic diseases.

2. **Psychotherapy**

Approaches derived from cognitive behavioral therapy have been shown to reduce the intensity and frequency of somatic complaints and to improve functioning in many somatizing patients

3. **Reattribution model**

This can be used by general practitioners after brief training and is based on a cognitive-orientated approach. The key principles are:

- To make the patient feel understood.
- Then to broaden the agenda.
- Finally, to negotiate a new understanding of the symptoms, including psychosocial factors.

4. Communication

5. Pharmacotherapy

- Antidepressants
- SSRIs low dose

Complications:

Complications may result from invasive testing and from multiple evaluations that are performed while looking for the cause of the symptoms. A dependency on pain relievers or sedatives may develop. A poor relationship with the healthcare provider seems to worsen the condition, as does evaluation by many providers.

There are a wide variety of psychiatric terms for patients who have symptoms for which no organic cause can be found:

Somatisation disorder:

- multiple physical symptoms present for at least 2 years
- patient refuses to accept reassurance or negative test result

Illness anxiety disorder (hypochondriasis):

- persistent belief in the presence of an underlying serious DISEASE, e.g. cancer
- patient again refuses to accept reassurance or negative test results

Conversion disorder:

- typically involves loss of motor or sensory function
- the patient doesn't consciously feign the symptoms (factitious disorder) or seek material gain (malingering)
- patients may be indifferent to their apparent disorder - la belle indifference - although this has not been backed up by some studies

Dissociative disorder:

- dissociation is a process of 'separating off' certain memories from normal consciousness
- in contrast to conversion disorder involves psychiatric symptoms e.g. Amnesia, fugue, stupor
- dissociative identity disorder (DID) is the new term for multiple personality disorder as is the most severe form of dissociative disorder

Factitious disorder:

- also known as Munchausen's syndrome
- the intentional production of physical or psychological symptoms

Malingering:

- fraudulent simulation or exaggeration of symptoms with the intention of financial or other gain.

TOPIC: Aphonia

Aphonia describes the inability to speak. Causes include:

- recurrent laryngeal nerve palsy (e.g. Post-thyroidectomy)
- psychogenic

TOPIC: Charles-Bonnet syndrome

Charles-Bonnet syndrome (CBS) is characterised by persistent or recurrent complex hallucinations (usually visual or auditory), occurring in clear consciousness. This is generally against a background of visual impairment (although visual impairment is not mandatory for a diagnosis). Insight is usually preserved. This must occur in the absence of any other significant neuropsychiatric disturbance.

Risk factors include:

- Advanced age
- Peripheral visual impairment
- Social isolation
- Sensory deprivation
- Early cognitive impairment

CBS is equally distributed between sexes and does not show any familial predisposition. The most common ophthalmological conditions associated with this syndrome are age-related macular degeneration, followed by glaucoma and cataract.

TOPIC: Cotard syndrome

Cotard syndrome is a rare mental disorder where the affected patient believes that they (or in some cases just a part of their body) is either dead or non-existent. This delusion is often difficult to treat and can result in significant problems due to patients stopping eating or drinking as they deem it not necessary.

Cotard syndrome is associated with severe depression and psychotic disorders.

TOPIC: De Clerambault's syndrome

De Clerambault's syndrome, also known as erotomania, is a form of paranoid delusion with an amorous quality. The patient, often a single woman, believes that a famous person is in love with her.

TOPIC: Othello's syndrome

Othello's syndrome is pathological jealousy where a person is convinced their partner is cheating on them without any real proof. This is accompanied by socially unacceptable behaviour linked to these claims.

TOPIC: Sleep paralysis

Sleep paralysis is a common condition characterized by transient paralysis of skeletal muscles which occurs when awakening from sleep or less often while falling asleep. It is thought to be related to the paralysis that occurs as a natural part of REM (rapid eye movement) sleep. Sleep paralysis is recognised in a wide variety of cultures

Features:

- paralysis - this occurs after waking up or shortly before falling asleep
- hallucinations - images or speaking that appear during the paralysis

Management:

- if troublesome clonazepam may be used

TOPIC: Delusional parasitosis

Delusional parasitosis is a relatively rare condition where a patient has a fixed, false belief (delusion) that they are infested by 'bugs' e.g. worms, parasites, mites, bacteria, fungus. This may occur in conjunction with other psychiatric conditions or may present by itself, with patients often otherwise quite functional despite the delusion.

TOPIC: Pseudo-hallucination

ICD10 definition of hallucination: false sensory perception in the absence of an external stimulus. Maybe organic, drug-induced or associated with mental disorder.
The definition of a pseudohallucination is harder to pin down. There is no mention of pseudohallucinations in either the ICD10 nor the DSM-5. However, there is a generally accepted definition that a pseudohallucination is a false sensory perception in the absence of external stimuli when the affected is aware that they are hallucinating.

There is disagreement among specialists about not only the definition but also the role in the treatment of pseudohallucinations. Many specialists feel that it is more appropriate to think about hallucinations on a spectrum from mild sensory disturbance to hallucinations to prevent symptoms from being mistreated or misdiagnosed.

An example of a pseudohallucination is a hypnagogic hallucination which occurs when transitioning from wakefulness to sleep. These are experienced vivid auditory or visual hallucinations which are fleeting in duration and may occur in anyone. These are pseudohallucinations as the affected person is able to determine that the hallucination was not real.

The relevance of pseudohallucinations in practice is that patients may need reassurance that these experiences are normal and do not mean that they will develop a mental illness.

Pseudohallucinations commonly occur in people who are grieving.

TOPIC: Post-concussion syndrome

Post-concussion syndrome is seen after even minor head trauma

Typical features include:
- Headache
- Fatigue
- Anxiety/depression
- Dizziness

Player should stop any game for one week.

TOPIC: Electroconvulsive Therapy

Electroconvulsive therapy is a useful treatment option for patients with severe depression refractory to medication (e.g., catatonia) those with psychotic symptoms.

Absolute contraindication

- raised intracranial pressure

Short-term side-effects

- headache
- nausea
- short term memory impairment
- memory loss of events prior to ECT
- cardiac arrhythmia

Long-term side-effects

- some patients report impaired memory

TOPIC: Cognitive Behavioral Therapy (CBT)

Cognitive behavioral therapy is a form of psychotherapy that combines both cognitive therapy and behavioral therapy. It is based on scientific evidence, problem-focused and practical therapy. CBT can be offered to individuals, couples, families or groups. It can be used alone or in conjunction with medication.

Terminology:

- Behavioral therapy: the therapist targets to change harmful and unhelpful behaviors that an individual may have.
- Cognitive therapy: the therapist identifies maladaptive beliefs and targets to correct them.

Conditions That Can be Treated With CBT:

- Depression
- Generalized Anxiety Disorder
- Obsessive convulsive disorder

- Body dysmorphic disorder
- Post-traumatic stress disorder

Sessions:

- The number of therapy sessions depends on the client's problems and needs.
- Typically, sessions usually last about an hour once a week.
- An average course consists of 10-15 sessions.
- Follow-up sessions are planned at the end of therapy to help maintain the progress.

TOPIC: Lithium

Lithium, a mood stabilizing drug, is used most commonly prophylactically in bipolar disorder but also as an adjunct in refractory depression.

Therapeutic range

- It has a very narrow therapeutic range (0.4-1.0 mmol/L)
- It has a long plasma half-life being
- Excreted primarily by the kidneys

Mechanism of action

- Not fully understood and there are two theories:

- interferes with inositol triphosphate formation
- interferes with cAMP formation

Adverse effects:

- nausea/vomiting, diarrhea
- fine tremor
- nephrotoxicity: polyuria, secondary to nephrogenic diabetes insipidus
- thyroid enlargement, may lead to hypothyroidism
- ECG: T wave flattening/inversion
- weight gain
- idiopathic intracranial hypertension
- leukocytosis
- hyperparathyroidism and resultant hypercalcemia

Monitoring of patients on lithium therapy:

- Inadequate monitoring of patients taking lithium is common - NICE and the National Patient Safety Agency (NPSA) have issued guidance to try and address this.
- When checking lithium levels, the sample should be taken 12 hours post-dose
- After starting lithium levels should be performed weekly and after each dose change until concentrations are stable
- Once established, lithium blood level should 'normally' be checked every 3 months
- After a change in dose, lithium levels should be taken a week later and weekly until the levels are stable.
- Thyroid and renal function should be checked every 6 months
- Patients should be issued with an information booklet, alert card and record book

TOPIC: Mirtazapine

Mirtazapine is an atypical antidepressant that works by blocking alpha2-adrenergic receptors, which increases the release of neurotransmitters. It is generally taken in the evening as it can be sedative.

Side effects:

Mirtazapine has fewer side effects and interactions than many other antidepressants and so is useful in older people who may be affected more or be taking other medications. Two side effects of mirtazapine

- Sedation
- increased appetite

Note: it can be beneficial to older people that are suffering from insomnia and poor appetite.

TOPIC: Selective Serotonin Reuptake Inhibitor (SSRI)

Selective serotonin reuptake inhibitors (SSRIs) are considered first-line treatment for the majority of patients with depression.

Preferred SSRI:

- citalopram (although see below re: QT interval) and fluoxetine are currently the preferred SSRIs
- sertraline is useful post myocardial infarction as there is more evidence for its safe use in this situation than other antidepressants
- SSRIs should be used with caution in children and adolescents. Fluoxetine is the drug of choice when an antidepressant is indicated

Adverse effects

- Gastrointestinal symptoms are the most common side-effect
- There is an increased risk of gastrointestinal bleeding in patients taking SSRIs. A proton pump inhibitor should be prescribed if a patient is also taking a NSAID
- Patients should be counselled to be vigilant for increased anxiety and agitation after starting a SSRI
- Hyponatremia
- Fluoxetine and paroxetine have a higher propensity for drug interactions
- Citalopram and the QT interval:
- the Medicines and Healthcare products Regulatory Agency (MHRA) released a warning on the use of citalopram in 2011
- it advised that citalopram and escitalopram are associated with dose-dependent QT interval prolongation and should not be used in those with: congenital long QT syndrome; known pre-existing QT interval prolongation; or in combination with other medicines that prolong the QT interval
- the maximum daily dose is now 40 mg for adults; 20 mg for patients older than 65 years; and 20 mg for those with hepatic impairment

Interactions with other medications:

- NSAIDs: NICE guidelines advise 'do not normally offer SSRIs', but if given co-prescribe a proton pump inhibitor

- warfarin / heparin: NICE guidelines recommend avoiding SSRIs and considering mirtazapine
- aspirin: see above
- triptans - increased risk of serotonin syndrome
- monoamine oxidase inhibitors (MAOIs) - increased risk of serotonin syndrome

Follow up after initiation:

- Following the initiation of antidepressant therapy patients should normally be reviewed by a doctor after 2 weeks.
- Patients under the age of 30 years or at increased risk of suicide should be reviewed after 1 week.
- If a patient makes a good response to antidepressant therapy, they should continue on treatment for at least 6 months after remission as this reduces the risk of relapse.

Withdrawing:

- When stopping SSRI, the dose should be gradually reduced over a 4-week period
- This is not necessary with fluoxetine.
- Note: Paroxetine has a higher incidence of discontinuation symptoms.

Discontinuation symptoms

- increased mood change
- restlessness
- difficulty sleeping
- unsteadiness
- sweating
- gastrointestinal symptoms: pain, cramping, diarrhea, vomiting
- Paresthesia

SSRIs and pregnancy:

- BNF says to weigh up benefits and risk when deciding whether to use in pregnancy.
- Use during the first trimester gives a small increased risk of congenital heart defects
- Use during the third trimester can result in persistent pulmonary hypertension of the newborn
- Paroxetine has an increased risk of congenital malformations, particularly in the first trimester

TOPIC: Tricyclic Antidepressants (TCA)

Tricyclic antidepressants (TCAs) are used less commonly now for depression due to their side-effects and toxicity in overdose. They are however used widely in the treatment of neuropathic pain, where smaller doses are typically required.

Common side-effects:

- drowsiness
- dry mouth
- blurred vision
- constipation
- urinary retention

- lengthening of QT interval monitor by doing ECG

Choice of tricyclic:

- low-dose amitriptyline is commonly used in the management of neuropathic pain and the prophylaxis of headache (both tension and migraine)
- lofepramine has a lower incidence of toxicity in overdose
- amitriptyline and dosulepin (dothiepin) are considered the most dangerous in overdose

More sedative	Less sedative
Amitriptyline Clomipramine Dosulepin Trazodone* - technically a 'tricyclic-related antidepressant'*	Imipramine Lofepramine Nortriptyline

TOPIC: Z Drugs

Z drugs have similar effects to benzodiazepines but are different structurally. They act on the α2-subunit of the GABA receptor.

They can be divided into 3 groups:

- Imidazopyridines: e.g., zolpidem
- Cyclopyrrolones: e.g., zopiclone
- Pyrazolopyrimidines: e.g., zaleplon

Adverse effects:

- similar to benzodiazepines
- increase the risk of falls in the elderly

Comparison:

- They are chemically different and share similar mechanisms of action

Benzodiazepines	Z Drugs
• Bind unselectively to GABA-alfa receptor • This makes benzodiazepine anxiolytics and hypnotics •	• Bind selectively at alfa-1 subunit of GABA receptor • This makes z drugs hypnotics without anxiolytic effect • Has quick onset with short duration
Both are CNS depressant	
• Can be used for anxiety disorders, epilepsy, depression, insomnia... etc.	• Can be used for insomnia • Can be used for anxiety disorder (certain type like buspirone and not as first line)